P

Packaging

The most common materials used in packaging are paper, plastics, and paper coated with plastics. *See* Environmental Accountability; Plastics; Pulp and Paper.

Paper (*see* Pulp and Paper)

Pipe (*see* Some Commonly Used Specifications, Codes, Standards, and Texts)

Plastics

This is an industry that has a vast network of products and producers, as well its own detailed handbooks. For the process engineer, two revolutionary developments on the horizon are worth a look, although neither has yet resulted in gross-scale production.

Today's* growing environmental awareness has forced many businesses to modify their operations and their marketing strategies in response to consumer demands. For instance, a wide range of products available in the market now claim to be "environmentally friendly."

Most kinds of plastic, however, can make no such claim. Widely used to make an almost unlimited number of everyday consumer goods, plastic is generally not biodegradable. As a result, plastic waste has built up, creating a growing disposal problem and posing a threat to the ecology. To make the problem even worse, some types of plastic release potentially harmful chemicals into the atmosphere when incinerated.

In the United States and Western Europe, recycling is becoming the preferred way of "getting rid of" plastic waste. Even in less powerful economies, such as Thailand, some companies have been recycling a limited quantity of plastic for the past few years. But each time plastic is recycled, its quality declines. This means that recycled plastic can mostly be used to manufacture relatively inexpensive, low-quality goods. Another problem with recycling is that it is often difficult to separate plastic waste from other types of refuse. For recycling to work, there must be an efficient garbage sorting system in place. Also, some types of plastic, most notably thermosetting plastics, cannot be recycled at all. This is why researchers have been at work developing another, more environmentally friendly type of plastic, one that is biodegradable.

The processes by which plastic degrades naturally can be divided into these four types:

1. *Environmental degradation.* Environmental degradation is the process by which plastic gradually degrades as a result of surrounding environmental factors, such as sunlight, heat, water, atmospheric conditions, and microorganisms.

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* Source: Adapted from extracts from Chantravekin, "Biodegradable Plastic: An Alternative for Better Environment," *NPC Focus*, Vol. 5, No. 24, February 1997.

2. *Photodegradation.* This is the process by which plastic slowly degrades in the presence of sunlight or ultraviolet rays.
3. *Oxidative degradation.* This is the degradation that results when plastic reacts chemically with oxygen or ozone in the air.
4. *Biodegradation.* Biodegradation occurs when microorganisms such as bacteria and fungi break down the carbon atoms in the plastic molecules.

The key factors that determine the speed at which plastic degrades include the molecular weight and structure of the plastic; its melting point and crystallinity; the volume of microorganisms, temperature; moisture; pH factor; and the quantity of nutrients in the surrounding environment. Generally, it has been found that plastics with low molecular weight, a low melting point, and a relatively low level of crystallinity or a straight molecular structure degrade more quickly than other types of plastic. Degradation occurs because microorganisms release an enzyme that is used to break down the carbon molecules contained in plastic into carbon dioxide and water in the case of aerobic respiration or into other organic substances by the temperature at which the enzyme can function most cost-effectively.

Generally speaking, naturally occurring biopolymers, such as polypeptides, polynucleotides, and polysaccharides are most subject to natural degradation, whereas synthetic polymers like polyethylene, polystyrene, and polyvinyl chloride are most resistant. However, there are two types of synthetic polymers that are biodegradable. These are aliphatic polyesters, including polycarpolactone, which is used in the manufacture of medical equipment, artificial organs, and contraceptive devices, and polyethylene glycol, which is used to produce lubricants and coating materials.

Polyethylene plastic bags that contain a starch filler are another attempt to produce goods with a minimal impact on the environment. But because polyethylene is highly resistant to the process of natural degradation, only the starch filler degrades when the bags are subjected to soil burial. This is why this particular type of plastic is not classified as biodegradable but rather as biodistegrable. Once the filler has degraded, small pieces of plastic remain, and unlike most polyethylene plastic, these pieces have greatly reduced mechanical properties, such as tensile strength and elongation. Just how reduced these properties are depends on the amount of filler used in the manufacture of the bags.

Biodegradable plastics are attracting considerable attention at present, but because they are still relatively expensive, their use is currently restricted almost exclusively to the medical field. Attempts to extend the use of biodegradable plastics to the production of everyday consumer items are being led by a U.S. company called Bioplastics Inc. The company is a joint project between Michigan State University and the Michigan Biotechnology Institute. The company has experimented with a reactive blending process combining cornstarch and polycarpolactone to produce a polymer alloy resin, which goes by the trade name ENVAR.

ENVAR's most outstanding feature is that it has mechanical properties comparable to those of low-density polyethylene (LDPE), thereby making it suitable for use in the manufacture of plastic film and sheeting. ENVAR can be used to produce consumer items such as plastic garbage bags and shopping bags. What distinguishes ENVAR from polyethylene, however, is that it is biodegradable. In soil burial tests, ENVAR was found to be 100 percent biodegradable, its carbon atoms having been converted into carbon dioxide after only three weeks. ENVAR's other principal advantage is that because cornstarch is an inexpensive commodity, the biodegradable plastic is cheaper than synthetic polycarpolactone. This should help to make the dream of using biodegradation plastics in everyday lives a reality.

Two chief advantages could be gained by adapting biodegradable plastics technology to, for instance, countries like Thailand. First, the problem of safety disposing of plastic waste would be greatly reduced. The country's landfills would be less severely taxed by the thousands of tons of plastic waste that are sent for burial every day. Second, the country would find an answer to the trade barriers erected by the European Community against Thai exports of tapioca starch. By using the native-grown starch in the production of biodegradable plastic, the country would add greatly to the value of this crop. To do this, it would be necessary first to submit the starch to a process of plasticization to obtain thermoplastic tapioca starch. This then would be combined with polycarpolactone in a reactive blending process to produce biodegradable plastic resin with similar properties to ENVAR resin. The major drawback to this proposal, however, is that because polycarpolactone is a specialty chemical, it is relatively expensive. This would, in turn, make the production of biodegradable plastic more expensive than other types of plastic.

Nevertheless, it will probably not be long before we begin to see biodegradable plastic goods in markets around the world. It will take cooperation between the private and government sectors and a joint commitment to improving environmental conditions.

Electric Plastics*

Engineers at AGFA in Köln, Germany, were facing a critical problem with the production of photofilm in the late 1980s. Static discharges were ruining the huge, costly rolls of the company's film; induced by friction, the little electric sparks generated big losses. The engineers' investigation showed that the inorganic salts AGFA traditionally used as an antistatic coating failed to work when the humidity dropped below 50 percent. These water-soluble ionic compounds also washed away after developing, again leaving the photofilm vulnerable to stray sparks.

AGFA turned to parent company Bayer AG in Krefeld, Germany, to see whether its central research arm could develop a new low-cost antistatic agent. The antistatic coating had to operate independent of air humidity, with a surface resistance greater than 10^8 ohms square; it had to be transparent and free of heavy metals; and it had to be produced from a waterborne solution.

The most promising candidate to fill these criteria was, surprisingly, an electrically conductive polymer material known as polythiophene. Such polymers have always had great commercial potential because of their unusual ability (for a plastic) to provide a path for electrons, but they had not found any wide commercial applications to that point.

Following a thorough development effort involving the selection of the ideal polythiophene derivative, its subsequent synthesis, and its polymerization, the Bayer research team succeeded in inventing an aqueous processing route for the plastic coating. As of early 1998, more than 10,000 m^2 of AGFA photographic film had been coated with the conductive polymer.

Now the chemical company is marketing the polythiophene under the trade name Baytron. The material could also be used to make plastics paintable by adding the conductive agent first, or in the electrodes of small, high-performance tantalum capacitors found in telecommunications, computer, and automotive products.

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* Source: Adapted from extracts from Ashley, "Electric Plastics," *Mechanical Engineering*, ASME, April 1998.

Another significant potential application is in the through-hole plating of circuit boards. The chemical process of depositing the initial layers of copper into these holes requires formaldehyde, a known carcinogen. Blasberg Oberflae-chentechnik in Soligen, Germany, has patented a method using polythiophene as the first coat instead of the electroless copper. The new plating technology has been licensed to several Japanese circuit-board makers and to Enthone Inc., a subsidiary of ASARCO Inc. in West Haven, Conn.

Long-time researchers on conductive polymers point to Bayer's Baytron polythiophene as the most notable success story in the field. As with most new materials, finding sufficient demand is the key to convincing manufacturers to go into full-scale production. Antistatic applications have a huge potential, but conductive polymers have yet to make many inroads. The once highly acclaimed technology has been reduced to the point that the only successful large application—antistatic coatings for AGFA photofilm—is for internal company use.

Companies from Alstom and AlliedSignal to Westinghouse and W.R. Grace have tried to make conductive polymers into a success, but they have reportedly curtailed or aborted their research. Even though one application for the material—flat-panel displays for televisions and computers—is starting to involve researchers again, much of the payoff for this technology lies in the future.

That future looked a lot brighter for conductive polymers in the 1980s. Probably the most significant commercialization of conductive polymers was for flexible, long-lived batteries that were produced in quantity by Bridgestone Corp. and Seiko Co. in Japan and by BASF/Varta in Germany. Fifteen years ago, when they first came to the market, interest in conductive-polymer batteries was high. In the end, though the batteries worked, they were difficult to sell because their costs were not significantly lower than those of the competition. So the battery products were withdrawn due to insufficient demand. (Researchers at the Johns Hopkins Applied Physics Laboratory in Baltimore recently developed a nontoxic, flexible, all-plastic battery made from another class of conductive plastics called fluorophenylthiophenes, but little is expected of the technology.)

Another once-promising product incorporating conductive polymers is Contex, a fiber that has been manufactured by Milliken & Co. in Spartanburg, S.C., since 1990. The fiber is coated with a conductive-polymer material called polypyrrole and can be woven to create an antistatic fabric. Milliken had been interested in using this type of antistatic technology for its carpet products.

The material's best chance for success was in military applications. Polypyrrole was approved for use in the U.S. Navy's A-12 stealth attack carrier aircraft. The polymer was to be used in edge cards—components that dissipate incoming radar energy by conducting electric charge across a gradient of increasing resistance that the plastic material produces. The A-12 program has been canceled, however.

Milliken also tried to market ultralight camouflage netting based on Contex to help conceal military equipment and personnel from near-infrared and radar detection, but the company lost a U.S. Army contract for conductive camouflage material in 1997. Despite a recent modest contract with NASA to produce conductive-polymer electromagnetic shielding for the space shuttle, Milliken's research program was in financial jeopardy by early 1998.

Despite ups and downs, electrically conductive polymers have attracted a substantial amount of attention since they were accidentally discovered two decades ago, when a Tokyo Institute of Technology student added too much catalyst to a batch of polyacetylene. When the resulting silvery film was later doped with various oxidizing agents at the University of Pennsylvania in Philadelphia, it became conductive, and the race was on to invent new conductive polymers.

Conductive polymers are long, carbon-based chains composed of simple repeating units called monomers. When the Japanese student made his fortuitous error, he

TABLE P-1 Electrical Conductivity of Conductive Polymers and Other Conducting Materials

Material (Year Conductivity Discovered)	Amperes Conducted per Volt Centimeter
Conducting Polymers	
Polyacetylene (1977)	1.7×10^5
Polypyrrole (1979)	7.5×10^3
Poly-*p*-phenylene (1979)	1×10^3
Poly-*p*-phenylenevinylene (1979)	5×10^3
Polyaniline (1980)	2×10^2
Polythiophene (1981)	1×10^3
Other Conducting Materials	
Copper—good conductor	5.8×10^7
Silicon—semiconductor	4×10^{-4}
Quartz—insulator	2×10^{-17}

converted the standard single-bond carbon chains to polymer backbones with alternating single and double bonds, a change that provided a pathway for free-electron-charge carriers. To make the altered polymer materials conductive, they are doped with atoms that donate negative or positive charges (oxidizing or reducing agents) to each unit, enabling current to travel down the chain. Depending on the dopant, conductive polymers exhibit either *p*- or *n*-type conductivity.

The most extensively studied conductive-polymer systems are based on polyaniline, polythiophene, polypyrrole, and polyacetylene. The principal attractions of these polymers over conventional conducting materials are their potential ease of processing, relative robustness, and light weight. Successful commercial applications require a fine balance of conductivity, processability, and stability, but until recently, materials researchers could not obtain all three properties simultaneously.

Conductive polymers are much more electrically conductive than standard polymers but much less than metals such as copper. (See Table P-1.) In practice, the conductivity of these materials is characterized by low-charge carrier mobility—a measure of how easily electric charge moves. This characteristic limits response speed in the case of a transistor, for example, making such a device rather inefficient.

Still, efforts to produce semiconductor devices from conductive polymers are proceeding. In 1994, a team at the Laboratory of Molecular Materials in Thais, France, made a field-effect transistor from polythiophene using printing techniques. Rolling up, bending, and twisting did not affect the transistor's electrical characteristics.

The opportunity to produce relatively low-cost semiconductor devices that are insensitive to mechanical deformation is an attractive one. Probably the most exciting development in this area is the intensifying effort to use conductive polymers to produce flat, flexible plastic screens for TVs and computers. See Fig. P-1. This screen technology emerged from the discovery that certain conductive polymers, such as poly-*p*-phenylenevinylene, emit light when sandwiched between oppositely charged electrodes, a configuration that fits in well with current flat-panel display designs.

The current leader in this work is Cambridge Display Technology (CDT) in Cambridge, England. CDT recently entered into a collaboration with Japanese electronics maker Seiko-Epson to develop light-emitting polymer screens. Philips Electronics NV in the Netherlands is also working on a portable telephone using such a display. Other licensees include Hoechst AG in Germany and Uniax Corp. in Santa Barbara, Calif. While it is likely to be some time before this technology

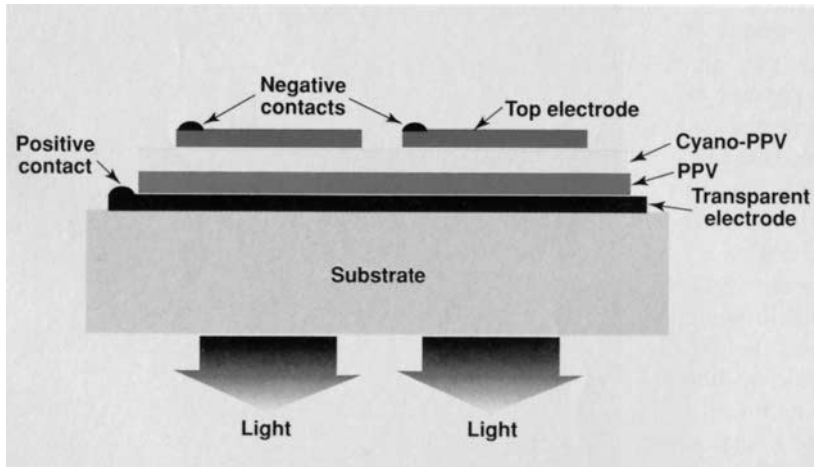


FIG. P-1 Flat-panel display technology for televisions and computers using poly-*p*-phenylenevinylene (PPV) has emerged as one of the most promising applications for conductive polymers. Cambridge Display Technologies in Cambridge, England, is the current leader in this area. (Source: Ashley.)

makes it to the market in flexible flat-panel screens, the development work has created new interest in conductive polymers.

Another promising application is in capacitor technology, where there has been good progress, due mainly to federal funding of ultracapacitors for future electric vehicles. Kemet Electronics Corp. in Greenville, S.C., is working on using polythiophene or polypyrrole to replace manganese dioxide counterelectrodes in tantalum surface-mount capacitors, which are widely used in the electronics industry. Conductive polymers can provide lower equivalent-series resistance (ESR). With the designers of mobile electronics constantly being pushed for space, the new capacitors can simultaneously be smaller and have a lower ESR.

Kemet is operating a small pilot line to produce the electrodes. Kemet predicted that a capacitor product using conductive polymer would be available shortly. High-volume production could follow.

Yet another emerging application for electrically conductive polymer materials is biosensors and chemical sensors, which can convert chemical information into a measurable electrical response. Abtech Scientific Inc. in Yardley, Pa., is making chemical transducers from mostly polyaniline as well as polythiophene and polypropylene for analytical applications in which one measures conductivity and as a result infers what the chemical composition is. In other words, a very small change in the redox composition brought about by small quantities of a range of chemicals can induce a large, rapid change in electrical conductivity.

The challenge for Abtech is how to confer specificity to these materials. One way is to build biopolymer/conductive-polymer complexes. Using this technique, Abtech has developed a range of enzyme biosensors. For example, immobilized glucose oxidase can be incorporated into this polymer transducer system, which acts like a glucose-sensitive biosensor, as the enzyme-catalyzed oxidation of the glucose produces an oxidant by-product that is measured indirectly. Levels of therapeutic drugs in patients can also be monitored in a similar way.

Abtech is developing the technology for point-of-care testing by physicians, a market that is of great interest to several major medical-product companies. The disposable point-of-care-testing product will be used to make many medical tests much cheaper.

An area with some further-off potential—smart membranes of conductive polymers—is being pursued by a team at Los Alamos National Laboratory in Los Alamos, N.M. The lab's Chemical Sciences and Technical Development Division has developed engineered porous-fiber materials with electrically controlled porosity using polyaniline. The technology could find use in gas separation, pharmaceutical separation, environmental cleanup, batteries, or capacitors. A spin-off company to develop the technology already has been established.

The list of potential applications for conductive polymers remains a long one, and includes antiradiation coatings, batteries, catalysts, deicer panels, electrochromic windows, electromechanical actuators, embedded-array antennas, fuel cells, lithographic resists, nonlinear optics, radar dishes, and wave guides. Just how big an impact the materials will make in these markets remains unclear, however. Most observers are putting their money on antistatic coatings and flat-panel displays. Abtech predicts that neither application is going to be a "big winner" initially because they are displacing other established approaches, but they do have promise, especially if they are not oversold.

Pollutants, Chemical; Pollutants, (from) Chemical Processes; Pollutant Indicators; Pollutants, Toxic; Pollutants, Toxic Chemicals (*see also* Emissions; Environmental Accountability)

Toxic chemical pollutants have two main damaging effects to life and the environment:

1. These pollutants generally have a long half-life and can remain unchanged for decades and, in some cases, centuries. They can then move from air to land to water and find their way into the systems of humans or other living organisms. During the winter, some toxins may condense on the ground, and then reenter the atmosphere when the ground warms up.
2. Toxic chemicals can accumulate in fatty tissue of humans and other animals. These chemical concentrations increase as you go higher up the food chain. So, for instance, if we consider polychlorinated biphenyls (PCBs) that migrate, airborne (and via the air they then contaminate water and land), to arctic latitudes, the level of PCBs is higher in polar bears and people (the Inuit) than it is in seals and fish. The bears and the humans are at the top of the food chain.

Toxins travel well, unfortunately. Toxaphene, a pesticide used in the cotton fields of the southern United States, is found in the arctic. So are lead, dioxin, DDT, chlordane, and lindane (industrial chemicals and pesticides). The bad news is that the presence of these compounds in the arctic means they are already well distributed in areas where more people live.

Pollutant indicators are measurements of concentrations of these pollutants in air, water, or soil. The governing environmental bodies in developed countries track these concentrations per indicator that is "tagged" for observation.

Pollutants know no national boundaries, as is illustrated well in the case of the Great Lakes, which are both in Canada and the United States.

The following material is provided to give the reader an indication of how vast the issue of pollutants is. It is relevant that this material—although not comprehensive in terms of the overall subject—be provided. This is because improvements in pollutant measurement technology and constant changes in legislation can make a process plant the potential recipient of an order to shut down or curtail operations. Massive changes in process systems, requiring retrofits and

reengineering, may result. Increasingly, the process engineer must also be, in part, an environmental—or at least an environmentally well intentioned—engineer. If this material raises more questions than it answers, the process engineer may eventually be better off, as this could prompt further reading (see additional reading recommended for this and other related entries).

Note also that the information source Environment Canada is an advisory entity and cannot itself prosecute plants not in compliance with legislated limits. Its information is good, easily available on request from the source, and an excellent benchmark in most cases for end users wanting to set up their own monitoring standards, as well as check on their own. Note also that the U.S. Environmental Protection Agency (EPA) is a regulatory body and can penalize and fine offending companies. The information provided in this book, therefore, should be used as a condensed basis for asking questions in the country the process engineer is working in—not as "complete" in itself.

Country Case Study (Canada)*

The extracts from the "NPRI Summary Report, 1994" that follow include some of the process industry data released by the National Pollutant Release Inventory (NPRI) by Environment Canada.

The NPRI was established to provide a national, publicly accessible database of pollutants released to the Canadian environment from industrial and transportation sources. This second Summary Report reflects the data reported to the NPRI for calendar year 1994 and estimated for various sectors.

Anyone in Canada who owns or operates a facility with 10 or more full-time employees in the reporting year and that manufactures, processes, or otherwise uses any of the NPRI-listed substances, in concentration equal to or greater than 1 percent and in quantities equal to or greater than 10 tonnes (10,000 kg), must file a report with Environment Canada and identify any releases or transfers in waste of those substances to air, water, or land.

What are releases?

A release is an on-site discharge of a substance to the environment. This includes emissions to air, discharges to surface waters, releases to land within the boundaries of the facility, and deep-well injection.

Releases are further subdivided as follows:

- Air
  - Stack/point
  - Storage/handling
  - Fugitive
  - Spills
  - Other nonpoint
- Surface water
  - Direct
  - Spills
  - Leaks

* Source: Environment Canada.

- Land
  - Landfill
  - Land treatment
  - Spills
  - Leaks
  - Other
- Underground injection

Land treatment, also called application farming, is a disposal method in which a waste containing a listed substance is applied or incorporated into soil for biological degradation. Landfills are sites in which wastes are buried. These two disposal methods are generally conducted under permit.

A leak differs from a spill in terms of the time required for an event. Spills normally occur over a period of hours to days, whereas leaks occur over periods of days to months.

Underground injection is another method of waste disposal. Wastes are injected into known geological formations, generally at great depths. This disposal method is subject to provincial regulation.

What are transfers?

A transfer is a shipment of a listed substance in waste to an off-site location. Facilities must provide the name and location of the off-site facility receiving the shipment. Waste is defined as material that is sent for final disposal or for treatment prior to final disposal. There are seven off-site disposal or treatment methods:

- Physical treatment such as encapsulation and vitrification
- Chemical treatment such as stabilization and neutralization
- Biological treatment such as bio-oxidation
- Incineration or thermal treatment
- Containment in a landfill and other storage
- Municipal sewage treatment plant
- Underground injection

Off-site transfers in waste are reported separately from on-site releases because:

- Off-site transfers represent a movement of the substance to a different geographic location than that of the facility
- Transfers off site may not necessarily represent entry of the substance into the environment
- Management of the substance becomes the responsibility of another owner or operator
- Reporting on off-site transfers completes information on the fate of the substance
- Wastes could be transferred a number of times leading to some double counting

The NPRI requires that only the quantity of the listed substance in the waste be reported. Waste materials, such as sludges, are often a mixture of many compounds associated with water and other inert material with a small quantity of potentially hazardous substances. As a result, the total reported to the NPRI may be smaller than the quantity reported in other inventories since only the net weight of a listed substance is reported.

What is recovery, re-use, and recycling?

As with transfers, recovery, re-use, and recycling (3Rs) represent a movement of the substance to an off-site facility, generally under the jurisdiction of another owner or operator. Facilities were required to report the name and the address of the receiving off-site facility.

The definition of waste for the 1994 reporting year has been modified from that of 1993. For 1994 reporting, waste excluded substances sent for the 3Rs. These 3Rs could be reported voluntarily under another section of the reporting form.

Generally, materials sent for 3Rs are those transferred to recyclers, such as metal shavings or turnings, those materials transferred off-site for processing, cleaning, or reclamation and returned to the facility, and those materials sent back to the suppliers for credit or payment.

Energy recovery is applicable only when recuperated energy from combustion is used as an alternative to fossil fuels or other forms of energy.

Extracts from Appendices 4 through 11 of NPRI Summary Report, 1994

Appendix 4: List of NPRI substances and releases (tonnes)

CAS #	Substance Name	Total Releases
75-07-0	Acetaldehyde	133.287
67-64-1	Acetone	3,841.890
75-05-8	Acetonitrile	79.130
79-06-1	Acrylamide	5.878
79-10-7	Acrylic acid	0.889
107-13-1	Acrylonitrile	19.556
7429-90-5	Aluminum (fume or dust)	60.326
1344-28-1	Aluminum oxide (fibrous forms)	34.609
7664-41-7	Ammonia	26,465.316
6484-52-2	Ammonium nitrate (solution)	1,900.121
7783-20-2	Ammonium sulfate (solution)	1,193.082
62-53-3	Aniline	0.005
120-12-7	Anthracene	2.517
NA	Antimony (and its compounds)	19.398
NA	Arsenic (and its compounds)	3,980.656
1332-21-4	Asbestos	352.184
71-43-2	Benzene	2,675.468
94-36-0	Benzoyl peroxide	0.000
100-44-7	Benzyl chloride	0.142
92-52-4	Biphenyl	15.435
103-23-1	*Bis*(2-ethylhexyl) adipate	7.276
117-81-7	*Bis*(2-ethylhexyl) phthalate	92.804
106-99-0	1,3-Butadiene	310.180
141-32-2	Butyl acrylate	1.141
78-83-1	*i*-Butyl alcohol	103.616
71-36-3	*n*-Butyl alcohol	1,248.821
78-92-2	*sec*-Butyl alcohol	0.000
75-65-0	*tert*-Butyl alcohol	91.512
85-68-7	Butyl benzyl phthalate	3.699
4680-78-8	C.I. Acid Green 3	0.000
NA	Cadmium (and its compounds)	96.041
156-62-7	Calcium cyanamide	0.000
75-15-0	Carbon disulfide	26.168
56-23-5	Carbon tetrachloride	18.135
7782-50-5	Chlorine	2,086.098
10049-04-4	Chlorine dioxide	1,735.537

CAS #	Substance Name	Total Releases
79-11-8	Chloroacetic acid	0.000
75-00-3	Chloroethane	336.931
67-66-3	Chloroform	0.172
74-87-3	Chloromethane	1,177.083
NA	Chromium (and its compounds)	800.859
NA	Cobalt (and its compounds)	29.182
NA	Copper (and its compounds)	16,069.629
1319-77-3	Cresol (mixed isomers)	2.224
95-48-7	*o*-Cresol	0.001
98-82-8	Cumene	28.859
80-15-9	Cumene hydroperoxide	0.020
NA	Cyanides (ionic)	17.008
110-82-7	Cyclohexane	2,730.773
1163-19-5	Decabromodiphenyl oxide	0.000
84-74-2	Dibutyl phthalate	12.258
95-50-1	*o*-Dichlorobenzene	0.480
106-46-7	*p*-Dichlorobenzene	10.400
107-06-2	1,2-Dichloroethane	7.579
75-09-2	Dichloromethane	2,222.089
111-42-2	Diethanolamine	790.216
84-66-2	Diethyl phthalate	0.000
77-78-1	Dimethyl sulfate	0.011
534-52-1	4,6-Dinitro-*o*-cresol	0.000
121-14-2	2,4-Dinitrotoluene	4.000
117-84-0	Di-*n*-octyl phthalate	0.480
123-91-1	1,4-Dioxane	9.144
106-89-8	Epichlorohydrin	0.133
110-80-5	2-Ethoxyethanol	2.354
111-15-9	2-Ethoxyethyl acetate	92.003
140-88-5	Ethyl acrylate	1.120
100-41-4	Ethylbenzene	537.561
74-85-1	Ethylene	2,558.541
107-21-1	Ethylene glycol	3,333.492
75-21-8	Ethylene oxide	51.067
50-00-0	Formaldehyde	1,116.417
302-01-2	Hydrazine	1.242
7647-01-0	Hydrochloric acid	2,293.897
74-90-8	Hydrogen cyanide	0.130
7664-39-3	Hydrogen fluoride	28.597
123-31-9	Hydroquinone	0.021
67-63-0	Isopropyl alcohol	12,132.317
80-05-7	*p*,*p*′-Isopropylidenediphenol	0.169
NA	Lead (and its compounds)	2,142.220
108-31-6	Maleic anhydride	27.609
NA	Manganese (and its compounds)	3,087.366
NA	Mercury (and its compounds)	3.806
67-56-1	Methanol	30,403.335
109-86-4	2-Methoxyethanol	17.000
96-33-3	Methyl acrylate	2.112
1634-04-4	Methyl *tert*-butyl ether	28.157
101-14-4	*p*,*p*′-Methylene*bis*(2-chloroaniline)	0.005
101-68-8	Methylene*bis*(phenylisocyanate)	1.914
78-93-3	Methyl ethyl ketone	4,448.431
108-10-1	Methyl isobutyl ketone	894.763
80-62-6	Methyl methacrylate	32.442
1313-27-5	Molybdenum trioxide	44.094
91-20-3	Naphthalene	84.592
NA	Nickel (and its compounds)	704.496
7697-37-2	Nitric acid	64.920
139-13-9	Nitrilotriacetic acid	1.001
55-63-0	Nitroglycerin	9.000
79-46-9	2-Nitropropane	0.125
79-21-0	Peracetic acid	0.000
108-95-2	Phenol	205.405

CAS #	Substance Name	Total Releases
90-43-7	*o*-Phenylphenol	0.000
7664-38-2	Phosphoric acid	820.110
7723-14-0	Phosphorus (yellow or white)	14.611
85-44-9	Phthalic anhydride	13.250
115-07-1	Propylene	1,188.704
75-56-9	Propylene oxide	11.071
NA	Selenium (and its compounds)	8.208
NA	Silver (and its compounds)	4.674
100-42-5	Styrene	1,792.518
96-09-3	Styrene oxide	0.130
7664-93-9	Sulfuric acid	22,294.515
127-18-4	Tetrachloroethylene	163.335
62-56-6	Thiourea	0.000
7550-45-0	Titanium tetrachloride	0.023
108-88-3	Toluene	7,723.317
584-84-9	Toluene-2,4-diisocyanate	0.148
26471-62-5	Toluenediisocyanate (mixed isomers)	1.033
79-00-5	1,1,2-Trichloroethane	0.000
79-01-6	Trichloroethylene	859.483
95-63-6	1,2,4-Trimethylbenzene	398.468
7440-62-2	Vanadium (fume or dust)	30.552
108-05-4	Vinyl acetate	126.078
75-01-4	Vinyl chloride	23.725
75-35-4	Vinylidene chloride	0.000
1330-20-7	Xylene (mixed isomers)	8,878.386
108-38-3	*m*-Xylene	18.764
95-47-6	*o*-Xylene	11.775
106-42-3	*p*-Xylene	16.274
NA	Zinc (and its compounds)	10,006.590
7440-66-6	Zinc (fume or dust)	656.034
	Total released	**190,263.945**

Appendix 5: NPRI substances released by two-digit SIC code (tonnes)

02—Service industries incidental to agriculture

Substance Name	Air	Water	Underground	Land	Total Releases*
Cobalt (and its compounds)	0.000	0.000	0.000	0.000	0.000
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.000
Manganese (and its compounds)	0.000	0.000	0.000	0.000	0.000
Selenium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

NOTE: See p. P-31 for footnote.

03—Fishing and trapping industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Chlorine	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

04—Logging industry

Substance Name	Air	Water	Underground	Land	Total Releases*
Methanol	77.900	0.000	0.000	2.800	80.700
Sulfuric acid	10.240	0.000	0.000	0.000	10.240
Chlorine dioxide	7.169	0.000	0.000	0.000	7.169
Ammonia	1.650	0.830	0.000	0.000	2.480
Chlorine	0.680	0.000	0.000	0.000	0.680
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Total	**97.639**	**0.830**	**0.000**	**2.800**	**101.269**

06—Mining industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Copper (and its compounds)	79.183	14,431.987	0.000	0.473	14,512.520
Zinc (and its compounds)	141.500	4,489.490	0.000	0.492	4,632.582
Arsenic (and its compounds)	38.296	45.489	3,800.000	0.000	3,883.785
Ammonia	570.560	436.503	0.000	0.000	1,007.064
Sulfuric acid	82.457	719.000	0.000	25.000	826.717
Phosphoric acid	0.000	774.000	0.000	0.000	774.000
Ammonium sulphate (solution)	95.000	577.000	0.000	0.000	672.720
Acetone	377.250	0.000	0.000	0.000	377.250
Lead (and its compounds)	174.654	154.108	0.000	0.025	329.454
Methanol	0.000	0.000	0.000	37.400	37.400
Ethylene glycol	27.070	0.000	0.000	6.400	33.470
Hydrochloric acid	0.000	0.000	7.000	13.050	20.240
Cyanides (ionic)	8.467	6.705	0.000	0.000	15.776
Cadmium (and its compounds)	6.665	6.581	0.000	0.000	13.246
Antimony (and its compounds)	9.600	3.300	0.000	0.000	12.900
Manganese (and its compounds)	0.202	11.812	0.000	0.000	12.014
Nitric acid	10.560	0.000	0.000	0.000	10.690
Selenium (and its compounds)	0.185	3.896	0.000	0.000	4.081
Mercury (and its compounds)	3.580	0.148	0.000	0.000	3.728
Silver (and its compounds)	0.130	3.320	0.000	0.000	3.670
Nickel (and its compounds)	0.601	2.911	0.000	0.000	3.512
Zinc (fume or dust)	3.350	0.032	0.000	0.000	3.493
Hydrogen fluoride	2.969	0.132	0.000	0.000	3.102
Ammonium nitrate (solution)	1.048	0.000	0.000	0.000	1.071
Asbestos	0.000	0.000	0.000	0.000	0.587
Chlorine	0.160	0.000	0.000	0.000	0.160
Acrylamide	0.000	0.000	0.000	0.000	0.130
Isopropyl alcohol	0.000	0.000	0.000	0.000	0.000
Molybdenum trioxide	0.000	0.000	0.000	0.000	0.000
Methyl isobutyl ketone	0.000	0.000	0.000	0.000	0.000
Chromium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Cobalt (and its compounds)	0.000	0.000	0.000	0.000	0.000
Carbon disulfide	0.000	0.000	0.000	0.000	0.000
Total	**1,633.487**	**21,666.414**	**3,807.000**	**82.840**	**27,195.362**

07—Crude petroleum and natural gas industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Hydrochloric acid	0.020	0.000	924.00	0.000	924.020
Diethanolamine	2.940	3.800	694.713	1.028	702.511
Methanol	18.269	0.000	496.854	4.222	522.120
Ethylene glycol	70.721	8.690	390.132	38.200	508.623

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluene	260.407	0.008	15.860	0.365	283.157
Xylene (mixed isomers)	234.722	0.009	1.043	0.757	242.664
Benzene	175.608	0.008	45.900	0.459	228.494
Ammonia	44.145	1.600	48.600	0.000	94.345
Propylene	79.161	0.000	0.000	0.000	79.291
Cyclohexane	46.891	0.000	0.000	0.084	53.171
Asbestos	0.000	0.000	0.000	51.000	51.000
Ethylene	50.660	0.000	0.000	0.000	50.790
1,2,4-Trimethylbenzene	45.543	0.000	0.000	0.552	49.921
Molybdenum trioxide	0.000	0.000	0.000	41.800	41.800
Nickel (and its compounds)	5.234	0.449	0.000	27.190	32.873
Ethylbenzene	30.870	0.000	0.000	0.158	32.638
Vanadium (fume or dust)	19.040	0.000	0.000	0.000	19.040
Chlorine	5.320	0.394	11.670	0.000	17.384
Isopropyl alcohol	0.130	0.000	2.702	0.000	2.832
Naphthalene	2.580	0.000	0.000	0.170	2.750
Cumene	2.240	0.000	0.000	0.044	2.284
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.880
Carbon disulphide	0.448	0.000	0.000	0.000	0.448
Sulfuric acid	0.130	0.000	0.000	0.000	0.220
o-Xylene	0.000	0.000	0.000	0.000	0.219
Methyl ethyl ketone	0.000	0.000	0.190	0.000	0.190
Manganese (and its compounds)	0.170	0.000	0.000	0.000	0.170
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.130
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	0.000	0.000
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Cobalt (and its compounds)	0.000	0.000	0.000	0.000	0.000
Total	**1,095.249**	**14.958**	**2,631.664**	**166.029**	**3,943.965**

09—Service industries incidental to mineral extraction

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	3.553	0.000	0.000	0.000	3.553
Methanol	1.247	0.000	0.000	0.000	1.247
Isopropyl alcohol	1.083	0.000	0.000	0.000	1.083
Toluene	0.783	0.000	0.000	0.000	0.783
Styrene	0.728	0.000	0.000	0.000	0.728
1,2,4-Trimethylbenzene	0.293	0.000	0.000	0.000	0.293
Ethylene glycol	0.197	0.000	0.000	0.000	0.197
Ethylbenzene	0.046	0.000	0.000	0.000	0.046
Total	**7.930**	**0.000**	**0.000**	**0.000**	**7.930**

10—Food industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Hydrochloric acid	0.000	16.900	0.000	0.000	17.835
Ammonia	10.000	0.000	0.000	0.000	10.130
Nitric acid	10.000	0.000	0.000	0.000	10.000
Phosphoric acid	0.000	0.000	0.000	0.000	0.130
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.000
Manganese (and its compounds)	0.000	0.000	0.000	0.000	0.000
Selenium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.000
Nickel (and its compounds)	0.000	0.000	0.000	0.000	0.000

Substance Name	Air	Water	Underground	Land	Total Releases*
Peracetic acid	0.000	0.000	0.000	0.000	0.000
Formaldehyde	0.000	0.000	0.000	0.000	0.000
Chlorine dioxide	0.000	0.000	0.000	0.000	0.000
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Trichloroethylene	0.000	0.000	0.000	0.000	0.000
Chlorine	0.000	0.000	0.000	0.000	0.000
Total	**20.000**	**16.900**	**0.000**	**0.000**	**38.095**

11—Beverage industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ammonia	5.200	0.000	0.000	0.000	5.200
Methanol	2.220	0.000	0.000	0.000	2.220
Isopropyl alcohol	1.480	0.000	0.000	0.000	1.480
Toluene	1.265	0.000	0.000	0.000	1.265
Acetone	0.910	0.000	0.000	0.000	0.910
Tetrachloroethylene	0.000	0.000	0.000	0.000	0.630
Dichloromethane	0.000	0.000	0.000	0.000	0.630
Methyl ethyl ketone	0.000	0.000	0.000	0.000	0.380
n-Butyl alcohol	0.000	0.000	0.000	0.000	0.380
Xylene (mixed isomers)	0.000	0.000	0.000	0.000	0.380
Chloroethane	0.000	0.000	0.000	0.000	0.130
Ethylene glycol	0.000	0.000	0.000	0.000	0.130
Trichloroethylene	0.000	0.000	0.000	0.000	0.130
Nitric acid	0.000	0.000	0.000	0.000	0.000
Formaldehyde	0.000	0.000	0.000	0.000	0.000
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Total	**11.075**	**0.000**	**0.000**	**0.000**	**13.865**

15—Rubber products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	1,273.236	0.000	0.000	0.000	1,273.236
Toluene	670.091	0.000	0.000	0.000	670.091
Methyl ethyl ketone	125.664	0.000	0.000	0.000	125.664
Trichloroethylene	97.863	0.000	0.000	0.000	97.863
Bis(2-ethylhexyl) phthalate	11.957	0.000	0.000	12.624	24.581
Zinc (and its compounds)	1.857	0.391	0.000	15.300	19.175
Methyl isobutyl ketone	16.941	0.000	0.000	0.000	16.941
Asbestos	0.000	0.000	0.000	10.800	10.800
Ethylbenzene	10.267	0.000	0.000	0.000	10.267
Sulfuric acid	2.200	0.000	0.000	0.000	2.200
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	1.854	1.854
Styrene	0.000	0.000	0.000	0.000	0.500
Ammonia	0.000	0.421	0.000	0.000	0.421
Lead (and its compounds)	0.298	0.000	0.000	0.000	0.298
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.026
Copper (and its compounds)	0.000	0.012	0.000	0.000	0.012
Cobalt (and its compounds)	0.000	0.000	0.000	0.000	0.000
Bis(2-ethylhexyl) adipate	0.000	0.000	0.000	0.000	0.000
Di-*n*-octyl phthalate	0.000	0.000	0.000	0.000	0.000
1,3-Butadiene	0.000	0.000	0.000	0.000	0.000
Acrylonitrile	0.000	0.000	0.000	0.000	0.000

Substance Name	Air	Water	Underground	Land	Total Releases*
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Zinc (fume or dust)	0.000	0.000	0.000	0.000	0.000
Epichlorohydrin	0.000	0.000	0.000	0.000	0.000
Total	**2,210.374**	**0.824**	**0.000**	**40.578**	**2,253.929**

16—Plastic products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Dichloromethane	1,356.821	0.000	0.000	0.000	1,356.821
Methyl ethyl ketone	1,195.662	0.000	0.000	0.000	1,195.917
Toluene	385.113	0.000	0.000	0.000	385.748
Isopropyl alcohol	303.485	0.000	0.000	0.000	303.610
Styrene	198.445	0.380	0.000	0.000	200.720
Acetone	177.062	0.000	0.000	0.000	177.317
Chloroethane	162.331	0.000	0.000	0.000	162.331
Methanol	109.702	1.970	0.000	0.000	111.802
n-Butyl alcohol	89.117	18.680	0.000	0.000	108.052
2-Ethoxyethyl acetate	92.000	0.000	0.000	0.000	92.000
Formaldehyde	0.173	85.450	0.000	0.000	85.753
Bis(2-ethylhexyl) phthalate	8.730	0.000	0.000	55.847	64.608
Trichloroethylene	33.995	0.000	0.000	0.000	33.995
Ammonia	0.090	17.880	0.000	0.000	18.126
1,3-Butadiene	17.529	0.000	0.000	0.000	17.529
Bis(2-ethylhexyl) adipate	0.000	5.600	0.000	1.660	7.260
i-Butyl alcohol	0.012	6.440	0.000	0.000	6.582
Methyl methacrylate	4.545	0.000	0.000	0.000	4.545
Xylene (mixed isomers)	0.749	0.752	0.000	0.000	2.136
Acrylonitrile	1.669	0.122	0.000	0.000	2.046
Antimony (and its compounds)	0.000	0.000	0.000	0.000	1.083
Methylene*bis*(phenylisocyanate)	0.001	0.000	0.000	0.300	1.043
Toluenediisocyanate (mixed isomers)	0.230	0.000	0.000	0.000	0.631
Phosphoric acid	0.000	0.000	0.000	0.000	0.510
Methyl isobutyl ketone	0.000	0.000	0.000	0.000	0.385
1,2,4-Trimethylbenzene	0.260	0.000	0.000	0.000	0.260
Vinyl chloride	0.000	0.000	0.000	0.000	0.255
Ethylene	0.000	0.000	0.000	0.000	0.255
Propylene	0.000	0.000	0.000	0.000	0.255
Phenol	0.023	0.000	0.000	0.000	0.153
Lead (and its compounds)	0.000	0.000	0.000	0.000	0.142
Nitric acid	0.000	0.000	0.000	0.000	0.130
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.130
Phthalic anhydride	0.000	0.000	0.000	0.000	0.130
Styrene oxide	0.000	0.000	0.000	0.000	0.130
Chromium (and its compounds)	0.000	0.000	0.000	0.000	0.130
Dibutyl phthalate	0.000	0.000	0.000	0.000	0.130
Sulfuric acid	0.000	0.000	0.000	0.000	0.130
Ethyl acrylate	0.000	0.000	0.000	0.000	0.130
Hydrogen cyanide	0.000	0.000	0.000	0.000	0.130
Nickel (and its compounds)	0.000	0.000	0.000	0.000	0.130
Propylene oxide	0.000	0.000	0.000	0.000	0.130
Hydrochloric acid	0.000	0.000	0.000	0.000	0.130
Ethylene glycol	0.000	0.000	0.000	0.000	0.125
Ammonium nitrate (solution)	0.000	0.000	0.000	0.000	0.125
Cyclohexane	0.000	0.000	0.000	0.000	0.125
2-Nitropropane	0.000	0.000	0.000	0.000	0.125
Epichlorohydrin	0.000	0.000	0.000	0.000	0.125
Di-*n*-octyl phthalate	0.000	0.000	0.000	0.000	0.104
Acrylic acid	0.000	0.000	0.000	0.000	0.010

Substance Name	Air	Water	Underground	Land	Total Releases*
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.007
Diethanolamine	0.000	0.000	0.000	0.000	0.002
Maleic anhydride	0.001	0.000	0.000	0.000	0.001
Toluene-2,4-diisocyanate	0.000	0.000	0.000	0.000	0.000
Butyl benzyl phthalate	0.000	0.000	0.000	0.000	0.000
Manganese (and its compounds)	0.000	0.000	0.000	0.000	0.000
Cadmium (and its compounds)	0.000	0.000	0.000	0.000	0.000
C.I. Acid Green 3	0.000	0.000	0.000	0.000	0.000
Total	**4,137.745**	**137.274**	**0.000**	**57.807**	**4,344.179**

17—Leather and allied products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluene	22.865	0.000	0.000	0.000	22.865
2-Methoxyethanol	17.000	0.000	0.000	0.000	17.000
Acetone	11.470	0.000	0.000	0.000	11.470
Xylene (mixed isomers)	10.200	0.000	0.000	0.000	10.200
Ammonium sulphate (solution)	0.000	0.000	0.000	0.000	0.000
Chromium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Bis(2-ethylhexyl) phthalate	0.000	0.000	0.000	0.000	0.000
Methylene*bis* (phenylisocyanate)	0.000	0.000	0.000	0.000	0.000
Total	**61.535**	**0.000**	**0.000**	**0.000**	**61.535**

18—Primary textile industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Sulfuric acid	0.000	85.000	0.000	0.000	85.000
Acetaldehyde	43.551	0.000	0.000	0.000	43.551
Xylene (mixed isomers)	30.324	0.000	0.000	0.000	30.324
Biphenyl	8.137	0.060	0.000	0.059	8.256
1,4-Dioxane	1.161	4.643	0.000	0.000	5.884
Chlorine	0.300	0.700	0.000	0.000	1.000
Methanol	0.000	1.000	0.000	0.000	1.000
Ethylene glycol	0.000	0.000	0.000	0.000	0.840
Tetrachloroethylene	0.000	0.000	0.000	0.000	0.700
Antimony (and its compounds)	0.000	0.000	0.000	0.000	0.355
Phosphoric acid	0.002	0.000	0.000	0.000	0.002
Phenol	0.000	0.000	0.000	0.000	0.000
Total	**83.475**	**91.403**	**0.000**	**0.059**	**176.912**

19—Textile products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Methyl ethyl ketone	186.098	0.000	0.000	0.000	186.098
Acetone	108.936	0.000	0.000	0.000	108.936
Sulfuric acid	0.000	0.000	0.000	0.000	0.493
Phenol	0.000	0.000	0.000	0.000	0.130
Formaldehyde	0.000	0.000	0.000	0.000	0.130
Lead (and its compounds)	0.009	0.000	0.000	0.000	0.009

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluenediisocyanate (mixed isomers)	0.001	0.000	0.000	0.000	0.001
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Ammonium sulphate (solution)	0.000	0.000	0.000	0.000	0.000
Total	**295.044**	**0.000**	**0.000**	**0.000**	**295.797**

25—Wood industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Formaldehyde	455.173	0.000	0.000	0.000	455.173
Methanol	149.151	0.000	0.000	0.000	149.151
Benzene	29.206	0.000	0.000	0.000	29.206
Phenol	16.354	0.000	0.000	0.000	16.354
Xylene (mixed isomers)	16.000	0.000	0.000	0.000	16.000
Toluene	14.398	0.000	0.000	0.000	14.398
Acetone	12.633	0.000	0.000	0.000	12.633
Isopropyl alcohol	5.000	0.000	0.000	0.000	5.000
Ammonia	3.221	0.000	0.000	0.000	3.221
Ethylene glycol	0.000	0.000	0.000	1.600	1.600
Methylene*bis*(phenylisocyanate)	0.173	0.000	0.000	0.000	0.303
Chromium (and its compounds)	0.000	0.020	0.000	0.000	0.170
Arsenic (and its compounds)	0.000	0.009	0.000	0.000	0.109
Copper (and its compounds)	0.000	0.007	0.000	0.000	0.078
Sulfuric acid	0.023	0.000	0.000	0.000	0.023
Anthracene	0.000	0.000	0.000	0.000	0.000
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Biphenyl	0.000	0.000	0.000	0.000	0.000
Naphthalene	0.000	0.000	0.000	0.000	0.000
Total	**701.332**	**0.036**	**0.000**	**1.600**	**703.419**

26—Furniture and fixture industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluene	292.430	0.000	0.000	0.000	292.430
Xylene (mixed isomers)	119.212	0.000	0.000	0.000	119.212
Methyl ethyl ketone	62.476	0.000	0.000	0.000	62.476
n-Butyl alcohol	18.681	0.000	0.000	0.000	18.681
Dichloromethane	18.207	0.000	0.000	0.000	18.207
1,2,4-Trimethylbenzene	13.143	0.000	0.000	0.000	13.143
Trichloroethylene	13.000	0.000	0.000	0.000	13.000
Methyl isobutyl ketone	11.160	0.000	0.000	0.000	11.160
Styrene	10.614	0.000	0.000	0.000	10.614
Toluenediisocyanate (mixed isomers)	0.003	0.000	0.000	0.000	0.003
Diethanolamine	0.000	0.000	0.000	0.000	0.000
Total	**558.926**	**0.000**	**0.000**	**0.000**	**558.926**

27—Paper and allied products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Methanol	8,229.524	11,973.449	0.000	125.928	20,329.031
Sulfuric acid	352.996	2,719.703	0.000	0.000	3,072.959
Chlorine	1,930.280	0.000	0.000	0.000	1,931.812

Substance Name	Air	Water	Underground	Land	Total Releases*
Chlorine dioxide	1,721.135	5.273	0.000	0.000	1,726.408
Toluene	1,242.286	0.000	0.000	0.000	1,242.286
Xylene (mixed isomers)	1,132.400	0.000	0.000	0.000	1,132.400
Methyl ethyl ketone	937.295	0.010	0.000	0.000	937.305
Hydrochloric acid	507.649	9.128	0.000	0.000	517.337
Isopropyl alcohol	398.601	0.440	0.000	0.000	399.041
Ammonia	101.345	188.077	0.000	0.000	290.369
Formaldehyde	19.912	221.560	0.000	0.130	241.602
Acetone	212.710	0.315	0.000	0.000	213.155
Phenol	92.400	2.430	0.000	0.000	94.830
Ethylene glycol	76.165	17.390	0.000	0.346	93.901
Ammonium nitrate (solution)	0.000	26.700	0.000	0.000	26.700
Phosphoric acid	0.000	25.000	0.000	0.000	25.130
Methyl isobutyl ketone	22.800	0.000	0.000	0.000	22.800
n-Butyl alcohol	14.000	0.000	0.000	0.000	14.000
Acrylamide	0.000	0.316	0.000	2.845	3.161
Vinyl acetate	3.000	0.000	0.000	0.000	3.000
Asbestos	0.000	0.000	0.000	3.000	3.000
Diethanolamine	1.200	0.000	0.000	0.000	1.200
Zinc (and its compounds)	0.090	0.000	0.000	0.000	0.090
Copper (and its compounds)	0.039	0.000	0.000	0.000	0.039
Manganese (and its compounds)	0.000	0.000	0.000	0.000	0.000
Tetrachloroethylene	0.000	0.000	0.000	0.000	0.000
Nitric acid	0.000	0.000	0.000	0.000	0.000
Aluminum (fume or dust)	0.000	0.000	0.000	0.000	0.000
Total	**16,995.844**	**15,189.791**	**0.000**	**132.249**	**32,321.573**

28—Printing, publishing and allied industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluene	706.528	5.181	0.000	0.000	711.709
Isopropyl alcohol	330.919	0.544	0.000	0.000	331.463
Methanol	144.961	0.000	0.000	0.000	144.961
Methyl ethyl ketone	100.487	0.000	0.000	0.000	100.487
Acetone	36.768	0.000	0.000	0.000	36.768
Cyclohexane	21.365	0.120	0.000	0.000	21.485
Xylene (mixed isomers)	15.936	0.077	0.000	0.000	16.013
Dibutyl phthalate	11.427	0.000	0.000	0.000	11.427
Naphthalene	10.000	0.000	0.000	0.000	10.000
Tetrachloroethylene	8.808	0.000	0.000	0.000	8.808
n-Butyl alcohol	2.620	0.000	0.000	0.000	2.620
Sulfuric acid	0.438	0.000	0.000	0.000	0.438
Nitric acid	0.386	0.000	0.000	0.000	0.386
Ammonium nitrate (solution)	0.000	0.000	0.000	0.000	0.000
Ethylene glycol	0.000	0.000	0.000	0.000	0.000
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Total	**1,390.643**	**5.922**	**0.000**	**0.000**	**1,396.565**

29—Primary metal industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Zinc (and its compounds)	1,026.974	33.898	0.000	3,835.094	4,898.773
Manganese (and its compounds)	132.340	40.585	0.000	2,744.702	2,918.725
Sulfuric acid	2,804.985	7.500	0.000	4.000	2,817.179

Substance Name	Air	Water	Underground	Land	Total Releases*
Lead (and its compounds)	889.192	2.388	0.000	859.628	1,753.169
Copper (and its compounds)	478.457	11.529	0.000	961.388	1,453.894
Benzene	1,110.960	0.007	0.000	0.130	1,111.097
Styrene	1,101.453	0.004	0.000	0.000	1,102.870
Ammonia	451.258	500.167	0.000	0.000	951.425
Chromium (and its compounds)	7.382	28.319	0.000	739.668	777.575
Nickel (and its compounds)	541.693	65.379	0.000	46.839	655.459
Zinc (fume or dust)	38.264	57.000	0.000	554.000	649.644
Trichloroethylene	419.483	0.000	0.000	0.000	419.483
Toluene	135.092	0.005	0.000	0.000	135.097
Tetrachloroethylene	122.180	0.000	0.000	0.000	122.180
Isopropyl alcohol	103.300	0.000	0.000	0.000	104.118
Hydrochloric acid	89.224	0.000	0.000	11.400	100.624
Arsenic (and its compounds)	94.634	1.706	0.000	0.000	96.472
Cadmium (and its compounds)	66.105	1.717	0.000	14.000	82.795
Chlorine	78.274	1.095	0.000	0.000	80.399
Ethylene glycol	8.365	54.056	0.000	13.564	76.945
Xylene (mixed isomers)	60.560	0.006	0.000	0.000	60.566
Ethylene	37.180	0.000	0.000	0.000	37.180
Aluminum (fume or dust)	20.443	4.790	0.000	11.900	37.133
Methanol	24.560	11.000	0.000	0.000	35.560
Cobalt (and its compounds)	11.172	0.990	0.000	8.800	21.092
Acetone	20.800	0.000	0.000	0.000	20.800
Naphthalene	19.054	0.011	0.000	0.010	19.075
Propylene	15.300	0.000	0.000	0.000	15.300
Phenol	6.260	7.585	0.000	0.000	13.845
Hydrogen fluoride	12.900	0.290	0.000	0.000	13.190
Formaldehyde	5.075	0.000	0.000	0.000	5.075
Nitric acid	1.295	3.500	0.000	0.000	5.055
Selenium (and its compounds)	0.220	3.560	0.000	0.000	4.127
Vanadium (fume or dust)	0.061	0.030	0.000	3.600	3.702
Antimony (and its compounds)	1.150	0.000	0.000	2.400	3.668
Ethylbenzene	1.460	0.000	0.000	0.000	1.460
Cyanides (ionic)	0.000	1.162	0.000	0.000	1.162
Anthracene	0.871	0.037	0.000	0.003	0.911
Silver (and its compounds)	0.206	0.016	0.000	0.008	0.874
Molybdenum trioxide	0.000	0.000	0.000	0.000	0.323
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	0.000	0.260
Phosphoric acid	0.000	0.000	0.000	0.000	0.130
Bis(2-ethylhexyl) adipate	0.000	0.000	0.000	0.000	0.010
Ammonium sulfate (solution)	0.000	0.000	0.000	0.004	0.004
Mercury (and its compounds)	0.000	0.000	0.000	0.000	0.002
Hydrazine	0.000	0.000	0.000	0.000	0.000
Decabromodiphenyl oxide	0.000	0.000	0.000	0.000	0.000
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.000
Cumene hydroperoxide	0.000	0.000	0.000	0.000	0.000
Dichloromethane	0.000	0.000	0.000	0.000	0.000
Thiourea	0.000	0.000	0.000	0.000	0.000
Total	**9,938.182**	**838.332**	**0.000**	**9,811.138**	**20,608.427**

30—Fabricated metal products industries (except machinery and trans. equipment industries)

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	594.745	0.000	0.000	0.000	594.954
n-Butyl alcohol	409.842	0.000	0.000	0.000	410.875
Methyl ethyl ketone	162.496	0.000	0.000	0.000	163.197

Substance Name	Air	Water	Underground	Land	Total Releases*
Trichloroethylene	148.430	0.000	0.000	0.000	148.430
Zinc (and its compounds)	92.328	0.263	0.000	20.995	116.578
Toluene	110.015	0.000	0.000	0.000	110.143
Methyl isobutyl ketone	93.574	0.000	0.000	0.000	94.347
Copper (and its compounds)	86.047	0.000	0.000	0.314	87.006
Hydrochloric acid	12.785	0.050	0.000	51.000	64.074
Dichloromethane	42.474	0.000	0.000	0.000	42.474
Acetone	33.984	0.000	0.000	0.000	33.984
Methanol	12.792	0.000	0.000	0.000	12.792
Sulfuric acid	6.163	0.400	0.000	0.150	8.198
Lead (and its compounds)	5.032	1.531	0.000	0.000	7.083
Isopropyl alcohol	6.370	0.000	0.000	0.000	6.370
Chromium (and its compounds)	2.935	0.021	0.000	0.176	3.865
Ethylene glycol	2.500	0.000	0.000	0.000	2.500
Zinc (fume or dust)	0.000	0.000	0.000	0.000	1.482
Manganese (and its compounds)	0.288	0.002	0.000	0.262	1.406
Phosphoric acid	0.000	0.000	0.000	0.000	1.260
Bis(2-ethylhexyl) phthalate	0.870	0.000	0.000	0.000	0.870
Nickel (and its compounds)	0.651	0.000	0.000	0.194	0.845
Aluminum (fume or dust)	0.130	0.000	0.000	0.630	0.760
Nitric acid	0.055	0.000	0.000	0.000	0.410
Ethylbenzene	0.000	0.000	0.000	0.000	0.369
Ammonia	0.000	0.000	0.000	0.000	0.131
Methylene*bis*(phenylisocyanate)	0.001	0.000	0.000	0.000	0.131
Formaldehyde	0.070	0.000	0.000	0.000	0.070
Methyl methacrylate	0.061	0.000	0.000	0.000	0.061
Hydrogen fluoride	0.010	0.000	0.000	0.000	0.010
Di-*n*-octyl phthalate	0.000	0.000	0.000	0.000	0.006
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	0.000	0.004
Chlorine	0.000	0.000	0.000	0.000	0.000
Antimony (and its compounds)	0.000	0.000	0.000	0.000	0.000
Cadmium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Total	**1,824.666**	**2.267**	**0.000**	**73.721**	**1,914.685**

31—Machinery industries (except electrical machinery)

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	35.946	0.000	0.000	0.000	35.946
1,2,4-Trimethylbenzene	34.785	0.000	0.000	0.000	34.785
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	31.600	31.600
Sulfuric acid	0.000	21.700	0.000	0.000	21.700
Vanadium (fume or dust)	3.600	0.000	0.000	1.600	5.200
Manganese (and its compounds)	0.134	0.000	0.000	0.630	1.050
Copper (and its compounds)	0.000	0.000	0.000	0.130	0.130
Chromium (and its compounds)	0.000	0.000	0.000	0.130	0.130
Dibutyl phthalate	0.000	0.000	0.000	0.000	0.130
Nickel (and its compounds)	0.000	0.000	0.000	0.130	0.130
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.000
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Toluene	0.000	0.000	0.000	0.000	0.000
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.000
Ethylene glycol	0.000	0.000	0.000	0.000	0.000
Total	**74.465**	**21.700**	**0.000**	**34.220**	**130.801**

32—Transportation equipment industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	3,187.581	0.000	0.000	0.000	3,187.581
Toluene	1,482.724	0.000	0.000	0.000	1,482.894
Acetone	629.204	0.000	0.000	0.000	629.204
Methyl ethyl ketone	566.682	0.000	0.000	0.000	566.682
n-Butyl alcohol	553.828	0.000	0.000	0.000	554.348
Methyl isobutyl ketone	571.276	0.000	0.000	0.000	517.276
Methanol	253.186	0.000	0.000	0.000	253.697
Isopropyl alcohol	245.095	0.000	0.000	0.000	245.095
Styrene	132.352	0.000	0.000	0.000	132.352
Asbestos	0.447	0.000	0.000	105.840	106.287
Ethylbenzene	106.270	0.000	0.000	0.001	106.271
i-Butyl alcohol	89.955	0.000	0.000	0.000	90.765
Trichloroethylene	89.334	0.000	0.000	0.000	89.334
1,2,4-Trimethylbenzene	41.413	0.000	0.000	0.000	41.413
Ammonia	39.488	0.000	0.000	0.000	39.488
Phenol	16.763	0.000	0.000	0.000	16.763
Aluminum (fume or dust)	6.615	0.000	0.000	5.752	12.367
Dichloromethane	11.627	0.000	0.000	0.000	11.627
Ethylene glycol	9.487	0.110	0.000	0.000	10.477
Zinc (and its compounds)	2.829	2.075	0.000	0.260	5.277
Copper (and its compounds)	3.530	0.006	0.000	0.000	3.698
Chromium (and its compounds)	0.049	0.000	0.000	0.000	2.878
Hydrochloric acid	1.288	0.000	0.000	0.000	1.662
Diethanolamine	1.107	0.000	0.000	0.000	1.607
Lead (and its compounds)	0.895	0.015	0.000	0.000	1.054
Manganese (and its compounds)	0.509	0.012	0.000	0.000	1.020
Phosphoric acid	0.555	0.000	0.000	0.000	0.555
o-Dichlorobenzene	0.380	0.000	0.000	0.000	0.380
Sulfuric acid	0.329	0.000	0.000	0.000	0.373
Nickel (and its compounds)	0.199	0.000	0.000	0.000	0.232
Cresol (mixed isomers)	0.130	0.000	0.000	0.000	0.130
Methylene*bis*(phenylisocyanate)	0.003	0.000	0.000	0.110	0.123
Nitric acid	0.018	0.000	0.000	0.000	0.018
Toluenediisocyanate (mixed isomers)	0.000	0.000	0.000	0.000	0.016
Zinc (fume or dust)	0.010	0.000	0.000	0.000	0.010
Butyl benzyl phthalate	0.002	0.000	0.000	0.000	0.002
Vinylidene chloride	0.000	0.000	0.000	0.000	0.000
Calcim cyanamide	0.000	0.000	0.000	0.000	0.000
Toluene-2,4-diisocyanate	0.000	0.000	0.000	0.000	0.000
Antimony (and its compounds)	0.000	0.000	0.000	0.000	0.000
Decabromodiphenyl oxide	0.000	0.000	0.000	0.000	0.000
Bis(2-ethylhexyl) phthalate	0.000	0.000	0.000	0.000	0.000
Total	**7,991.16**	**2.218**	**0.000**	**111.963**	**8,112.956**

33—Electrical and electronic products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	237.340	0.000	0.000	0.000	237.940
Isopropyl alcohol	58.915	0.000	0.000	0.000	58.915
Toluene	50.297	0.000	0.000	0.000	50.297
Trichloroethylene	33.309	0.022	0.000	0.000	33.331
Tetrachloroethylene	22.070	0.000	0.000	0.000	22.070
1,2,4-Trimethylbenzene	18.430	0.000	0.000	0.000	18.430
Ethylene glycol	16.260	0.000	0.000	0.000	16.260
n-Butyl alcohol	13.904	0.000	0.000	0.000	13.904
Ammonia	0.327	10.120	0.000	0.000	10.447

Substance Name	Air	Water	Underground	Land	Total Releases*
Copper (and its compounds)	0.013	0.036	0.000	5.044	6.517
Lead (and its compounds)	2.597	1.079	0.000	0.669	5.366
Phenol	2.840	0.000	0.000	0.000	2.840
Cresol (mixed isomers)	1.960	0.000	0.000	0.000	1.960
Hydrochloric acid	1.100	0.000	0.000	0.000	1.100
Zinc (and its compounds)	0.171	0.089	0.000	0.535	0.795
Nitric acid	0.000	0.000	0.000	0.000	0.690
Cumene	0.000	0.000	0.000	0.000	0.630
Sulfuric acid	0.000	0.000	0.000	0.000	0.610
Antimony (and its compounds)	0.000	0.000	0.000	0.000	0.490
Manganese (and its compounds)	0.130	0.000	0.000	0.000	0.130
Formaldehyde	0.130	0.000	0.000	0.000	0.130
Toluenediisocyanate (mixed isomers)	0.000	0.000	0.000	0.000	0.112
Cobalt (and its compounds)	0.000	0.000	0.000	0.000	0.000
Nickel (and its compounds)	0.000	0.000	0.000	0.000	0.000
Zinc (fume or dust)	0.000	0.000	0.000	0.000	0.000
Aluminum (fume or dust)	0.000	0.000	0.000	0.000	0.000
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.000
Chromium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Hydrogen fluoride	0.000	0.000	0.000	0.000	0.000
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Total	**459.793**	**11.346**	**0.000**	**6.248**	**482.964**

35—Nonmetallic mineral products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluene	661.297	0.000	0.000	0.000	661.427
Xylene (mixed isomers)	601.563	0.000	0.000	0.000	601.693
Ammonia	239.798	0.036	0.000	0.000	239.834
Isopropyl alcohol	175.479	45.415	0.000	0.000	220.894
Methanol	146.324	0.000	0.000	0.000	146.324
Zinc (and its compounds)	127.685	0.040	0.130	9.200	137.185
Formaldehyde	134.520	1.276	0.000	0.630	136.426
n-Butyl alcohol	73.070	0.000	0.000	0.000	73.070
Manganese (and its compounds)	70.000	0.000	0.000	0.630	71.760
Dichloromethane	66.000	0.000	0.000	0.000	66.130
Methyl ethyl ketone	50.335	0.000	0.000	0.000	50.335
Lead (and its compounds)	30.000	0.000	0.000	5.700	35.708
Phenol	21.484	0.000	0.000	0.000	21.484
1,2,4-Trimethylbenzene	17.970	0.000	0.000	0.000	17.970
Acetone	5.134	0.000	0.000	0.000	5.134
Cyclohexane	4.107	0.000	0.000	0.000	4.107
Chromium (and its compounds)	1.723	0.010	0.000	2.300	4.037
Ammonium sulphate (solution)	3.898	0.000	0.000	0.000	3.898
Ethylene glycol	0.000	0.000	0.000	0.000	0.140
Tetrachloroethylene	0.000	0.000	0.000	0.000	0.130
Chloroform	0.000	0.000	0.000	0.000	0.130
1,2-Dichloroethane	0.000	0.000	0.000	0.000	0.130
Trichloroethylene	0.000	0.000	0.000	0.000	0.130
Phosphoric acid	0.000	0.000	0.000	0.000	0.021
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Cobalt (and its compounds)	0.000	0.000	0.000	0.000	0.000
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Total	**2,430.387**	**46.777**	**0.130**	**18.460**	**2,498.097**

36—Refined petroleum and coal products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ammonia	335.477	138.887	5,021.400	0.020	5,495.784
Toluene	947.616	0.793	19.160	1.525	969.333
Methanol	820.260	0.000	0.000	0.000	820.468
Xylene (mixed isomers)	753.650	0.976	13.590	1.100	769.866
Benzene	498.697	0.582	20.390	0.745	520.634
Propylene	516.889	0.000	0.000	0.000	516.889
Ethylene	422.360	0.000	0.000	0.000	423.370
Methyl ethyl ketone	247.280	0.000	0.000	0.000	247.280
Cyclohexane	205.556	0.067	0.050	1.367	207.670
1,2,4-Trimethylbenzene	159.210	0.087	0.610	0.017	160.339
Ethylbenzene	154.430	0.213	1.870	0.442	157.259
Methyl isobutyl ketone	116.830	0.000	0.000	0.000	116.830
Sulfuric acid	81.469	0.000	0.000	0.000	81.469
Diethanolamine	44.435	21.400	0.000	0.000	66.685
Isopropyl alcohol	41.060	0.000	0.000	0.000	41.710
Naphthalene	30.663	0.395	0.045	0.084	31.187
Cumene	24.627	0.020	0.029	0.000	25.645
Acetone	24.710	0.081	0.000	0.000	24.791
Hydrochloric acid	0.000	0.000	22.680	0.000	22.680
1,3-Butadiene	22.200	0.000	0.000	0.000	22.289
Manganese (and its compounds)	0.198	0.000	0.000	22.000	22.198
Methyl *tert*-butyl ether	19.231	0.400	0.000	0.000	19.631
Ethylene glycol	18.175	1.200	0.000	0.000	19.375
Phenol	9.223	5.221	1.980	0.000	16.554
m-Xylene	13.150	0.000	0.000	0.000	13.150
p-Xylene	8.080	0.000	0.000	0.000	8.080
o-Xylene	7.791	0.000	0.000	0.000	7.791
Zinc (and its compounds)	0.059	0.607	0.077	5.910	6.653
Styrene	6.190	0.000	0.000	0.000	6.190
Biphenyl	1.729	0.000	0.000	0.000	1.729
Anthracene	1.030	0.000	0.000	0.000	1.476
Chlorine	1.260	0.063	0.000	0.000	1.323
Nickel (and its compounds)	0.123	0.000	0.000	0.850	1.003
Copper (and its compounds)	0.100	0.000	0.000	0.730	0.830
2-Ethoxyethanol	0.490	0.000	0.000	0.000	0.490
Chromium (and its compounds)	0.000	0.000	0.000	0.450	0.450
Molybdenum trioxide	0.319	0.000	0.000	0.000	0.441
Nitric acid	0.300	0.000	0.000	0.000	0.396
Arsenic (and its compounds)	0.000	0.000	0.000	0.290	0.290
Lead (and its compounds)	0.000	0.000	0.000	0.140	0.259
Phosphoric acid	0.150	0.000	0.000	0.000	0.226
Tetrachloroethylene	0.130	0.064	0.000	0.000	0.194
Dichloromethane	0.000	0.000	0.000	0.000	0.100
Hydrogen fluoride	0.000	0.000	0.000	0.000	0.039
o-Dichlorobenzene	0.020	0.000	0.000	0.000	0.020
Mercury (and its compounds)	0.000	0.000	0.000	0.011	0.011
Vanadium (fume or dust)	0.010	0.000	0.000	0.000	0.010
Cresol (mixed isomers)	0.000	0.000	0.000	0.000	0.004
Carbon tetrachloride	0.000	0.000	0.000	0.000	0.000
Asbestos	0.000	0.000	0.000	0.000	0.000
Cobalt (and tis compounds)	0.000	0.000	0.000	0.000	0.000
Butyl benzyl phthalate	0.000	0.000	0.000	0.000	0.000
Vinyl chloride	0.000	0.000	0.000	0.000	0.000
Trichloroethylene	0.000	0.000	0.000	0.000	0.000
Total	**5,535.177**	**171.056**	**5,101.881**	**35.681**	**10,851.091**

37—Chemical and chemical products industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ammonia	16,062.078	335.649	998.000	3.475	17,403.510
Sulfuric acid	108.269	15,002.700	0.000	2.196	15,116.683
Methanol	7,099.540	43.716	400.000	0.025	7,548.896
Cyclohexane	2,443.454	0.500	0.000	0.100	2,444.054
Acetone	1,984.247	38.550	84.000	49.400	2,159.373
Ethylene	2,046.103	0.000	0.000	0.000	2,046.946
Ammonium nitrate (solution)	334.100	663.100	828.000	47.023	1,872.225
Chloromethane	1,176.020	0.880	0.000	0.000	1,176.900
Benzene	776.256	0.455	7.600	1.577	786.018
Dichloromethane	668.759	0.000	0.000	0.039	670.490
Hydrochloric acid	566.804	54.000	0.000	0.662	623.788
Toluene	579.833	0.348	0.310	3.691	590.656
Propylene	576.964	0.000	0.000	0.000	576.969
Ammonium sulphate (solution)	43.900	454.300	0.000	18.260	516.460
Methyl ethyl ketone	155.973	0.000	280.000	0.129	440.391
Xylene (mixed isomers)	355.526	0.131	0.000	0.423	365.759
Isopropyl alcohol	322.428	0.130	0.000	0.000	328.446
Styrene	309.701	0.020	0.185	0.196	314.172
1,3-Butadiene	270.302	0.058	0.000	0.002	270.362
Ethylbenzene	196.770	0.101	0.000	0.175	197.046
Formaldehyde	102.345	2.504	69.920	0.000	178.702
Chloroethane	174.100	0.370	0.000	0.000	174.470
Ethylene glycol	145.933	9.487	0.000	0.084	159.636
Vinyl acetate	111.339	0.000	7.700	0.695	121.578
Asbestos	0.130	0.000	0.000	98.380	98.510
tert-Butyl alcohol	48.410	1.100	42.000	0.002	91.512
Acetaldehyde	70.506	13.200	6.000	0.030	89.736
Methyl isobutyl ketone	79.696	0.000	0.000	0.155	80.631
Acetonitrile	79.000	0.130	0.000	0.000	79.130
1,2,4-Trimethylbenzene	60.712	0.146	0.000	0.000	61.034
Manganese (and its compounds)	0.133	58.000	0.000	0.000	58.893
n-Butyl alcohol	48.593	0.000	0.000	0.100	52.459
Ethylene oxide	50.842	0.000	0.000	0.000	51.067
Nitric acid	0.179	32.700	0.000	1.600	35.038
Methyl methacrylate	27.289	0.000	0.000	0.000	27.829
Maleic anhydride	26.662	0.000	0.000	0.000	27.478
Carbon disulphide	13.435	10.405	0.000	0.000	25.720
Vinyl chloride	23.146	0.164	0.000	0.000	23.340
Phenol	19.595	0.599	0.000	0.000	22.452
Naphthalene	20.442	0.000	0.000	0.237	21.580
Diethanolamine	17.378	0.000	0.100	0.000	17.821
Acrylonitrile	16.600	0.040	0.000	0.000	17.510
Carbon tetrachloride	12.773	4.732	0.000	0.000	17.505
Chlorine	8.430	3.160	0.000	0.000	15.832
Phosphorus (yellow or white)	0.025	4.586	0.000	10.000	14.611
Phthalic anhydride	11.115	0.000	0.000	0.000	13.120
Hydrogen fluoride	12.200	0.000	0.000	0.000	12.256
Propylene oxide	10.941	0.000	0.000	0.000	10.941
p-Dichlorobenzene	9.900	0.000	0.000	0.500	10.400
Nickel (and its compounds)	5.194	3.610	0.000	0.000	10.311
Phosphoric acid	2.841	2.212	0.000	3.000	10.037
Nitroglycerin	0.000	9.000	0.000	0.000	9.000
Methyl *tert*-butyl ether	8.396	0.000	0.000	0.000	8.396
p-Xylene	8.194	0.000	0.000	0.000	8.194
Cobalt (and its compounds)	6.840	1.120	0.000	0.000	8.090
Lead (and its compounds)	5.370	0.360	0.000	0.000	7.545
1,2-Dichloroethane	6.886	0.523	0.000	0.040	7.449
Tetrachloroethylene	6.440	0.010	0.000	0.000	7.448
Zinc (and its compounds)	1.300	2.369	0.000	0.000	6.504
Aluminum (fume or dust)	0.001	0.000	0.380	4.810	6.156
i-Butyl alcohol	4.457	0.000	0.000	0.000	6.137

Substance Name	Air	Water	Underground	Land	Total Releases*
Chromium (and its compounds)	1.700	0.708	0.000	0.000	5.694
m-Xylene	5.604	0.000	0.000	0.000	5.614
Biphenyl	5.000	0.000	0.000	0.320	5.450
Copper (and its compounds)	1.340	2.509	0.000	0.530	4.515
2,4-Dinitrotoluene	0.000	4.000	0.000	0.000	4.000
o-Xylene	3.765	0.000	0.000	0.000	3.765
Vanadium (fume or dust)	0.000	2.600	0.000	0.000	2.600
Methyl acrylate	2.112	0.000	0.000	0.000	2.112
Acrylamide	0.897	0.000	0.000	0.000	1.977
Chloride dioxide	1.960	0.000	0.000	0.000	1.960
2-Ethoxyethanol	1.415	0.000	0.000	0.000	1.864
Molybdenum trioxide	1.400	0.000	0.000	0.000	1.530
Zinc (fume or dust)	1.405	0.000	0.000	0.000	1.405
Butyl acrylate	0.435	0.000	0.000	0.000	1.141
Nitrilotriacetic acid	0.000	0.000	0.000	0.000	1.001
Ethyl acrylate	0.130	0.000	0.000	0.000	0.990
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	0.000	0.890
Acrylic acid	0.139	0.000	0.000	0.000	0.879
1,4-Dioxane	0.860	0.000	0.000	0.000	0.860
Antimony (and its compounds)	0.009	0.003	0.000	0.000	0.772
Trichloroethylene	0.000	0.000	0.000	0.000	0.702
Dibutyl phthalate	0.309	0.000	0.000	0.000	0.569
Bis(2-ethylhexyl) phthalate	0.244	0.000	0.000	0.161	0.415
Di-*n*-octyl phthalate	0.180	0.000	0.000	0.000	0.310
Cumene	0.300	0.000	0.000	0.000	0.300
p,*p*′-Isopropylidenediphenol	0.000	0.000	0.000	0.000	0.169
Toluene-2,4-diisocyanate	0.000	0.000	0.000	0.000	0.148
Toluenediisocyanate (mixed isomers)	0.011	0.000	0.000	0.000	0.147
Benzyl chloride	0.012	0.000	0.000	0.000	0.142
Anthracene	0.000	0.000	0.000	0.000	0.130
Cresol (mixed isomers)	0.000	0.000	0.000	0.000	0.130
Butyl benzyl phthalate	0.130	0.000	0.000	0.000	0.130
o-Dichlorobenzene	0.000	0.000	0.000	0.000	0.080
Cyanides (ionic)	0.000	0.000	0.000	0.000	0.070
Mercury (and its compounds)	0.057	0.008	0.000	0.000	0.065
Chloroform	0.039	0.002	0.000	0.001	0.042
Methylene*bis*(phenylisocyanate)	0.026	0.000	0.000	0.000	0.028
Titanium tetrachloride	0.023	0.000	0.000	0.000	0.023
Hydroquinone	0.000	0.000	0.000	0.000	0.021
Cumene hydroperoxide	0.000	0.000	0.000	0.000	0.020
Dimethyl sulphate	0.011	0.000	0.000	0.000	0.011
Epichlorohydrin	0.000	0.000	0.000	0.000	0.008
Bis(2-ethylhexyl) adipate	0.000	0.000	0.000	0.000	0.006
p,*p*′-Methylene*bis*(2-chloroaniline)	0.000	0.000	0.000	0.000	0.005
Aniline	0.005	0.000	0.000	0.000	0.005
2-Ethoxyethyl acetate	0.003	0.000	0.000	0.000	0.003
o-Cresol	0.000	0.000	0.000	0.000	0.001
2-Methoxyethanol	0.000	0.000	0.000	0.000	0.000
Cadmium (and its compounds)	0.000	0.000	0.000	0.000	0.000
4,6-Dinitro-*o*-cresol	0.000	0.000	0.000	0.000	0.000
Chloroacetic acid	0.000	0.000	0.000	0.000	0.000
sec-Butyl alcohol	0.000	0.000	0.000	0.000	0.000
1,1,2-Trichloroethane	0.000	0.000	0.000	0.000	0.000
Peracetic acid	0.000	0.000	0.000	0.000	0.000
Hydrazine	0.000	0.000	0.000	0.000	0.000
Benzoyl peroxide	0.000	0.000	0.000	0.000	0.000
Decabromodiphenyl oxide	0.000	0.000	0.000	0.000	0.000
o-Phenylphenol	0.000	0.000	0.000	0.000	0.000
Total	**37,364.316**	**16,764.995**	**2,724.195**	**248.018**	**57,207.901**

39—Other manufacturing industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Methyl ethyl ketone	321.720	0.000	0.000	0.000	321.720
Methanol	95.900	0.000	0.000	0.000	96.195
Trichloroethylene	22.748	0.000	0.000	0.000	22.748
Styrene	12.030	0.000	0.000	0.000	12.030
Xylene (mixed isomers)	10.858	0.000	0.000	0.000	10.858
Methyl isobutyl ketone	10.042	0.000	0.000	0.000	10.042
Phosphoric acid	7.420	0.000	0.000	0.130	7.550
Chromium (and its compounds)	0.000	0.000	0.000	5.930	5.930
Butyl benzyl phthalate	3.550	0.000	0.000	0.000	3.550
Acetone	2.850	0.000	0.000	0.000	2.852
Bis(2-ethylhexyl) phthalate	2.330	0.000	0.000	0.000	2.330
Lead (and its compounds)	1.603	0.000	0.000	0.000	1.993
Nitric acid	1.800	0.000	0.000	0.000	1.800
1,2,4-Trimethylbenzene	0.880	0.000	0.000	0.000	0.880
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.270
Silver (and its compounds)	0.000	0.000	0.000	0.000	0.130
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.130
Ethylene glycol	0.000	0.000	0.000	0.000	0.130
Antimony (and its compounds)	0.000	0.000	0.000	0.000	0.130
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.130
Vinyl chloride	0.000	0.000	0.000	0.000	0.130
Toluenediisocyanate (mixed isomers)	0.000	0.000	0.000	0.000	0.123
Nickel (and its compounds)	0.000	0.000	0.000	0.000	0.001
Isopropyl alcohol	0.000	0.000	0.000	0.000	0.001
Aluminum oxide (fibrous forms)	0.000	0.000	0.000	0.001	0.001
Naphthalene	0.000	0.000	0.000	0.000	0.000
1,1,2-Trichloroethane	0.000	0.000	0.000	0.000	0.000
Benzene	0.000	0.000	0.000	0.000	0.000
Ethylene	0.000	0.000	0.000	0.000	0.000
Aluminum (fume or dust)	0.000	0.000	0.000	0.000	0.000
p-Dichlorobenzene	0.000	0.000	0.000	0.000	0.000
Toluene	0.000	0.000	0.000	0.000	0.000
Methyl *tert*-butyl ether	0.000	0.000	0.000	0.000	0.000
Ammonia	0.000	0.000	0.000	0.000	0.000
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Chlorine	0.000	0.000	0.000	0.000	0.000
Dichloromethane	0.000	0.000	0.000	0.000	0.000
Ethylene oxide	0.000	0.000	0.000	0.000	0.000
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Total	**493.731**	**0.000**	**0.000**	**6.061**	**501.654**

41—Industrial and heavy (engineering) construction industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ammonia	111.798	13.170	0.000	585.200	710.168
Sulfuric acid	0.000	114.382	0.000	0.000	114.382
Asbestos	0.000	0.000	0.000	82.000	82.000
Chlorine	1.558	0.390	0.000	0.000	1.948
Acrylamide	0.000	0.610	0.000	0.000	0.610
Hydrazine	0.000	0.000	0.000	0.000	0.345
Xylene (mixed isomers)	0.000	0.000	0.000	0.000	0.300
Lead (and its compounds)	0.000	0.000	0.000	0.000	0.000
Nickel (and its compounds)	0.000	0.000	0.000	0.000	0.000
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.000
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.000
Chromium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Antimony (and its compounds)	0.000	0.000	0.000	0.000	0.000

Substance Name	Air	Water	Underground	Land	Total Releases*
Bis(2-ethylhexyl) phthalate	0.000	0.000	0.000	0.000	0.000
Aluminum (fume or dust)	0.000	0.000	0.000	0.000	0.000
Ethylene glycol	0.000	0.000	0.000	0.000	0.000
Vinyl acetate	0.000	0.000	0.000	0.000	0.000
Chlorine dioxide	0.000	0.000	0.000	0.000	0.000
Nitric acid	0.000	0.000	0.000	0.000	0.000
Vanadium (fume or dust)	0.000	0.000	0.000	0.000	0.000
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Total	**113.356**	**128.552**	**0.000**	**667.200**	**909.753**

42—Trade contracting industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	72.092	0.000	0.000	0.000	72.092
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.130
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Total	**72.092**	**0.000**	**0.000**	**0.000**	**72.222**

44—Service industries incidental to construction

Substance Name	Air	Water	Underground	Land	Total Releases*
Ethylene glycol	0.000	0.000	0.000	0.000	0.000
Methanol	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

45—Transportation industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ethylene glycol	2.180	0.000	0.000	2,224.068	2,226.378
Styrene	0.000	10.000	0.000	0.000	10.000
Propylene	0.000	0.000	0.000	0.000	0.000
Total	**2.180**	**10.000**	**0.000**	**2,224.068**	**2,236.378**

46—Pipeline transport industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Xylene (mixed isomers)	75.200	0.000	0.000	0.000	75.200
Phosphoric acid	0.000	0.000	0.000	0.000	0.000
Methylene*bis*(phenylisocyanate)	0.000	0.000	0.000	0.000	0.000
Total	**75.200**	**0.000**	**0.000**	**0.000**	**75.200**

47—Storage and warehousing industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Methanol	1.600	0.000	0.000	0.000	1.610
Vinyl acetate	1.500	0.000	0.000	0.000	1.500
n-Butyl alcohol	0.000	0.000	0.000	0.000	0.000
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Methyl ethyl ketone	0.000	0.000	0.000	0.000	0.000
Isopropyl alcohol	0.000	0.000	0.000	0.000	0.000
Toluene	0.000	0.000	0.000	0.000	0.000
Xylene (mixed isomers)	0.000	0.000	0.000	0.000	0.000
Methyl isobutyl ketone	0.000	0.000	0.000	0.000	0.000
Total	**3.100**	**0.000**	**0.000**	**0.000**	**3.110**

48—Communication industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Isopropyl alcohol	10,032.000	0.000	0.000	0.000	10,032.000
Total	**10,032.000**	**0.000**	**0.000**	**0.000**	**10,032.000**

49—Other utility industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ammonia	0.000	183.013	0.000	0.000	183.013
Sulfuric acid	0.000	129.641	0.000	0.000	129.641
Chlorine	0.000	35.300	0.000	0.000	35.300
Hydrazine	0.000	0.000	0.000	0.000	0.897
Manganese (and its compounds)	0.000	0.000	0.000	0.000	0.000
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Ethylene glycol	0.000	0.000	0.000	0.000	0.000
Chlorine dioxide	0.000	0.000	0.000	0.000	0.000
Vanadium (fume or dust)	0.000	0.000	0.000	0.000	0.000
Asbestos	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**347.954**	**0.000**	**0.000**	**348.851**

52—Food, beverage, drug, and tobacco industries, wholesale

Substance Name	Air	Water	Underground	Land	Total Releases*
Isopropyl alcohol	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

55—Motor vehicle, parts, and accessories industries, wholesale

Substance Name	Air	Water	Underground	Land	Total Releases*
Toluene	92.200	0.000	0.000	0.000	92.200
Methyl ethyl ketone	41.620	0.000	0.000	0.000	41.620
Isopropyl alcohol	40.830	0.000	0.000	0.000	40.830
Ethylbenzene	32.200	0.000	0.000	0.000	32.200
Methyl isobutyl ketone	23.400	0.000	0.000	0.000	23.400
Xylene (mixed isomers)	15.800	0.000	0.000	0.000	15.800

Substance Name	Air	Water	Underground	Land	Total Releases*
Acetone	7.000	0.000	0.000	0.000	7.000
Aluminum (fume or dust)	0.000	0.000	0.000	3.030	3.030
1,4-Dioxane	2.400	0.000	0.000	0.000	2.400
Formaldehyde	0.000	0.000	0.000	0.000	0.640
Carbon tetrachloride	0.000	0.000	0.000	0.000	0.630
Methanol	0.000	0.000	0.000	0.000	0.350
i-Butyl alcohol	0.000	0.000	0.000	0.000	0.110
Zinc (and its compounds)	0.000	0.000	0.000	0.000	0.000
Phenol	0.000	0.000	0.000	0.000	0.000
Total	**255.450**	**0.000**	**0.000**	**3.030**	**260.210**

56—Metals, hardware, plumbing, heating, and building materials industries, wholesale

Substance Name	Air	Water	Underground	Land	Total Releases*
Sulfuric acid	4.730	0.000	0.000	0.000	4.730
Lead (and its compounds)	0.000	0.140	0.000	0.000	0.140
Zinc (and its compounds)	0.000	0.031	0.000	0.000	0.031
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.000
Hydrochloric acid	0.000	0.000	0.000	0.000	0.000
Arsenic (and its compounds)	0.000	0.000	0.000	0.000	0.000
Chromium (and its compounds)	0.000	0.000	0.000	0.000	0.000
Total	**4.730**	**0.171**	**0.000**	**0.000**	**4.901**

59—Other products and industries, wholesale

Substance Name	Air	Water	Underground	Land	Total Releases*
Zinc (and its compounds)	0.000	0.000	0.000	181.790	181.790
Methanol	104.890	0.000	0.000	0.000	107.811
Dichloromethane	55.480	0.000	0.000	0.000	55.610
Acetone	18.064	0.000	0.000	0.000	20.281
Formaldehyde	12.456	0.000	0.000	0.000	12.716
Isopropyl alcohol	6.624	0.000	0.000	0.130	9.397
Methyl ethyl ketone	6.541	0.000	0.000	0.000	8.657
Toluene	4.115	0.000	0.000	0.000	6.520
Xylene (mixed isomers)	2.294	0.000	0.000	0.000	3.701
Tetrachloroethylene	1.014	0.000	0.000	0.000	1.175
Sulfuric acid	0.130	0.000	0.000	0.000	1.130
Methyl isobutyl ketone	0.456	0.000	0.000	0.000	0.951
Aluminum (fume or dust)	0.000	0.000	0.000	0.000	0.880
Ethylene glycol	0.141	0.000	0.000	0.000	0.795
Phosphoric acid	0.014	0.000	0.000	0.000	0.559
n-Butyl alcohol	0.133	0.000	0.000	0.000	0.432
Hydrochloric acid	0.000	0.000	0.000	0.000	0.407
Diethanolamine	0.000	0.000	0.000	0.000	0.390
Trichloroethylene	0.001	0.000	0.000	0.000	0.337
Nitric acid	0.168	0.000	0.000	0.000	0.307
Chlorine	0.000	0.000	0.000	0.000	0.260
Chloromethane	0.000	0.000	0.000	0.000	0.183
Cyclohexane	0.026	0.000	0.000	0.000	0.161
Ammonia	0.030	0.000	0.000	0.000	0.160
Maleic anhydride	0.130	0.000	0.000	0.000	0.130
Methyl *tert*-butyl ether	0.000	0.000	0.000	0.000	0.130
Di-*n*-octyl phthalate	0.000	0.000	0.000	0.000	0.060
Styrene	0.000	0.000	0.000	0.000	0.048
i-Butyl alcohol	0.001	0.000	0.000	0.000	0.022
Methyl methacrylate	0.000	0.000	0.000	0.000	0.007

Substance Name	Air	Water	Underground	Land	Total Releases*
Dibutyl phthalate	0.000	0.000	0.000	0.000	0.002
Butyl benzyl phthalate	0.000	0.000	0.000	0.000	0.000
Diethyl phthalate	0.000	0.000	0.000	0.000	0.000
Ethylbenzene	0.000	0.000	0.000	0.000	0.000
Lead (and its compounds)	0.000	0.000	0.000	0.000	0.000
Copper (and its compounds)	0.000	0.000	0.000	0.000	0.000
sec-Butyl alcohol	0.000	0.000	0.000	0.000	0.000
Bis(2-ethylhexyl) phthalate	0.000	0.000	0.000	0.000	0.000
2-Ethoxyethanol	0.000	0.000	0.000	0.000	0.000
2-Ethoxyethyl acetate	0.000	0.000	0.000	0.000	0.000
Bis(2-ethylhexyl) adipate	0.000	0.000	0.000	0.000	0.000
2-Methoxyethanol	0.000	0.000	0.000	0.000	0.000
Total	**212.708**	**0.000**	**0.000**	**181.920**	**415.009**

77—Business service industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Chlorine	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

81—Federal government service industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Ethylene glycol	0.000	0.000	0.000	181.970	181.970
Isopropyl alcohol	0.000	0.000	0.000	0.000	0.032
Acetone	0.000	0.000	0.000	0.000	0.032
Methyl ethyl ketone	0.000	0.000	0.000	0.000	0.032
Benzene	0.000	0.000	0.000	0.000	0.019
Toluene	0.000	0.000	0.000	0.000	0.018
Xylene (mixed isomers)	0.000	0.000	0.000	0.000	0.012
Ethylbenzene	0.000	0.000	0.000	0.000	0.005
Styrene	0.000	0.000	0.000	0.000	0.005
Total	**0.000**	**0.000**	**0.000**	**181.970**	**182.125**

97—Personal and household service industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

99—Other service industries

Substance Name	Air	Water	Underground	Land	Total Releases*
Styrene	2.289	0.000	0.000	0.000	2.289
Sulfuric acid	0.000	0.000	0.000	0.000	0.000
Total	**2.289**	**0.000**	**0.000**	**0.000**	**2.289**

* Total releases may be greater than the sum of the releases by environmental medium since releases of less than one tonne could be reported as an undifferentiated total release.

Appendix 6: Sectorial releases of toxic,* carcinogenic,† or potentially carcinogenic substances in alphabetical order (tonnes)

Acetaldehyde

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	70.506	13.200	6.000	0.030	89.736
18	Primary textile industries	43.551	0.000	0.000	0.000	43.551
	Total	**114.057**	**13.200**	**6.000**	**0.030**	**133.287**

NOTE: See p. P-40 for footnotes.

Acrylamide

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
27	Paper and allied products industries	0.000	0.316	0.000	2.845	3.161
37	Chemical and chemical products industries	0.897	0.000	0.000	0.000	1.977
41	Industrial and heavy (engineering) construction industries	0.000	0.610	0.000	0.000	0.610
06	Mining industries	0.000	0.000	0.000	0.000	0.130
	Total	**0.897**	**0.926**	**0.000**	**2.845**	**5.878**

Acrylonitrile

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	16.600	0.040	0.000	0.000	17.510
16	Plastic products industries	1.669	0.122	0.000	0.000	2.046
15	Rubber products industries	0.000	0.000	0.000	0.000	0.000
	Total	**18.269**	**0.162**	**0.000**	**0.000**	**19.556**

Arsenic (and its compounds)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
06	Mining industries	38.296	45.489	3,800.000	0.000	3,883.785
29	Primary metal industries	94.634	1.706	0.000	0.000	96.472
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.290	0.290
25	Wood industries	0.000	0.009	0.000	0.000	0.109
56	Metals, hardware, plumbing, heating, and building materials industries, wholesale	0.000	0.000	0.000	0.000	0.000
	Total	**132.930**	**47.204**	**3,800.000**	**0.290**	**3,980.656**

Asbestos

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
32	Transportation equipment industries	0.447	0.000	0.000	105.840	106.287
37	Chemical and chemical products industries	0.130	0.000	0.000	98.380	98.510
41	Industrial and heavy (engineering) construction industries	0.000	0.000	0.000	82.000	82.000

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
07	Crude petroleum and natural gas industries	0.000	0.000	0.000	51.000	51.000
15	Rubber products industries	0.000	0.000	0.000	10.800	10.800
27	Paper and allied products industries	0.000	0.000	0.000	3.000	3.000
06	Mining industries	0.000	0.000	0.000	0.000	0.587
49	Other utility industries	0.000	0.000	0.000	0.000	0.000
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.000	0.000
	Total	**0.557**	**0.000**	**0.000**	**351.020**	**352.184**

Benzene

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	1,110.960	0.007	0.000	0.130	1,111.097
37	Chemical and chemical products industries	776.256	0.455	7.600	1.577	786.018
36	Refined petroleum and coal products industries	498.697	0.582	20.390	0.745	520.634
07	Crude petroleum and natural gas industries	175.608	0.008	45.900	0.459	228.494
25	Wood industries	29.206	0.000	0.000	0.000	29.206
81	Federal government service industries	0.000	0.000	0.000	0.000	0.019
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.000
	Total	**2,590.727**	**1.052**	**73.890**	**2.911**	**2,675.468**

***Bis*(2-ethylhexyl) phthalate**

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
16	Plastic products industries	8.730	0.000	0.000	55.874	64.608
15	Rubber products industries	11.957	0.000	0.000	12.624	24.581
39	Other manufacturing industries	2.330	0.000	0.000	0.000	2.330
30	Fabricated metal products industries (except machinery and trans. equipment industries)	0.870	0.000	0.000	0.000	0.870
37	Chemical and chemical products industries	0.244	0.000	0.000	0.161	0.415
32	Transportation equipment industries	0.000	0.000	0.000	0.000	0.000
17	Leather and allied products industries	0.000	0.000	0.000	0.000	0.000
59	Other products and industries, wholesale	0.000	0.000	0.000	0.000	0.000
41	Industrial and heavy (engineering) construction industries	0.000	0.000	0.000	0.000	0.000
	Total	**24.131**	**0.000**	**0.000**	**68.632**	**92.804**

1,3-Butadiene

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	270.302	0.058	0.000	0.002	270.362
36	Refined petroleum and coal products industries	22.200	0.000	0.000	0.000	22.289
16	Plastic products industries	17.529	0.000	0.000	0.000	17.529
15	Rubber products industries	0.000	0.000	0.000	0.000	0.000
	Total	**310.031**	**0.058**	**0.000**	**0.002**	**310.180**

Cadmium (and its compounds)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	66.105	1.717	0.000	14.000	82.795
06	Mining industries	6.665	6.581	0.000	0.000	13.246
16	Plastic products industries	0.000	0.000	0.000	0.000	0.000
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	0.000
30	Fabricated metal products industries (except machinery and trans. equipment industries)	0.000	0.000	0.000	0.000	0.000
	Total	**72.770**	**8.298**	**0.000**	**14.000**	**96.041**

Carbon tetrachloride

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	12.773	4.732	0.000	0.000	17.505
55	Motor vehicle, parts, and accessories industries, wholesale	0.000	0.000	0.000	0.000	0.630
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.000	0.000
	Total	**12.773**	**4.732**	**0.000**	**0.000**	**18.135**

Chloroform

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
35	Nonmetallic mineral products industries	0.000	0.000	0.000	0.000	0.130
37	Chemical and chemical products industries	0.039	0.002	0.000	0.001	0.042
	Total	**0.039**	**0.002**	**0.000**	**0.001**	**0.172**

Chromium (and its compounds)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	7.382	28.319	0.000	739.668	777.575
39	Other manufacturing industries	0.000	0.000	0.000	5.930	5.930
37	Chemical and chemical products industries	1.700	0.708	0.000	0.000	5.694
35	Nonmetallic mineral products industries	1.723	0.010	0.000	2.300	4.037
30	Fabricated metal products industries (except machinery and trans. equipment industries)	2.593	0.021	0.000	0.176	3.865
32	Transportation equipment industries	0.049	0.000	0.000	0.000	2.878
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.450	0.450
25	Wood industries	0.000	0.020	0.000	0.000	0.170
16	Plastic products industries	0.000	0.000	0.000	0.000	0.130
31	Machinery industries (except electrical machinery)	0.000	0.000	0.000	0.130	0.130
17	Leather and allied products industries	0.000	0.000	0.000	0.000	0.000
33	Electrical and electronic products industries	0.000	0.000	0.000	0.000	0.000
41	Industrial and heavy (engineering) construction industries	0.000	0.000	0.000	0.000	0.000
06	Mining industries	0.000	0.000	0.000	0.000	0.000
56	Metals, hardware, plumbing, heating, and building materials industries, wholesale	0.000	0.000	0.000	0.000	0.000
	Total	**13.807**	**29.078**	**0.000**	**748.654**	**800.859**

_p_-Dichlorobenzene

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
37	Chemical and chemical products industries	9.900	0.000	0.000	0.500	10.400
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.000
	Total	**9.900**	**0.000**	**0.000**	**0.500**	**10.400**

1,2-Dichloroethane

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
37	Chemical and chemical products industries	6.886	0.523	0.000	0.040	7.449
35	Nonmetallic mineral products industries	0.000	0.000	0.000	0.000	0.130
	Total	**6.886**	**0.523**	**0.000**	**0.040**	**7.579**

Dichloromethane

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
16	Plastic products industries	1,356.821	0.000	0.000	0.000	1,356.821
37	Chemical and chemical products industries	668.759	0.000	0.000	0.039	670.490
35	Nonmetallic mineral products industries	66.000	0.000	0.000	0.000	66.130
59	Other products and industries, wholesale	55.480	0.000	0.000	0.000	55.610
30	Fabricated metal products industries (except machinery and trans. equipment industries)	42.474	0.000	0.000	0.000	42.474
26	Furniture and fixture industries	18.207	0.000	0.000	0.000	18.207
32	Transportation equipment industries	11.627	0.000	0.000	0.000	11.627
11	Beverage industries	0.000	0.000	0.000	0.000	0.630
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.000	0.100
29	Primary metal industries	0.000	0.000	0.000	0.000	0.000
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.000
	Total	**2,219.368**	**0.000**	**0.000**	**0.039**	**2,222.089**

Dimethyl sulfate

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
37	Chemical and chemical products industries	0.011	0.000	0.000	0.000	0.011
	Total	**0.011**	**0.000**	**0.000**	**0.000**	**0.011**

1,4-Dioxane

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
18	Primary textile industries	1.161	4.643	0.000	0.000	5.884
55	Motor vehicle, parts, and accessories industries, wholesale	2.400	0.000	0.000	0.000	2.400
37	Chemical and chemical products industries	0.860	0.000	0.000	0.000	0.860
	Total	**4.421**	**4.643**	**0.000**	**0.000**	**9.144**

Epichlorohydrin

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
16	Plastic products industries	0.000	0.000	0.000	0.000	0.125
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	0.008
15	Rubber products industries	0.000	0.000	0.000	0.000	0.000
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.133**

Ethyl acrylate

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
37	Chemical and chemical products industries	0.130	0.000	0.000	0.000	0.990
16	Plastic products industries	0.000	0.000	0.000	0.000	0.130
	Total	**0.130**	**0.000**	**0.000**	**0.000**	**1.120**

Ethylene oxide

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
37	Chemical and chemical products industries	50.842	0.000	0.000	0.000	51.067
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.000
	Total	**50.842**	**0.000**	**0.000**	**0.000**	**51.067**

Formaldehyde

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
25	Wood industries	455.173	0.000	0.000	0.000	455.173
27	Paper and allied products industries	19.912	221.560	0.000	0.130	241.602
37	Chemical and chemical products industries	102.345	2.504	69.920	0.000	178.702
35	Nonmetallic mineral products industries	134.520	1.276	0.000	0.630	136.426
16	Plastic products industries	0.173	85.450	0.000	0.000	85.753
59	Other products and industries, wholesale	12.456	0.000	0.000	0.000	12.716
29	Primary metal industries	5.075	0.000	0.000	0.000	5.075
55	Motor vehicle, parts, and accessories industries, wholesale	0.000	0.000	0.000	0.000	0.640
19	Textile products industries	0.000	0.000	0.000	0.000	0.130
33	Electrical and electronic products industries	0.130	0.000	0.000	0.000	0.130
30	Fabricated metal products industries (except machinery and trans. equipment industries)	0.070	0.000	0.000	0.000	0.070
11	Beverage industries	0.000	0.000	0.000	0.000	0.000
10	Food industries	0.000	0.000	0.000	0.000	0.000
	Total	**729.854**	**310.790**	**69.920**	**0.760**	**1,116.417**

Hydrazine

SIC #	Sector Name	Air	Water	Underground	Land	Total Releases‡
49	Other utility industries	0.000	0.000	0.000	0.000	0.897
41	Industrial and heavy (engineering) construction industries	0.000	0.000	0.000	0.000	0.345
29	Primary metal industries	0.000	0.000	0.000	0.000	0.000
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	0.000
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**1.242**

Lead (and its compounds)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	889.192	2.388	0.000	859.628	1,753.169
06	Mining industries	174.654	154.108	0.000	0.025	329.454
35	Nonmetallic mineral products industries	30.000	0.000	0.000	5.700	35.708
37	Chemical and chemical products industries	5.370	0.360	0.000	0.000	7.545
30	Fabricated metal products industries (except machinery and trans. equipment industries)	5.032	1.531	0.000	0.000	7.083
33	Electrical and electronic products industries	2.597	1.079	0.000	0.669	5.366
39	Other manufacturing industries	1.603	0.000	0.000	0.000	1.993
32	Transportation equipment industries	0.895	0.015	0.000	0.000	1.054
15	Rubber products industries	0.298	0.000	0.000	0.000	0.298
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.140	0.259
16	Plastic products industries	0.000	0.000	0.000	0.000	0.142
56	Metals, hardware, plumbing, heating, and building materials industries, wholesale	0.000	0.140	0.000	0.000	0.140
19	Textile products industries	0.009	0.000	0.000	0.000	0.009
41	Industrial and heavy (engineering) construction industries	0.000	0.000	0.000	0.000	0.000
59	Other products and industries, wholesale	0.000	0.000	0.000	0.000	0.000
	Total	**1,109.650**	**159.621**	**0.000**	**866.162**	**2,142.220**

Mercury (and its compounds)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
06	Mining industries	3.580	0.148	0.000	0.000	3.728
37	Chemical and chemical products industries	0.057	0.008	0.000	0.000	0.065
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.011	0.011
29	Primary metal industries	0.000	0.000	0.000	0.000	0.002
	Total	**3.637**	**0.156**	**0.000**	**0.011**	**3.806**

***p,p,*-Methylene*bis*(2-chloroaniline)**

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	0.005
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.005**

Nickel (and its compounds)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	541.693	65.379	0.000	46.839	655.459
07	Crude petroleum and natural gas industries	5.234	0.449	0.000	27.190	32.873
37	Chemical and chemical products industries	5.194	3.610	0.000	0.000	10.311
06	Mining industries	0.601	2.911	0.000	0.000	3.512
36	Refined petroleum and coal products industries	0.123	0.000	0.000	0.850	1.003
30	Fabricated metal products industries (except machinery and trans. equipment industries)	0.651	0.000	0.000	0.194	0.845
32	Transportation equipment industries	0.199	0.000	0.000	0.000	0.232
31	Machinery industries (except electrical machinery)	0.000	0.000	0.000	0.130	0.130
16	Plastic products industries	0.000	0.000	0.000	0.000	0.130
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.001
33	Electrical and electronic products industries	0.000	0.000	0.000	0.000	0.000
41	Industrial and heavy (engineering) construction industries	0.000	0.000	0.000	0.000	0.000
10	Food industries	0.000	0.000	0.000	0.000	0.000
	Total	**553.695**	**72.349**	**0.000**	**75.203**	**704.496**

Nitrilotriacetic acid

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	1.001
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**1.001**

2-Nitropropane

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
16	Plastic products industries	0.000	0.000	0.000	0.000	0.125
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.125**

Propylene oxide

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	10.941	0.000	0.000	0.000	10.941
16	Plastic products industries	0.000	0.000	0.000	0.000	0.130
	Total	**10.941**	**0.000**	**0.000**	**0.000**	**11.071**

Styrene

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	1,101.453	0.004	0.000	0.000	1,102.870
37	Chemical and chemical products industries	309.701	0.020	0.185	0.196	314.172
16	Plastic products industries	198.445	0.380	0.000	0.000	200.720
32	Transportation equipment industries	132.352	0.000	0.000	0.000	132.352
39	Other manufacturing industries	12.030	0.000	0.000	0.000	12.030
26	Furniture and fixture industries	10.614	0.000	0.000	0.000	10.614
45	Transportation industries	0.000	10.000	0.000	0.000	10.000
36	Refined petroleum and coal products industries	6.190	0.000	0.000	0.000	6.190
99	Other service industries	2.289	0.000	0.000	0.000	2.289
09	Service industries incidental to mineral extraction	0.728	0.000	0.000	0.000	0.728
15	Rubber products industries	0.000	0.000	0.000	0.000	0.500
59	Other products and industries, wholesale	0.000	0.000	0.000	0.000	0.048
81	Federal government service industries	0.000	0.000	0.000	0.000	0.005
	Total	**1,773.802**	**10.404**	**0.185**	**0.196**	**1,792.518**

Styrene oxide

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
16	Plastic products industries	0.000	0.000	0.000	0.000	0.130
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.130**

Tetrachloroethylene

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	122.180	0.000	0.000	0.000	122.180
33	Electrical and electronic products industries	22.070	0.000	0.000	0.000	22.070
28	Printing, publishing, and allied industries	8.808	0.000	0.000	0.000	8.808
37	Chemical and chemical products industries	6.440	0.010	0.000	0.000	7.448
59	Other products and industries, wholesale	1.014	0.000	0.000	0.000	1.175
18	Primary textile industries	0.000	0.000	0.000	0.000	0.700
11	Beverage industries	0.000	0.000	0.000	0.000	0.630
36	Refined petroleum and coal products industries	0.130	0.064	0.000	0.000	0.194
35	Nonmetallic mineral products industries	0.000	0.000	0.000	0.000	0.130
27	Paper and allied products industries	0.000	0.000	0.000	0.000	0.000
	Total	**160.642**	**0.074**	**0.000**	**0.000**	**163.335**

Thiourea

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	0.000	0.000	0.000	0.000	0.000
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

Toluene-2,4-diisocyanate

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	0.148
32	Transportation equipment industries	0.000	0.000	0.000	0.000	0.000
16	Plastic products industries	0.000	0.000	0.000	0.000	0.000
	Total	**0.000**	**0.000**	**0.000**	**0.000**	**0.148**

Toluenediisocyanate (mixed isomers)

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
16	Plastic products industries	0.230	0.000	0.000	0.000	0.631
37	Chemical and chemical products industries	0.011	0.000	0.000	0.000	0.147
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.123
33	Electrical and electronic products industries	0.000	0.000	0.000	0.000	0.112
32	Transportation equipment industries	0.000	0.000	0.000	0.000	0.016
26	Furniture and fixture industries	0.003	0.000	0.000	0.000	0.003
19	Textile products industries	0.001	0.000	0.000	0.000	0.001
	Total	**0.245**	**0.000**	**0.000**	**0.000**	**1.033**

Trichloroethylene

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
29	Primary metal industries	419.483	0.000	0.000	0.000	419.483
30	Fabricated metal products industries (except machinery and trans. equipment industries)	148.430	0.000	0.000	0.000	148.430
15	Rubber products industries	97.863	0.000	0.000	0.000	97.863
32	Transportation equipment industries	89.334	0.000	0.000	0.000	89.334
16	Plastic products industries	33.995	0.000	0.000	0.000	33.995

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
33	Electrical and electronic products industries	33.309	0.022	0.000	0.000	33.331
39	Other manufacturing industries	22.748	0.000	0.000	0.000	22.748
26	Furniture and fixture industries	13.000	0.000	0.000	0.000	13.000
37	Chemical and chemical products industries	0.000	0.000	0.000	0.000	0.702
59	Other products and industries, wholesale	0.001	0.000	0.000	0.000	0.337
35	Nonmetallic mineral products industries	0.000	0.000	0.000	0.000	0.130
11	Beverage industries	0.000	0.000	0.000	0.000	0.130
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.000	0.000
10	Food industries	0.000	0.000	0.000	0.000	0.000
	Total	**858.163**	**0.022**	**0.000**	**0.000**	**859.483**

Vinyl chloride

SIC #	Sector Name	Air	Water	Under-ground	Land	Total Releases‡
37	Chemical and chemical products industries	23.146	0.164	0.000	0.000	23.340
16	Plastic products industries	0.000	0.000	0.000	0.000	0.255
39	Other manufacturing industries	0.000	0.000	0.000	0.000	0.130
36	Refined petroleum and coal products industries	0.000	0.000	0.000	0.000	0.000
	Total	**23.146**	**0.164**	**0.000**	**0.000**	**23.725**

* Toxic includes Schedule 1 and CEPA-Toxic.
† Carcinogenic and potentially carcinogenic include IARC 1, 2a, 2b and NTP classification K and P substances.
‡ Total releases may be greater than the sum of the releases by environmental medium since releases of less than one tonne could be reported as an undifferentiated total release.

Appendix 7: Surface waters receiving NPRI substances

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
Alberni Inlet, BC	NA	Zinc (and its compounds)	1.500	1
	NA	Copper (and its compounds)	0.130	1
	7664-41-7	Ammonia	0.130	1
		Total	1.760	3
Amikougami Creek, PQ	NA	Cyanides (ionic)	0.058	1
		Total	0.058	1
Assiniboine River, MB	6484-52-2	Ammonium nitrate (solution)	6.900	1
	7783-20-2	Ammonium sulfate (solution)	2.300	1
	7664-41-7	Ammonia	0.100	1
		Total	9.300	3
Athabasca River, AB	7664-41-7	Ammonia	3.847	3
	111-42-2	Diethanolamine	3.800	1
	107-21-1	Ethylene glycol	3.700	2
	NA	Nickel (and its compounds)	0.449	1
		Total	11.796	7
Baker Creek, NT	NA	Arsenic (and its compounds)	0.630	1
		Total	0.630	1
Balmer Lake, ON	NA	Cyanides (ionic)	0.660	1
	7440-66-6	Zinc (fume or dust)	0.032	1
		Total	0.692	2
Balmer River, ON	NA	Arsenic (and its compounds)	2.126	1
	NA	Cyanides (ionic)	1.311	1
		Total	3.437	2
Battle River, AB	7664-41-7	Ammonia	58.683	1
		Total	58.683	1

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
Baie de Chaleur, NB	7440-62-2	Vanadium (fume or dust)	2.600	1
	NA	Cadmium (and its compounds)	1.716	1
	NA	Lead (and its compounds)	1.034	1
	NA	Arsenic (and its compounds)	0.630	1
	NA	Silver (and its compounds)	0.001	1
		Total	5.981	5
Bay of Grand-Pabos, PQ	7664-38-2	Phosphoric acid	25.000	1
		Total	25.000	1
Bearskin Lake, BC	NA	Cyanides (ionic)	0.003	1
		Total	0.003	1
Bourlamaque River, PQ	NA	Cyanides (ionic)	0.130	1
	NA	Lead (and its compounds)	0.130	1
	NA	Zinc (and its compounds)	0.130	1
		Total	0.390	3
Bousquet River, PQ	NA	Copper (and its compounds)	85.940	1
		Total	85.940	1
Bow River, AB	6484-52-2	Ammonium nitrate (solution)	0.700	1
	7664-41-7	Ammonia	0.100	1
		Total	0.800	2
Burntwood River, MB	NA	Nickel (and its compounds)	16.319	1
	NA	Arsenic (and its compounds)	0.725	1
	NA	Cobalt (and its compounds)	0.555	1
	NA	Copper (and its compounds)	0.425	1
		Total	18.024	4
Burrard Inlet, BC	7664-41-7	Ammonia	15.880	1
	100-42-5	Styrene	10.000	1
	7782-50-5	Chlorine	6.600	1
	111-42-2	Diethanolamine	1.660	1
	108-95-2	Phenol	1.010	1
	107-21-1	Ethylene glycol	0.437	1
	107-06-2	1,2-Dichloroethane	0.432	1
	NA	Zinc (and its compounds)	0.160	1
		Total	36.179	8
Canagagique Creek, ON	108-88-3	Toluene	0.130	1
		Total	0.130	1
Columbia River, BC	NA	Zinc (and its compounds)	4,444.000	1
	NA	Copper (and its compounds)	898.000	1
	7664-38-2	Phosphoric acid	774.000	1
	7664-93-9	Sulfuric acid	719.000	1
	7783-20-2	Ammonium sulfate (solution)	577.000	1
	7664-41-7	Ammonia	403.450	2
	NA	Lead (and its compounds)	149.000	1
	NA	Arsenic (and its compounds)	41.500	1
	NA	Cadmium (and its compounds)	6.000	1
	NA	Silver (and its compounds)	3.320	1
	NA	Antimony (and its compounds)	3.300	1
	NA	Manganese (and its compounds)	1.400	1
	NA	Mercury (and its compounds)	0.140	1
		Total	8,020.110	14
Coniston Creek, ON	NA	Nickel (and its compounds)	1.589	1
	NA	Copper (and its compounds)	0.077	1
	7440-62-2	Vanadium (fume or dust)	0.030	1
	NA	Zinc (and its compounds)	0.008	1
	NA	Cobalt (and its compounds)	0.008	1
	NA	Manganese (and its compounds)	0.004	1
	NA	Arsenic (and its compounds)	0.003	1
	NA	Chromium (and its compounds)	0.003	1
	NA	Lead (and its compounds)	0.003	1
	NA	Cadmium (and its compounds)	0.001	1
		Total	1.726	10
Contwoyto Lake, NT	NA	Zinc (and its compounds)	0.209	1
	NA	Arsenic (and its compounds)	0.039	1
	NA	Cyanides (ionic)	0.010	1
	NA	Lead (and its compounds)	0.004	1
		Total	0.262	4

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
Demontigny Lake, PQ	NA	Copper (and its compounds)	0.102	1
	NA	Cyanides (ionic)	0.017	1
	NA	Lead (and its compounds)	0.019	1
		Total	0.138	3
Detroit River, ON	7664-41-7	Ammonia	158.500	1
	7440-66-6	Zinc (fume or dust)	57.000	1
	NA	Manganese (and its compounds)	6.200	1
	108-95-2	Phenol	5.500	1
	7429-90-5	Aluminum (fume or dust)	1.400	1
	NA	Zinc (and its compounds)	0.200	1
	107-21-1	Ethylene glycol	0.110	1
	NA	Lead (and its compounds)	0.003	1
		Total	228.913	8
Discovery Passage, BC	7647-01-0	Hydrochloric acid	0.350	1
	7664-41-7	Ammonia	0.160	1
		Total	0.510	2
Dodds Creek, ON	NA	Zinc (and its compounds)	0.110	1
	7664-93-9	Sulfuric acid	0.000	1
	107-21-1	Ethylene glycol	0.000	1
	7664-38-2	Phosphoric acid	0.000	1
		Total	0.110	4
Dormenan Creek, PQ	NA	Cyanides (ionic)	0.880	1
	NA	Zinc (and its compounds)	0.380	1
	NA	Copper (and its compounds)	0.380	1
	NA	Lead (and its compounds)	0.130	1
		Total	1.770	4
Estuary Pond, NB	7664-41-7	Ammonia	16.133	1
		Total	16.133	1
Fox Creek, ON	NA	Zinc (and its compounds)	7.435	1
	NA	Copper (and its compounds)	0.522	1
		Total	7.957	2
Frank Lake, ON	NA	Cyanides (ionic)	0.380	1
		Total	0.380	1
Fraser River, BC	7664-41-7	Ammonia	18.000	1
	67-56-1	Methanol	1.438	1
	7664-93-9	Sulfuric acid	0.200	1
	50-00-0	Formaldehyde	0.130	1
	7647-01-0	Hydrochloric acid	0.050	1
		Total	19.818	5
Gold Bar Creek, AB	107-21-1	Ethylene glycol	2.780	1
	NA	Manganese (and its compounds)	0.026	1
	NA	Zinc (and its compounds)	0.007	1
	NA	Lead (and its compounds)	0.004	1
	NA	Nickel (and its compounds)	0.003	1
	NA	Chromium (and its compounds)	0.002	1
		Total	2.822	6
Grande Rivière du Loup, PQ	67-63-0	Isopropyl alcohol	0.130	1
		Total	0.130	1
Great Slave Lake, NT	NA	Cyanides (ionic)	0.263	1
	NA	Copper (and its compounds)	0.262	1
		Total	0.525	2
Halifax Harbour, NS	7664-41-7	Ammonia	11.830	5
	108-88-3	Toluene	0.502	4
	NA	Zinc (and its compounds)	0.447	2
	1330-20-7	Xylene (mixed isomers)	0.404	4
	91-20-3	Naphthalene	0.369	3
	71-43-2	Benzene	0.290	4
	95-63-6	1,2,4-Trimethylbenzene	0.087	1
	100-41-4	Ethylbenzene	0.075	3
	110-82-7	Cyclohexane	0.067	2
	98-82-8	Cumene	0.020	1
	108-95-2	Phenol	0.010	1
		Total	14.101	30

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
Hamilton Harbour, ON	7664-41-7	Ammonia	35.500	1
	67-56-1	Methanol	11.000	1
	107-21-1	Ethylene glycol	7.400	1
	NA	Zinc (and its compounds)	2.430	1
	NA	Lead (and its compounds)	0.280	1
		Total	56.610	5
Harricana River, PQ	NA	Cyanides (ionic)	0.130	1
	NA	Lead (and its compounds)	0.130	1
		Total	0.260	2
Hesse Lake, PQ	6484-52-2	Ammonium nitrate (solution)	5.000	1
		Total	5.000	1
Hillsborough River, PE	7664-93-9	Sulfuric acid	21.700	1
	67-64-1	Acetone	8.050	1
	67-56-1	Methanol	7.090	1
	108-88-3	Toluene	0.077	1
		Total	36.917	4
Hour Lake, ON	NA	Cyanides (ionic)	0.030	1
		Total	0.030	1
Humber River, ON	108-88-3	Toluene	5.181	1
	110-82-7	Cyclohexane	0.120	1
	1330-20-7	Xylene (mixed isomers)	0.077	1
		Total	5.378	3
Junction Creek, ON	NA	Nickel (and its compounds)	25.144	3
	NA	Copper (and its compounds)	7.274	3
	NA	Cobalt (and its compounds)	0.427	2
	NA	Lead (and its compounds)	0.202	2
	NA	Arsenic (and its compounds)	0.108	1
	NA	Silver (and its compounds)	0.015	1
		Total	33.170	12
Kaministikwia River, ON	107-21-1	Ethylene glycol	2.900	1
	7664-41-7	Ammonia	0.922	1
		Total	3.822	2
Kitimat Arm of the Douglas Channel, BC	67-56-1	Methanol	11.000	1
	7664-41-7	Ammonia	7.300	1
		Total	18.300	2
Kitimat River, BC	108-95-2	Phenol	2.430	1
		Total	2.430	1
Kootenay River, BC	6484-52-2	Ammonium nitrate (solution)	20.100	1
	67-56-1	Methanol	9.000	1
		Total	29.100	2
Lachine Canal, PQ	NA	Zinc (and its compounds)	0.080	1
	NA	Lead (and its compounds)	0.020	1
		Total	0.100	2
LaHave River, NS	7664-41-7	Ammonia	0.421	1
	NA	Zinc (and its compounds)	0.133	1
	NA	Copper (and its compounds)	0.012	1
		Total	0.566	3
Lake Erie, ON	7647-01-0	Hydrochloric acid	16.900	1
	107-21-1	Ethylene glycol	10.950	1
	7664-41-7	Ammonia	1.715	1
	NA	Manganese (and its compounds)	1.675	1
	NA	Zinc (and its compounds)	1.257	1
	7782-50-5	Chlorine	1.095	1
	NA	Copper (and its compounds)	0.233	1
	NA	Chromium (and its compounds)	0.070	1
	71-43-2	Benzene	0.004	1
	1330-20-7	Xylene (mixed isomers)	0.004	1
	100-42-5	Styrene	0.004	1
	91-20-3	Naphthalene	0.004	1
	108-88-3	Toluene	0.004	1
		Total	33.915	13
Lake Huron, ON	7664-93-9	Sulfuric acid	129.641	1
		Total	129.641	1

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
Lake Ontario, ON	7664-93-9	Sulfuric acid	112.382	3
	107-21-1	Ethylene glycol	27.740	1
	NA	Manganese (and its compounds)	9.300	1
	NA	Zinc (and its compounds)	9.145	2
	123-91-1	1,4-Dioxane	4.643	1
	NA	Copper (and its compounds)	1.064	4
	67-56-1	Methanol	1.000	1
	7664-41-7	Ammonia	0.990	2
	7782-50-5	Chlorine	0.700	1
	NA	Nickel (and its compounds)	0.659	1
	NA	Chromium (and its compounds)	0.638	1
	NA	Lead (and its compounds)	0.205	3
	7664-39-3	Hydrogen fluoride	0.132	1
	92-52-4	Biphenyl	0.060	1
	120-12-7	Anthracene	0.037	1
	91-20-3	Naphthalene	0.007	1
	NA	Antimony (and its compounds)	0.003	1
	71-43-2	Benzene	0.002	1
	1330-20-7	Xylene (mixed isomers)	0.000	1
	100-41-4	Ethylbenzene	0.000	1
Lake Ontario, ON (continued)	100-42-5	Styrene	0.000	1
	1163-19-5	Decabromodiphenyl oxide	0.000	1
	108-88-3	Toluene	0.000	1
		Total	168.707	32
Lake Superior, ON	67-56-1	Methanol	2,271.000	1
	67-64-1	Acetone	0.315	1
	78-93-3	Methyl ethyl ketone	0.010	1
		Total	2,271.325	3
La Lièvre River, PQ	67-56-1	Methanol	192.600	1
	7664-38-2	Phosphoric acid	2.175	1
	NA	Chromium (and its compounds)	0.003	1
		Total	194.778	3
Lim Lake, ON	NA	Cyanides (ionic)	0.878	2
	NA	Copper (and its compounds)	0.395	1
		Total	1.273	3
Little River, NB	67-56-1	Methanol	64.400	1
	7664-41-7	Ammonia	43.000	1
		Total	107.400	2
Little South Tomogonops River, NB	NA	Zinc (and its compounds)	13.880	1
	NA	Copper (and its compounds)	1.425	1
	NA	Lead (and its compounds)	0.732	1
		Total	16.037	3
Magusi River, ON	NA	Cyanides (ionic)	0.006	1
		Total	0.006	1
Malaspina Strait, BC	10049-04-4	Chlorine dioxide	4.800	1
		Total	4.800	1
McLarens Creek, ON	NA	Nickel (and its compounds)	0.095	1
	NA	Chromium (and its compounds)	0.027	1
		Total	0.122	2
Muchalat Inlet, BC	7664-41-7	Ammonia	2.500	1
		Total	2.500	1
Myra Creek, BC	NA	Zinc (and its compounds)	3.330	1
	NA	Copper (and its compounds)	0.380	1
	NA	Lead (and its compounds)	0.000	1
		Total	3.710	3
Nepisiguit River, NB	7664-93-9	Sulfuric acid	51.900	1
	7664-41-7	Ammonia	3.900	1
		Total	55.800	2
Neroutsos Inlet, BC	67-56-1	Methanol	1,600.000	1
		Total	1,600.000	1
Niagara River, ON	7782-50-5	Chlorine	0.390	6
	NA	Manganese (and its compounds)	0.002	1
		Total	0.392	7
Nith River, ON	50-00-0	Formaldehyde	1.248	1
		Total	1.248	1

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
North Saskatchewan River, AB	7783-20-2	Ammonium sulfate (solution)	450.000	2
	6484-52-2	Ammonium nitrate (solution)	441.000	2
	NA	Nickel (and its compounds)	3.610	1
	7664-41-7	Ammonia	3.597	7
	NA	Cobalt (and its compounds)	1.120	1
	NA	Zinc (and its compounds)	0.780	1
	NA	Chromium (and its compounds)	0.600	1
	NA	Copper (and its compounds)	0.270	1
	7664-93-9	Sulfuric acid	0.200	1
	107-21-1	Ethylene glycol	0.050	1
	91-20-3	Naphthalene	0.026	1
	75-01-4	Vinyl chloride	0.003	1
	67-66-3	Chloroform	0.002	1
		Total	901.258	21
North Unnamed Creek, BC	NA	Cyanides (ionic)	0.857	1
		Total	0.857	1
Northumberland Channel, BC	7664-41-7	Ammonia	4.340	1
	7782-50-5	Chlorine	1.982	3
		Total	6.322	4
Onaping River, ON	NA	Copper (and its compounds)	0.380	1
	NA	Cyanides (ionic)	0.136	1
	NA	Nickel (and its compounds)	1.160	1
		Total	1.676	3
Otonabee River, ON	7664-41-7	Ammonia	122.600	1
	7782-50-5	Chlorine	2.600	1
		Total	125.200	2
Ottawa River, PQ/ON	7664-93-9	Sulfuric acid	2,500.000	1
	7664-41-7	Ammonia	77.900	1
	75-07-0	Acetaldehyde	13.200	1
	50-00-0	Formaldehyde	1.767	1
	108-95-2	Phenol	0.481	1
		Total	2,593.348	5
Peace River, AB/BC	107-21-1	Ethylene glycol	9.680	2
	7664-41-7	Ammonia	2.305	2
	7782-50-5	Chlorine	0.394	1
	1330-20-7	Xylene (mixed isomers)	0.009	1
	108-88-3	Toluene	0.008	1
	71-43-2	Benzene	0.008	1
		Total	12.404	8
Pell Creek, ON	NA	Chromium (and its compounds)	0.010	1
		Total	0.010	1
Porcupine River, ON	7664-41-7	Ammonia	32.304	1
	NA	Zinc (and its compounds)	4.341	1
	NA	Manganese (and its compounds)	4.282	1
	NA	Selenium (and its compounds)	3.896	1
	NA	Arsenic (and its compounds)	0.824	1
	NA	Lead (and its compounds)	0.781	1
	NA	Copper (and its compounds)	0.283	1
	NA	Cadmium (and its compounds)	0.201	1
		Total	46.912	8
Quevillon River, PQ	67-56-1	Methanol	57.380	1
		Total	57.380	1
Red Deer River, AB	107-21-1	Ethylene glycol	9.000	1
		Total	9.000	1
Richelieu River, PQ	6484-52-2	Ammonium nitrate (solution)	170.000	1
	67-64-1	Acetone	30.000	1
	7664-41-7	Ammonia	18.000	1
	7647-01-0	Hydrochloric acid	10.000	1
	NA	Lead (and its compounds)	0.140	1
	NA	Zinc (and its compounds)	0.131	2
		Total	228.271	7
Rouyn Lake, PQ	NA	Zinc (and its compounds)	4.420	1
	NA	Manganese (and its compounds)	3.660	1
	7429-90-5	Aluminum (fume or dust)	3.390	1
	NA	Copper (and its compounds)	1.850	1

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
	NA	Nickel (and its compounds)	0.660	1
	NA	Lead (and its compounds)	0.600	1
	NA	Arsenic (and its compounds)	0.240	1
		Total	14.820	7
Rupert Inlet, BC	NA	Copper (and its compounds)	13,439.526	1
		Total	13,439.526	1
Saint John River, NB	67-56-1	Methanol	3,129.039	1
	7664-41-7	Ammonia	51.360	1
	7647-01-0	Hydrochloric acid	6.442	1
		Total	3,186.841	3
Saskatchewan River, MB	6484-52-2	Ammonium nitrate (solution)	3.600	1
		Total	3.600	1
Shawinigan River, PQ	50-00-0	Formaldehyde	221.430	1
		Total	221.430	1
Similkameen River, BC	NA	Copper (and its compounds)	0.003	1
		Total	0.003	1
South Little River, NB	NA	Zinc (and its compounds)	4.800	1
	NA	Lead (and its compounds)	0.880	1
	NA	Copper (and its compounds)	0.380	1
		Total	6.060	3
South Porcupine River, ON	NA	Cyanides (ionic)	0.446	1
		Total	0.446	1
South Saskatchewan River, SK/AB	7664-93-9	Sulfuric acid	87.000	1
	7664-41-7	Ammonia	31.250	2
		Total	118.250	3
St. Charles River, PQ	55-63-0	Nitroglycerin	9.000	1
	7697-37-2	Nitric acid	7.700	1
	121-14-2	2,4-Dinitrotoluene	4.000	1
	67-64-1	Acetone	0.500	1
	NA	Zinc (and its compounds)	0.128	1
		Total	21.328	5
St. Clair River, ON	6484-52-2	Ammonium nitrate (solution)	39.500	1
	7664-41-7	Ammonia	38.421	5
	67-56-1	Methanol	3.188	1
	NA	Zinc (and its compounds)	1.400	1
	75-65-0	*tert*-Butyl alcohol	1.100	1
	74-87-3	Chloromethane	0.880	1
	110-82-7	Cyclohexane	0.500	2
	1634-04-4	Methyl *tert*-butyl ether	0.400	1
	75-00-3	Chloroethane	0.370	1
	NA	Lead (and its compounds)	0.360	1
	75-01-4	Vinyl chloride	0.160	1
	1330-20-7	Xylene (mixed isomers)	0.140	3
	75-05-8	Acetonitrile	0.130	1
	71-43-2	Benzene	0.128	5
	100-41-4	Ethylbenzene	0.106	3
	107-06-2	1,2-Dichloroethane	0.091	1
	108-88-3	Toluene	0.075	4
	7782-50-5	Chlorine	0.063	1
	105-99-0	1,3-Butadiene	0.058	1
	107-13-1	Acrylonitrile	0.040	1
	100-42-5	Styrene	0.020	2
		Total	87.130	38
St. François River, PQ	67-56-1	Methanol	22.000	1
	7783-20-2	Ammonium sulphate (solution)	2.000	1
	NA	Lead (and its compounds)	1.000	1
		Total	25.000	3
St. Germain River, PQ	67-63-0	Isopropyl alcohol	0.544	1
		Total	0.544	1
St. Lawrence River, PQ/ON	7664-93-9	Sulfuric acid	15,010.200	5
	67-56-1	Methanol	460.970	3
	7664-41-7	Ammonia	159.302	8
	50-00-0	Formaldehyde	85.457	2
	NA	Manganese (and its compounds)	77.500	2

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
	7647-01-0	Hydrochloric acid	44.000	2
	7697-37-2	Nitric acid	28.500	2
	NA	Chromium (and its compounds)	27.435	3
	NA	Nickel (and its compounds)	20.300	1
	111-42-2	Diethanolamine	19.740	1
	71-36-3	*n*-Butyl alcohol	18.680	1
	NA	Zinc (and its compounds)	14.989	5
	75-15-0	Carbon disulphide	10.405	1
	78-83-1	*i*-Butyl alcohol	6.440	1
	103-23-1	*Bis*(2-ethylhexyl) adipate	5.600	1
	56-23-5	Carbon tetrachloride	4.732	2
	108-95-2	Phenol	4.319	3
	NA	Selenium (and its compounds)	3.560	1
	NA	Copper (and its compounds)	2.751	3
	107-21-1	Ethylene glycol	1.382	2
	1330-20-7	Xylene (mixed isomers)	1.315	3
	7782-50-5	Chlorine	1.178	2
	71-43-2	Benzene	0.619	3
	79-06-1	Acrylamide	0.610	1
	100-42-5	Styrene	0.380	1
	108-88-3	Toluene	0.357	3
	7664-39-3	Hydrogen fluoride	0.290	1
	95-63-6	1,2,4-Trimethylbenzene	0.146	1
	100-41-4	Ethylbenzene	0.133	2
	107-13-1	Acrylonitrile	0.122	1
	67-64-1	Acetone	0.081	1
	127-18-4	Tetrachloroethylene	0.074	2
	NA	Lead (and its compounds)	0.060	2
	7723-14-0	Phosphorus (yellow or white)	0.030	1
	NA	Mercury (and its compounds)	0.008	1
		Total	16,011.665	74
St. Mary River, BC	NA	Zinc (and its compounds)	2.566	1
	NA	Lead (and its compounds)	0.489	1
		Total	3.055	2
St. Mary's River, ON	7664-41-7	Ammonia	448.952	1
	NA	Zinc (and its compounds)	1.274	1
	NA	Cyanides (ionic)	1.162	1
	1330-20-7	Xylene (mixed isomers)	0.002	1
	108-88-3	Toluene	0.001	1
	71-43-2	Benzene	0.001	1
		Total	451.392	6
St. Maurice River, PQ	67-56-1	Methanol	3,174.800	1
	79-06-1	Acrylamide	0.316	1
	NA	Zinc (and its compounds)	0.130	1
		Total	3,175.246	3
Strait of Canso, NS	67-56-1	Methanol	895.230	1
	7664-93-9	Sulfuric acid	137.800	1
	7647-01-0	Hydrochloric acid	2.336	1
	10049-04-4	Chlorine dioxide	0.473	1
		Total	1,035.839	4
Sunday Lake, ON	NA	Cyanides (ionic)	0.510	1
	NA	Lead (and its compounds)	0.130	1
		Total	0.640	2
Sydney Harbour, NS	107-21-1	Ethylene glycol	5.000	1
		Total	5.000	1
Thornbrough Channel of Howe Sound, B.C.	7664-41-7	Ammonia	0.100	1
		Total	0.100	1
Thousand Islands River, PQ	NA	Zinc (and its compounds)	1.757	1
	7664-41-7	Ammonia	1.095	1
	50-00-0	Formaldehyde	0.730	1
	7664-38-2	Phosphoric acid	0.037	1
		Total	3.619	4
Twelve Mile Creek, ON	107-21-1	Ethylene glycol	5.000	1
	7664-41-7	Ammonia	0.630	1

Surface Water Name and Province	CAS #	Substance Name	Releases (Tonnes)	# of Reports
	NA	Manganese (and its compounds)	0.012	1
		Total	5.642	3
Twin Lakes Creek, NT	NA	Zinc (and its compounds)	0.016	1
		Total	0.016	1
Vanze Creek, PQ	NA	Zinc (and its compounds)	0.803	1
	NA	Arsenic (and its compounds)	0.158	1
	NA	Lead (and its compounds)	0.103	1
	NA	Copper (and its compounds)	0.095	1
	NA	Nickel (and its compounds)	0.064	1
		Total	1.223	5
Wabamun Ash Lagoons, AB	7664-41-7	Ammonia	12.200	1
		Total	12.200	1
Wabigoon River, ON	107-21-1	Ethylene glycol	4.800	1
	6484-52-2	Ammonium nitrate (solution)	3.000	1
		Total	7.800	2
Wapiti River, AB	7664-41-7	Ammonia	0.830	1
		Total	0.830	1
Wascana Creek, SK	7782-50-5	Chlorine	26.100	1
		Total	26.100	1
Wawagosic River, PQ	NA	Zinc (and its compounds)	1.300	1
		Total	1.300	1
Wekusko Lake, MB	7664-41-7	Ammonia	8.135	1
	NA	Zinc (and its compounds)	3.288	3
	NA	Copper (and its compounds)	0.199	3
		Total	11.622	7
Welland Canal, ON	108-95-2	Phenol	2.085	1
	NA	Manganese (and its compounds)	0.220	2
	NA	Zinc (and its compounds)	0.067	1
	NA	Chromium (and its compounds)	0.019	1
	107-21-1	Ethylene glycol	0.004	1
		Total	2.395	6
Welland River, ON	7664-41-7	Ammonia	19.807	1
	7723-14-0	Phosphorus (yellow or white)	4.556	1
	NA	Nickel (and its compounds)	0.610	1
	NA	Zinc (and its compounds)	0.360	1
	NA	Chromium (and its compounds)	0.230	1
	75-01-4	Vinyl chloride	0.001	1
		Total	25.564	6
Whitesand River, ON	67-56-1	Methanol	120.000	1
	7664-93-9	Sulfuric acid	30.003	1
	NA	Zinc (and its compounds)	0.089	1
	NA	Copper (and its compounds)	0.024	1
		Total	150.116	4
Wolf Lake, SK	7664-41-7	Ammonia	0.044	1
		Total	0.044	1
Winnipeg River, ON/MB	67-56-1	Methanol	120.000	1
	7664-93-9	Sulfuric acid	30.003	1
		Total	150.003	2
Wye River, ON	7664-41-7	Ammonia	10.120	1
	NA	Zinc (and its compounds)	0.089	1
	NA	Lead (and its compounds)	0.079	1
	79-01-6	Trichloroethylene	0.022	1
		Total	10.310	4
York River, PQ	NA	Manganese (and its compounds)	6.130	1
	NA	Copper (and its compounds)	3.561	1
	NA	Nickel (and its compounds)	1.687	1
	NA	Lead (and its compounds)	1.580	1
	NA	Zinc (and its compounds)	1.423	1
	NA	Cadmium (and its compounds)	0.380	1
	NA	Arsenic (and its compounds)	0.212	1
	NA	Mercury (and its compounds)	0.008	1
		Total	14.981	8

Appendix 8: Substances transferred by category (tonnes)

CAS #	Substance Name	Physical	Chemical	Biological	Incineration	Landfill	Storage	MSTP*	Underground	Total Transfers	# of Reports
117-81-7	*Bis*(2-ethylhexyl) phthalate	32.073	0.000	0.000	6.858	14,852.205	0.003	0.000	0.000	14,891.139	33
7429-90-5	Aluminum (fume or dust)	1.000	0.000	0.000	8.940	7,756.462	0.000	0.000	0.000	7,766.402	35
NA	Copper (and its compounds)	6,652.737	58.004	0.000	17.881	293.937	9.712	5.376	0.772	7,038.419	243
NA	Zinc (and its compounds	880.377	260.877	0.000	46.238	4,069.363	15.229	14.761	0.000	5,286.846	283
67-56-1	Methanol	107.263	0.000	2,613.660	596.741	59.956	0.000	58.175	960.070	4,395.865	284
NA	Manganese (and its compounds)	8.017	1,174.712	0.000	0.000	2,517.297	87.725	4.023	0.000	3,791.774	192
7664-93-9	Sulfuric acid	49.710	1,138.948	0.012	15.150	368.306	250.263	100.785	800.000	2,723.174	387
108-88-3	Toluene	326.132	65.262	8.228	1,485.159	14.153	10.313	1.047	7.455	1,917.749	313
107-21-1	Ethylene glycol	14.598	46.060	801.546	148.792	13.479	7.748	550.352	76.846	1,659.420	232
1330-20-7	Xylene (mixed isomers)	155.860	0.000	0.688	1,156.631	44.562	21.719	56.705	3.157	1,439.323	285
7440-66-6	Zinc (fume or dust)	1.280	0.000	0.000	10.140	1,420.836	0.000	0.032	0.000	1,432.288	36
NA	Lead (and its compounds)	558.036	4.737	0.000	3.246	733.318	0.000	2.314	0.000	1,301.650	160
7647-01-0	Hydrochloric acid	52.600	225.508	0.000	0.731	26.427	22.310	681.770	0.000	1,009.346	211
NA	Chromium (and its compounds)	85.806	410.816	0.000	2.156	296.718	199.058	8.092	0.000	1,002.646	189
1332-21-4	Asbestos	0.000	0.000	0.000	0.000	935.063	0.000	0.000	0.000	935.063	37
67-63-0	Isopropyl alcohol	12.024	37.723	1.914	521.907	2.104	208.731	64.898	0.024	849.325	195
78-93-3	Methyl ethyl ketone	301.218	2.000	6.321	379.595	2.965	0.253	0.181	0.000	692.533	141
7664-38-2	Phosphoric acid	0.031	57.550	0.000	0.375	183.755	255.000	63.088	0.000	559.799	177
67-64-1	Acetone	16.235	0.200	0.000	431.260	48.851	0.010	43.611	0.000	540.167	118
NA	Nickel (and its compounds)	4.220	183.751	0.000	0.000	193.905	121.513	3.529	0.000	506.918	114
108-05-4	Vinyl acetate	0.000	0.000	0.000	464.347	0.001	0.000	0.678	0.000	465.026	14
111-42-2	Diethanolamine	0.080	0.000	0.000	3.904	2.501	0.015	20.337	384.370	411.207	75
7664-41-7	Ammonia	0.000	211.400	20.400	21.271	1.376	0.000	138.607	0.000	393.054	161
108-95-2	Phenol	0.000	136.200	124.770	58.744	22.091	0.000	49.785	0.000	391.590	50
100-42-5	Styrene	0.000	1.700	0.002	254.314	54.084	0.000	0.528	0.065	310.694	78
80-62-6	Methyl methacrylate	0.000	0.000	0.000	258.100	5.200	0.000	0.006	0.000	263.306	13

CAS #	Substance Name	Physical	Chemical	Biological	Incineration	Landfill	Storage	MSTP*	Underground	Total Transfers	# of Reports
71-36-3	*n*-Butyl alcohol	25.075	3.000	15.200	185.152	7.074	0.603	13.530	0.000	249.634	79
7697-37-2	Nitric acid	0.000	133.245	0.000	6.600	16.125	0.000	65.870	10.000	231.840	101
50-00-0	Formaldehyde	0.000	0.000	1.345	164.220	41.993	0.000	4.893	0.000	212.451	80
7440-62-2	Vanadium (fume or dust)	0.000	181.640	0.000	0.000	25.512	0.000	1.790	0.000	208.942	11
108-10-1	Methyl isobutyl ketone	20.618	0.000	0.500	127.300	21.223	0.000	0.000	0.000	169.641	71
71-43-2	Benzene	0.000	0.000	0.000	124.286	0.948	0.000	0.078	26.277	151.589	95
106-99-0	1,3-Butadiene	0.000	0.000	0.000	128.687	2.100	0.000	0.524	0.000	131.311	13
110-82-7	Cyclohexane	0.000	0.000	0.000	116.903	0.154	0.000	0.000	0.000	117.057	82
NA	Selenium (and its compounds)	0.000	0.000	0.000	0.000	93.305	0.000	4.565	0.000	97.870	7
127-18-4	Tetrachloroethylene	28.830	0.000	0.000	26.097	39.000	0.000	0.002	0.000	93.929	29
1344-28-1	Aluminum oxide (fibrous forms)	0.000	0.000	0.340	9.560	46.920	0.000	0.000	0.000	56.820	14
7783-20-2	Ammonium sulphate (solution)	0.000	0.000	0.000	0.000	0.000	0.000	52.058	0.000	52.058	19
95-63-6	1,2,4-Trimethylbenzene	0.404	0.000	0.000	50.828	0.074	0.602	0.043	0.014	51.966	75
100-41-4	Ethylbenzene	0.020	0.000	0.000	33.891	1.058	0.039	0.015	1.149	36.171	68
75-09-2	Dichloromethane	0.156	15.390	0.000	19.759	0.002	0.000	0.002	0.000	35.309	50
85-68-7	Butyl benzyl phthalate	0.000	0.000	0.000	25.099	8.188	0.000	0.000	0.000	33.287	11
NA	Mercury (and its compounds)	0.000	0.000	0.000	0.000	32.633	0.000	0.000	0.000	32.633	5
101-68-8	Methylene*bis* (phenylisocyanate)	20.800	2.046	0.000	0.383	2.890	6.200	0.000	0.000	32.319	48
78-83-1	*i*-Butyl alcohol	0.000	0.000	0.000	28.173	2.910	0.000	0.000	0.050	31.133	25
6484-52-2	Ammonium nitrate (solution)	0.000	0.000	0.000	0.000	0.000	30.000	0.030	0.000	30.030	22
NA	Arsenic (and its compounds)	2.200	0.000	0.000	0.000	26.625	0.100	0.192	0.000	29.117	45
91-20-3	Naphthalene	0.000	0.017	0.000	3.292	25.768	0.000	0.007	0.000	29.084	40
156-62-7	Calcium cyanamide	0.000	28.740	0.000	0.000	0.000	0.000	0.290	0.000	29.030	1
79-01-6	Trichloroethylene	5.000	13.355	0.000	8.652	0.040	0.000	0.181	0.000	27.228	45
95-50-1	*o*-Dichlorobenzene	0.000	0.000	0.000	22.700	0.000	0.000	0.000	0.000	22.700	3
107-13-1	Acrylonitrile	0.000	0.000	0.000	18.930	0.000	0.000	0.000	0.000	18.930	8
73-35-4	Vinylidene chloride	0.000	0.000	0.000	0.000	11.400	0.000	0.000	0.000	11.400	1

CAS #	Substance Name	Physical	Chemical	Biological	Incineration	Landfill	Storage	MSTP*	Underground	Total Transfers	# of Reports
56-23-5	Carbon tetrachloride	0.000	0.000	0.000	11.107	0.000	0.000	0.000	0.000	11.107	6
1319-77-3	Cresol (mixed isomers)	0.000	0.000	0.000	9.200	0.000	0.000	0.000	0.000	9.200	4
7664-39-3	Hydrogen fluoride	0.000	7.950	0.000	0.000	0.000	0.000	0.000	0.000	7.950	19
103-23-1	*Bis*(2-ethylhexyl) adipate	0.000	0.000	0.000	4.545	2.471	0.003	0.000	0.000	7.019	14
NA	Antimony (and its compounds)	0.000	0.000	0.000	0.000	5.603	0.485	0.000	0.000	6.088	30
NA	Cobalt (and its compounds)	2.919	0.000	0.000	0.000	2.718	0.000	0.029	0.000	5.666	23
84-74-2	Dibutyl phthalate	0.000	0.004	0.000	5.137	0.460	0.000	0.000	0.000	5.601	17
117-84-0	Di-*n*-octyl phthalate	0.000	0.000	0.000	3.001	2.088	0.000	0.047	0.000	5.136	14
96-33-3	Methyl acrylate	0.000	0.000	0.000	4.900	0.000	0.000	0.000	0.000	4.900	2
85-44-9	Phthalic anhydride	0.000	0.000	0.000	2.330	1.424	0.001	0.000	0.000	3.755	15
120-12-7	Anthracene	0.000	0.000	0.000	0.000	3.466	0.000	0.000	0.000	3.466	12
92-52-4	Biphenyl	0.000	0.000	0.000	2.899	0.154	0.000	0.000	0.000	3.052	10
75-56-9	Propylene oxide	0.000	0.000	0.000	2.906	0.000	0.000	0.000	0.000	2.906	4
79-06-1	Acrylamide	0.000	0.000	0.000	0.000	2.845	0.000	0.000	0.000	2.845	10
26471-62-5	Toluene-diisocyanate (mixed isomers)	0.000	0.720	0.268	1.496	0.207	0.000	0.000	0.000	2.691	31
62-53-3	Aniline	0.000	0.000	2.610	0.023	0.000	0.000	0.000	0.000	2.633	2
NA	Cadmium (and its compounds)	0.804	0.000	0.000	0.000	1.635	0.000	0.045	0.000	2.484	18
108-31-6	Maleic anhydride	0.000	0.000	0.000	2.460	0.008	0.000	0.000	0.000	2.468	15
109-86-4	2-Methoxyethanol	0.000	0.000	0.000	2.121	0.000	0.000	0.000	0.000	2.121	4
1163-19-5	Decabromo-diphenyl oxide	0.000	0.000	0.000	0.000	1.995	0.001	0.000	0.000	1.996	3
584-84-9	Toluene-2,4-diisocyanate	0.000	0.000	0.000	1.590	0.000	0.000	0.000	0.000	1.590	3
108-38-3	*m*-Xylene	0.000	0.000	0.000	0.000	0.004	0.000	1.348	0.000	1.352	7
75-01-4	Vinyl chloride	0.000	0.000	0.000	0.130	0.827	0.000	0.000	0.000	0.957	10
75-21-8	Ethylene oxide	0.000	0.000	0.000	0.877	0.000	0.000	0.000	0.000	0.877	12
110-80-5	2-Ethoxyethanol	0.000	0.000	0.000	0.782	0.000	0.000	0.000	0.000	0.782	7
75-15-0	Carbon disulphide	0.000	0.000	0.000	0.350	0.230	0.000	0.000	0.100	0.680	7
1313-27-5	Molybdenum trioxide	0.000	0.223	0.000	0.000	0.280	0.000	0.169	0.000	0.672	15

CAS #	Substance Name	Physical	Chemical	Biological	Incineration	Landfill	Storage	MSTP*	Underground	Total Transfers	# of Reports
95-47-6	*o*-Xylene	0.000	0.000	0.000	0.019	0.007	0.000	0.526	0.000	0.552	7
139-13-9	Nitrilotriacetic acid	0.000	0.004	0.000	0.000	0.050	0.000	0.495	0.000	0.549	11
106-42-3	*p*-Xylene	0.000	0.000	0.000	0.000	0.004	0.000	0.516	0.000	0.520	4
106-46-7	*p*-Dichlorobenzene	0.000	0.000	0.000	0.000	0.500	0.000	0.000	0.000	0.500	5
7550-45-0	Titanium tetrachloride	0.000	0.000	0.000	0.442	0.018	0.000	0.000	0.000	0.460	4
107-06-2	1,2-Dichloroethane	0.000	0.000	0.000	0.220	0.000	0.000	0.000	0.000	0.220	4
74-85-1	Ethylene	0.000	0.000	0.000	0.000	0.000	0.000	0.160	0.000	0.160	43
141-32-2	Butyl acrylate	0.000	0.000	0.000	0.111	0.000	0.000	0.031	0.000	0.142	9
1634-04-4	Methyl *tert*-butyl ether	0.000	0.000	0.000	0.100	0.000	0.000	0.000	0.000	0.100	9
NA	Silver (and its compounds)	0.000	0.000	0.000	0.000	0.066	0.000	0.033	0.000	0.099	11
NA	Cyanides (ionic)	0.000	0.087	0.000	0.000	0.000	0.000	0.000	0.000	0.087	38
7782-50-5	Chlorine	0.000	0.000	0.000	0.000	0.000	0.000	0.065	0.000	0.065	175
75-05-8	Acetonitrile	0.000	0.000	0.000	0.064	0.000	0.000	0.000	0.000	0.064	3
534-52-1	4,6-Dinitro-*o*-cresol	0.000	0.000	0.037	0.000	0.000	0.000	0.000	0.000	0.037	1
123-31-9	Hydroquinone	0.000	0.000	0.000	0.017	0.000	0.000	0.006	0.000	0.023	3
79-10-7	Acrylic acid	0.000	0.000	0.000	0.021	0.000	0.000	0.000	0.000	0.021	9
98-82-8	Cumene	0.000	0.000	0.000	0.000	0.017	0.000	0.000	0.000	0.017	16
80-05-7	*p,p′*-Isopropylidene diphenol	0.000	0.000	0.000	0.000	0.014	0.000	0.000	0.000	0.014	2
75-65-0	*tert*-Butyl alcohol	0.000	0.000	0.000	0.000	0.010	0.000	0.000	0.000	0.010	2
67-66-3	Chloroform	0.000	0.000	0.000	0.008	0.000	0.000	0.000	0.000	0.008	3
111-15-9	2-Ethoxyethyl acetate	0.000	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.002	4
75-07-0	Acetaldehyde	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.001	5
	Total	**9,366.123**	**4,401.869**	**3,597.841**	**7,049.850**	**34,351.959**	**1,247.636**	**2,016.220**	**2,270.349**	**64,301.847**	**5,786**

* Municipal sewage treatment plant.

Appendix 9: Anticipated transfers in 1994 (tonnes)

CAS #	Substance Name	Reported 1994	Anticipated 1995	Anticipated 1996	Anticipated 1997	# of Reports
117-81-7	*Bis*(2-ethylhexyl) phthalate	14,891.139	7,383.282	7,744.311	8,127.039	33
7429-90-5	Aluminum (fume or dust)	7,766.402	7,773.203	7,769.403	7,770.303	35
NA	Copper (and its compounds)	7,038.419	7,426.786	7,394.882	7,391.904	243
NA	Zinc (and its compounds)	5,286.846	7,294.211	6,991.567	7,105.072	283
67-56-1	Methanol	4,395.865	4,348.491	4,220.336	4,162.771	284
NA	Manganese (and its compounds)	3,791.774	3,972.976	3,989.268	4,036.078	192
7664-93-9	Sulfuric acid	2,723.174	5,250.951	5,047.393	5,047.903	387
108-88-3	Toluene	1,917.749	2,090.693	2,042.001	1,965.132	313
107-21-1	Ethylene glycol	1,659.420	1,049.766	999.990	987.329	232
1330-20-7	Xylene (mixed isomers)	1,439.323	1,479.167	1,385.135	1,336.726	285
7440-66-6	Zinc (fume or dust)	1,432.288	1,410.502	1,414.558	1,330.558	36
NA	Lead (and its compounds)	1,301.650	1,492.703	1,444.971	1,391.900	160
7647-01-0	Hydrochloric acid	1,009.346	987.275	872.275	877.375	211
NA	Chromium (and its compounds)	1,002.646	1,031.374	887.663	800.124	189
1332-21-4	Asbestos	935.063	1,147.136	850.819	755.625	37
67-63-0	Isopropyl alcohol	849.325	779.929	741.555	744.371	195
78-93-3	Methyl ethyl ketone	692.533	623.174	633.857	623.986	141
7664-38-2	Phosphoric acid	559.799	502.368	496.678	494.053	177
67-64-1	Acetone	540.167	398.755	363.591	354.327	118
NA	Nickel (and its compounds)	506.918	558.001	505.214	448.592	114
108-05-4	Vinyl acetate	465.026	401.551	401.151	401.051	14
111-42-2	Diethanolamine	411.207	372.741	372.853	372.978	75
7664-41-7	Ammonia	393.054	360.819	380.819	402.819	161
108-95-2	Phenol	391.590	231.801	228.103	228.606	50
100-42-5	Styrene	310.694	306.770	220.213	219.570	78
80-62-6	Methyl methacrylate	263.306	78.020	63.840	59.860	13
71-36-3	*n*-Butyl alcohol	249.634	237.669	229.335	198.472	79
7697-37-2	Nitric acid	231.840	206.622	151.122	150.722	101
50-00-0	Formaldehyde	212.451	105.004	105.826	101.167	80
7440-62-2	Vanadium (fume or dust)	208.942	203.400	203.400	203.400	11
108-10-1	Methyl isobutyl ketone	169.641	171.643	168.627	171.291	71
71-43-2	Benzene	151.589	159.322	112.332	76.342	95
106-99-0	1,3-Butadiene	131.311	87.500	12.600	12.210	13
110-82-7	Cyclohexane	117.057	117.987	117.988	117.989	82
NA	Selenium (and its compounds)	97.870	27.800	27.800	27.800	7
127-18-4	Tetrachloroethylene	93.929	98.394	98.394	99.193	29
1344-28-1	Aluminum oxide (fibrous forms)	59.820	54.630	54.430	54.230	14
7783-20-2	Ammonium sulphate (solution)	52.058	56.000	56.000	56.000	19
95-63-6	1,2,4-Trimethylbenzene	51.966	53.514	53.424	53.423	75
100-41-4	Ethylbenzene	36.171	46.750	47.037	46.831	68
75-09-2	Dichloromethane	35.309	49.897	57.345	63.848	50
85-68-7	Butyl benzyl phthalate	33.287	21.111	21.111	21.111	11
NA	Mercury (and its compounds)	32.633	16.000	16.000	6.100	5
101-68-8	Methylene*bis*(phenylisocyanate)	32.319	27.038	26.398	25.908	48
78-83-1	*i*-Butyl alcohol	31.133	23.574	24.863	14.959	25
6484-52-2	Ammonium nitrate (solution)	30.030	30.030	0.030	0.030	22
NA	Arsenic (and its compounds)	29.117	27.852	25.485	25.104	45
91-20-3	Naphthalene	29.084	18.136	11.336	10.136	40
156-62-7	Calcium cyanamide	29.030	30.000	30.000	30.000	1
79-01-6	Trichloroethylene	27.228	31.103	30.627	24.201	45
95-50-1	*o*-Dichlorobenzene	22.700	11.350	11.350	11.350	3
107-13-1	Acrylonitrile	18.930	8.000	2.500	2.300	8
75-35-4	Vinylidene chloride	11.400	0.000	0.000	0.000	1
56-23-5	Carbon tetrachloride	11.107	8.900	6.000	6.000	6
1319-77-3	Cresol (mixed isomers)	9.200	4.640	4.640	4.640	4
7664-39-3	Hydrogen fluoride	7.950	5.100	0.100	0.050	19
103-23-1	*Bis*(2-ethylhexyl) adipate	7.019	8.074	8.174	5.075	14
NA	Antimony (and its compounds)	6.088	4.488	3.671	3.551	30
NA	Cobalt (and its compounds)	5.666	5.496	5.358	5.158	23
84-74-2	Dibutyl phthalate	5.601	5.541	5.702	5.866	17

CAS #	Substance Name	Reported 1994	Anticipated 1995	Anticipated 1996	Anticipated 1997	# of Reports
117-84-0	Di-*n*-octyl phthalate	5.136	6.423	6.667	6.935	14
96-33-3	Methyl acrylate	4.900	3.400	3.200	3.000	2
85-44-9	Phthalic anhydride	3.755	1.215	1.204	1.293	15
120-12-7	Anthracene	3.466	1.866	0.066	0.066	12
92-52-4	Biphenyl	3.052	2.802	2.802	2.802	10
75-56-9	Propylene oxide	2.906	2.750	2.500	0.000	4
79-06-1	Acrylamide	2.845	2.845	2.845	2.845	10
26471-62-5	Toluenediisocyanate (mixed isomers)	2.691	2.321	2.401	2.501	31
62-53-3	Aniline	2.633	3.140	3.140	3.140	2
NA	Cadmium (and its compounds)	2.484	0.921	0.789	0.685	18
108-31-6	Maleic anhydride	2.468	2.110	2.110	2.110	15
109-86-4	2-Methoxyethanol	2.121	1.253	1.253	1.253	4
1163-19-5	Decabromodiphenyl oxide	1.996	1.075	0.157	0.157	3
584-84-9	Toluene-2,4-diisocyanate	1.590	0.150	0.150	0.170	3
108-38-3	*m*-Xylene	1.352	5.404	5.404	5.404	7
75-01-4	Vinyl chloride	0.957	0.830	0.830	0.830	10
75-21-8	Ethylene oxide	0.877	15.000	9.500	0.000	12
110-80-5	2-Ethoxyethanol	0.782	1.638	1.638	1.638	7
75-15-0	Carbon disulphide	0.680	1.100	0.100	0.100	7
1313-27-5	Molybdenum trioxide	0.672	0.700	0.700	0.710	15
95-47-6	*o*-Xylene	0.552	2.125	2.123	2.123	7
139-13-9	Nitrilotriacetic acid	0.549	0.100	0.100	0.100	11
106-42-3	*p*-Xylene	0.520	2.104	2.104	2.104	4
106-46-7	*p*-Dichlorobenzene	0.500	0.000	0.000	0.000	5
7550-45-0	Titanium tetrachloride	0.460	0.270	0.310	0.230	4
107-06-2	1,2-Dichloroethane	0.220	0.200	0.200	0.200	4
74-85-1	Ethylene	0.160	0.160	0.160	0.160	43
141-32-2	Butyl acrylate	0.142	0.200	0.200	0.200	9
1634-04-4	Methyl *tert*-butyl ether	0.100	0.100	0.100	0.100	9
NA	Silver (and its compounds)	0.099	0.036	0.030	0.030	11
NA	Cyanides (ionic)	0.087	0.100	0.100	0.100	38
7782-50-5	Chlorine	0.065	0.695	0.695	0.695	175
75-05-8	Acetonitrile	0.064	1.064	1.064	1.064	3
534-52-1	4,6-Dinitro-*o*-cresol	0.037	0.037	0.037	0.037	1
123-31-9	Hydroquinone	0.023	0.024	0.024	0.024	3
79-10-7	Acrylic acid	0.021	0.001	0.001	0.001	9
98-82-8	Cumene	0.017	0.017	0.017	0.017	16
80-05-7	*p*,*p*′-Isopropylidenediphenol	0.014	0.020	0.040	0.050	2
75-65-0	*tert*-Butyl alcohol	0.010	0.010	0.010	0.010	2
67-66-3	Chloroform	0.008	0.008	0.008	0.008	3
111-15-9	2-Ethoxyethyl acetate	0.002	0.002	0.002	0.002	4
75-07-0	Acetaldehyde	0.001	0.001	0.001	0.001	5
	Total	**64,301.847**	**60,779.227**	**59,246.233**	**59,113.404**	**5,786**

Appendix 10: Substances sent for the "3Rs" and energy recovery (tonnes)

CAS #	Substance Name	"3Rs"	Energy Recovery	Total	# of Reports
7664-93-9	Sulfuric acid	65,776.565	0.000	65,776.565	20
7647-01-0	Hydrochloric acid	41,922.140	0.000	41,922.140	9
NA	Manganese (and its compounds)	36,706.039	0.669	36,706.708	57
67-63-0	Isopropyl alcohol	33,784.297	696.065	34,480.362	50
117-84-0	Di-*n*-octyl phthalate	20,013.426	52.900	20,066.326	6
NA	Copper (and its compounds)	17,513.498	0.643	17,514.141	101
NA	Lead (and its compounds)	13,959.820	1.144	13,960.964	61

CAS #	Substance Name	"3Rs"	Energy Recovery	Total	# of Reports
115-07-1	Propylene	13,856.940	0.000	13,856.940	2
NA	Zinc (and its compounds)	13,594.043	93.108	13,687.151	88
108-88-3	Toluene	5,891.989	1,239.520	7,131.509	94
1330-20-7	Xylene (mixed isomers)	5,412.974	972.188	6,385.162	83
78-93-3	Methyl ethyl ketone	3,336.456	440.351	3,776.807	48
NA	Chromium (and its compounds)	3,445.765	0.000	3,445.765	45
NA	Nickel (and its compounds)	2,995.830	0.000	2,995.830	41
NA	Cobalt (and its compounds)	1,190.480	2.358	1,192.838	4
67-64-1	Acetone	656.112	386.565	1,042.677	34
107-21-1	Ethylene glycol	700.670	113.045	813.715	32
67-56-1	Methanol	324.031	447.362	771.393	35
7440-62-2	Vanadium (fume or dust)	712.810	0.000	712.810	4
7664-38-2	Phosphoric acid	603.566	0.000	603.566	7
7429-90-5	Aluminum (fume or dust)	590.819	4.410	595.229	12
NA	Arsenic (and its compounds)	515.607	0.000	515.607	2
108-10-1	Methyl isobutyl ketone	194.649	120.276	314.925	27
7697-37-2	Nitric acid	297.832	0.000	297.832	7
127-18-4	Tetrachloroethylene	236.241	30.340	266.581	8
NA	Antimony (and its compounds)	224.008	0.000	224.008	10
79-01-6	Trichloroethylene	188.980	10.738	199.718	24
100-42-5	Styrene	0.696	182.300	182.996	8
100-41-4	Ethylbenzene	151.512	27.442	178.954	10
71-36-3	*n*-Butyl alcohol	59.243	116.577	175.820	26
1313-27-5	Molybdenum trioxide	171.092	0.000	171.092	2
75-09-2	Dichloromethane	160.145	5.050	165.195	10
7440-66-6	Zinc (fume or dust)	110.780	0.000	110.780	6
NA	Cadmium (and its compounds)	102.937	0.650	103.587	4
95-63-6	1,2,4-Trimethylbenzene	62.242	28.311	90.553	8
7664-41-7	Ammonia	85.272	0.000	85.272	3
1344-28-1	Aluminum oxide (fibrous forms)	66.940	0.000	66.940	2
111-42-2	Diethanolamine	46.700	16.504	63.204	4
110-82-7	Cyclohexane	61.474	0.059	61.533	4
NA	Selenium (and its compounds)	58.770	0.000	58.770	1
117-81-7	*Bis*(2-ethylhexyl) phthalate	48.772	0.000	48.772	4
108-38-3	*m*-Xylene	14.666	5.424	20.090	1
108-05-4	Vinyl acetate	0.000	20.000	20.000	1
108-95-2	Phenol	15.246	1.573	16.819	5
101-68-8	Methylene*bis*(phenylisocyanate)	7.870	0.000	7.870	3
78-83-1	*i*-Butyl alcohol	4.724	2.380	7.104	3
71-43-2	Benzene	6.800	0.003	6.803	2
84-74-2	Dibutyl phthalate	0.304	3.656	3.960	3
85-68-7	Butyl benzyl phthalate	1.392	2.290	3.682	4
75-56-9	Propylene oxide	0.000	2.906	2.906	1
50-00-0	Formaldehyde	1.668	0.095	1.763	5
1319-77-3	Cresol (mixed isomers)	1.620	0.000	1.620	2
75-21-8	Ethylene oxide	0.000	0.877	0.877	1
98-82-8	Cumene	0.000	0.500	0.500	1
26471-62-5	Toluenediisocyanate (mixed isomers)	0.500	0.000	0.500	1
141-32-2	Butyl acrylate	0.000	0.500	0.500	1
110-80-5	2-Ethoxyethanol	0.369	0.008	0.377	2
140-88-5	Ethyl acrylate	0.000	0.150	0.150	1
1634-04-4	Methyl *tert*-butyl ether	0.100	0.000	0.100	1
80-62-6	Methyl methacrylate	0.000	0.100	0.100	1
109-86-4	2-Methoxyethanol	0.000	0.071	0.071	1
78-92-2	*sec*-Butyl alcohol	0.000	0.057	0.057	1
NA	Silver (and its compounds)	0.008	0.000	0.008	1
	Total	**285,887.429**	**5,029.165**	**290,916.594**	**1,045**

Appendix 11: Anticipated "3Rs" and energy recovery (tonnes)

CAS #	Substance Name	Reported 1994	Anticipated 1995	Anticipated 1996	Anticipated 1997	# of Reports
7664-93-9	Sulfuric acid	65,776.565	57,696.771	55,258.803	51,798.853	20
7647-01-0	Hydrochloric acid	41,922.140	41,069.500	41,073.000	41,068.000	9
NA	Manganese (and its compounds)	36,706.708	39,708.548	42,853.007	46,650.335	57
67-63-0	Isopropyl alcohol	34,480.362	21,183.130	20,996.355	20,998.630	50
117-84-0	Di-*n*-octyl phthalate	20,066.326	20,073.766	20,073.966	20,074.166	6
NA	Copper (and its compounds)	17,514.141	17,108.145	17,235.073	17,204.156	101
NA	Lead (and its compounds)	13,960.964	4,250.265	3,641.740	3,605.310	61
115-07-1	Propylene	13,856.940	14,560.000	14,560.000	14,560.000	2
NA	Zinc (and its compounds)	13,687.151	13,612.899	13,815.493	13,885.835	88
108-88-3	Toluene	7,131.509	6,270.499	4,773.476	4,745.719	94
1330-20-7	Xylene (mixed isomers)	6,385.162	4,961.039	4,508.673	4,315.652	83
78-93-3	Methyl ethyl ketone	3,776.807	3,178.159	2,052.329	2,027.049	48
NA	Chromium (and its compounds)	3,445.765	3,281.401	3,332.946	2,264.866	45
NA	Nickel (and its compounds)	2,995.830	2,882.309	2,914.951	2,942.554	41
NA	Cobalt (and its compounds)	1,192.838	1,175.000	1,175.000	1,175.000	4
67-64-1	Acetone	1,042.677	865.275	859.342	865.191	34
107-21-1	Ethylene glycol	813.715	719.572	682.096	671.216	32
67-56-1	Methanol	771.393	750.190	660.610	661.556	35
7440-62-2	Vanadium (fume or dust)	712.810	600.000	600.000	600.000	4
7664-38-2	Phosphoric acid	603.566	253.876	253.889	158.901	7
7429-90-5	Aluminum (fume or dust)	595.229	594.189	598.869	604.749	12
NA	Arsenic (and its compounds)	515.607	515.000	515.000	515.000	2
108-10-1	Methyl isobutyl ketone	314.925	224.780	205.720	202.790	27
7697-37-2	Nitric acid	297.832	249.000	249.000	249.000	7
127-18-4	Tetrachloroethylene	266.581	223.801	160.111	113.201	8
NA	Antimony (and its compounds)	224.008	220.718	223.014	221.864	10
79-01-6	Trichloroethylene	199.718	152.326	113.737	69.079	24
100-42-5	Styrene	182.996	200.620	201.220	200.420	8
100-41-4	Ethylbenzene	178.954	86.900	84.400	81.200	10
71-36-3	*n*-Butyl alcohol	175.820	173.060	168.870	166.670	26
1313-27-5	Molybdenum trioxide	171.092	170.000	170.000	170.000	2
75-09-2	Dichloromethane	165.195	172.050	177.000	183.000	10
7440-66-6	Zinc (fume or dust)	110.780	65.100	70.100	70.100	6
NA	Cadmium (and its compounds)	103.587	101.730	101.730	101.730	4
95-63-6	1,2,4-Trimethylbenzene	90.553	93.500	93.500	91.100	8
7664-41-7	Ammonia	85.272	84.870	81.370	77.870	3
1344-28-1	Aluminum oxide (fibrous forms)	66.940	0.000	0.000	0.000	2
111-42-2	Diethanolamine	63.204	56.500	56.500	56.500	4
110-82-7	Cyclohexane	61.533	73.000	73.000	73.000	4
NA	Selenium (and its compounds)	58.770	58.000	58.000	58.000	1
117-81-7	*Bis*(2-ethylhexyl) phthalate	48.772	65.673	65.673	65.673	4
108-38-3	*m*-Xylene	20.090	14.750	14.750	14.750	1
108-05-4	Vinyl acetate	20.000	10.000	5.000	5.000	1
108-95-2	Phenol	16.819	22.848	15.779	15.417	5
101-68-8	Methylene*bis*(phenylisocyanate)	7.870	4.950	4.950	4.950	3
78-83-1	*i*-Butyl alcohol	7.104	3.569	2.409	2.409	3
71-43-2	Benzene	6.803	7.000	7.000	7.000	2
84-74-2	Dibutyl phthalate	3.960	0.304	0.334	0.364	3
85-68-7	Butyl benzyl phthalate	3.682	3.641	3.641	3.641	4
75-56-9	Propylene oxide	2.906	0.000	0.000	0.000	1
50-00-0	Formaldehyde	1.763	4.723	1.336	1.392	5
1319-77-3	Cresol (mixed isomers)	1.620	1.500	1.500	1.500	2
75-21-8	Ethylene oxide	0.877	0.000	0.000	0.000	1
141-32-2	Butyl acrylate	0.500	0.000	0.000	0.000	1
98-82-8	Cumene	0.500	0.500	0.500	0.500	1
26471-62-5	Toluenediisocyanate (mixed isomers)	0.500	0.000	0.000	0.000	1
110-80-5	2-Ethoxyethanol	0.377	0.000	0.000	0.000	2
140-88-5	Ethyl acrylate	0.150	0.000	0.000	0.000	1
1634-04-4	Methyl *tert*-butyl ether	0.100	0.000	0.000	0.000	1

CAS #	Substance Name	Reported 1994	Anticipated 1995	Anticipated 1996	Anticipated 1997	# of Reports
80-62-6	Methyl methacrylate	0.100	0.000	0.000	0.000	1
109-86-4	2-Methoxyethanol	0.071	0.000	0.000	0.000	1
78-92-2	*sec*-Butyl alcohol	0.057	0.000	0.000	0.000	1
NA	Silver (and its compounds)	0.008	0.000	0.000	0.000	1
	Total	**290,916.594**	**257,854.946**	**254,845.762**	**253,702.858**	**1,045**

Air emissions

Air emissions data are presented in several formats. Tables P-2 and P-3 are examples of how air criteria contamination data for specific provinces are tabulated.

Data from Environment Canada indicate that the annual average level of airborne particles at all NAPS monitoring sites decreased by 46 percent between 1974 and 1986. This significant improvement in urban air quality relative to particulate emissions reductions reflects the fact that most major industrial sources of particulate matter are located outside urban areas. The average levels at more than 90 percent of the sites were well within the desirable range of air quality in 1986 (Fig. P-2).

Short-term exposure to high levels of particulate matter continues to be a problem, however. For example, the maximum acceptable level objective for a 24-hour exposure period was exceeded at least 10 percent of the time at some NAPS sites in Sydney, Rouyn, Windsor, Edmonton, Hamilton, and Calgary. As well, the 24-hour maximum tolerable level ambient air quality objective was exceeded once in 1986 at NAPS monitoring sites in each of Edmonton, Calgary, and Yellowknife. High short-term particulate levels can be attributed in part to natural windblown dust, as well as to construction and industrial activity and the increasing number of motor vehicles on city streets. Over the long term, economic and social development—which has led to the use of cleaner energy sources (e.g., natural gas instead of coal and wood), modernization of older city centers, cleaner streets, more grass and asphalt cover, and the upgrading of industrial and commercial facilities, including installation of equipment to control pollution—appears to be leading to better air quality in urban areas with respect to particulate matter.

Environment Canada has collected inhalable particles (<10 μm in diameter) in 15 Canadian cities since May 1984. Results for the period from May 1984 to December 1987 show that average inhalable particulate levels ranged from 17 $\mu g/m^3$ in St. John's to 49 $\mu g/m^3$ at a site in Montreal. Analysis has shown that the finest of the inhalable particles—those with a diameter less than about 2.5 μm—are different in origin and composition from the coarser particles in the 2.5- to 10-μm-diameter range. The coarse particles are mostly of natural origin (minerals from Earth's crust, sea salt, and plant material), whereas the fine particles consist of lead, sulfates, nitrates, carbon, and a variety of organic compounds, mainly resulting from man-made pollution. At eastern Canadian sites, fine particulate matter accounted for more than 60 percent of the inhalable particles; at sites in the Prairie provinces, the fine fraction was usually less than 40 percent of the inhalable particles.

The levels of inhalable particles measured in Canada are below the recently introduced U.S. air quality standards for this size of particle, set at 150 $\mu g/m^3$ for exposure over one hour and 50 $\mu g/m^3$ for exposure over one year. Canadian air quality objectives for inhalable particulate matter are being developed.

TABLE P-2 Case Study: 1990 Emissions of Criteria Contaminants for Ontario

	Emissions in Tonnes				
Category/Sector	Part	SO_2	NO_x	VOC	CO
Industrial Processes					
Abrasives manufacture	126	21	10	42	1
Aluminum production	235	5	49	5	1
Asphalt production	11,255	255	131	2,176	61
Bakeries			4	1,374	
Cement and concrete manufacture	5,422	12,373	9,603	155	503
Clay products	4,900	2	98	13	31
Crude oil production				46	
Grain industries	12,178				
Iron and steel production	22,562	48,125	21,719	25,275	548,265
Mining and rock quarrying	53,550	897	946	73	455
Nonferrous mining and smelting	6,264	692,302	52,668	277	26
Other chemicals	6,185	834	1,233	401	1,018
Paint & varnish manufacturing	40	1	9	912	2
Petrochemical industry	1,033	1,012	3,397	11,802	9,529
Petroleum refining	3,226	63,687	14,360	35,047	5,020
Plastics & synthetic resins fabrication	128	296	316	5,964	814
Pulp and paper industry	22,024	29,398	9,632	7,913	22,158
Wood industry	8,680	279	2,073	6,557	522
Other industries*	80,635	75,051	35,023	57,455	27,069
Category total	**238,440**	**924,539**	**151,269**	**155,489**	**615,474**
Fuel Combustion					
Commercial fuel combustion	517	4,050	8,323	317	1,657
Electric power generation	8,208	195,101	77,235	396	3,353
Residential fuel combustion	467	8,146	13,800	952	5,305
Residential fuelwood combustion	28,779	348	2,342	114,664	206,411
Category total	**37,971**	**207,645**	**101,700**	**116,329**	**216,725**
Transportation					
Aircraft	328	409	5,527	2,157	13,116
Heavy duty diesel vehicles	9,336	14,646	116,814	13,164	52,846
Heavy duty gasoline trucks	45	37	3,241	3,159	41,696
Light duty diesel trucks	35	1,146	789	324	622
Light duty diesel vehicles	31	1,012	605	207	490
Light duty gasoline trucks	392	1,134	32,226	44,150	422,853
Light duty gasoline vehicles	1,366	3,094	113,666	165,492	1,497,666
Marine	1,737	19,316	10,526	12,628	38,406
Motor cycles	6	7	217	1,358	3,612
Off-road use of diesel	5,785	6,970	70,651	6,335	21,296
Off-road use of gasoline	1,728	336	10,957	34,812	388,211
Propane powered vehicles	101	7	2,803	1,674	2,601
Railroads	1,783	2,663	33,042	1,604	13,192
Tire wear	15,088			305	
Category total	**37,759**	**50,775**	**401,065**	**287,369**	**2,496,607**
Incineration					
Other incineration	52	750	962	1,104	2,012
Wood waste incineration	2,034	35	351	3,856	45,565
Category total	**2,086**	**785**	**1,313**	**4,959**	**47,577**
Miscellaneous					
Application of surface coatings	16		4	121,628	
Cigarette smoking	1,203				480
Dry cleaning			1	5,933	
Fuel marketing				35,309	
General solvent use	4		9	124,225	2
Pesticides and fertilizer application	4,883		1	9,482	
Structural fires	2,062		424	2,062	4,123
Category total	**8,168**		**440**	**298,639**	**4,606**
TOTAL FOR ONTARIO	**324,424**	**1,183,745**	**655,786**	**862,784**	**3,380,990**

NOTE: Due to rounding, totals may not add exactly.

* Industrial sectors such as Coal Industry, Iron Ore Mining and Beneficiation, Carbon Black, Ferrous Foundries, and Other Petroleum and Coal Products were included in the Other Industries sector to protect the confidentiality of the information.

TABLE P-3 Case Study: 1990 Emissions of Criteria Contaminants for Quebec

Category/Sector	Emissions in Tonnes				
	Part	SO_2	NO_x	VOC	CO
Industrial Processes					
Aluminum production	14,790	25,669	602	691	240,024
Asbestos production	88	963	220	1	20
Asphalt production	1,629				
Bakeries				1,094	
Cement and concrete manufacture	3,829	14,061	3,941	13	825
Coal industry	1,729		156	2,041	2,236
Crude oil production				1,093	
Ferrous foundries	1,027	1,846	15	1,053	515
Grain industries	4,346				
Iron and steel production	24,547	19,477	5,465	1,142	21,313
Mining and rock quarrying	33,266				
Nonferrous mining and smelting	2,306	194,158	561	7	119
Other chemicals	1,945	11,470	2,985	185	232
Petrochemical industry	105	862	1,238	6,835	247
Petroleum refining	893	15,630	5,059	20,100	5,217
Plastics & synthetic resins fabrication	44	127	29	4,520	4
Pulp and paper industry	21,736	41,945	17,061	2,449	13,731
Wood industry	13,034	104	51	22	37
Other industries	18,662	7,244	5,218	3,130	4,466
Category total	**143,976**	**333,556**	**42,600**	**44,377**	**288,986**
Fuel combustion					
Commercial fuel combustion	354	4,271	3,930	145	800
Electric power generation	932	11,313	3,527	39	291
Residential fuel combustion	624	8,088	5,237	207	1,318
Residential fuelwood combustion	63,377	515	1,393	51,188	274,476
Category total	**65,287**	**24,186**	**14,087**	**51,579**	**276,884**
Transportation					
Aircraft	220	291	4,302	1,176	8,715
Heavy duty diesel vehicles	5,765	7,753	72,139	8,130	32,635
Heavy duty gasoline trucks	26	39	1,917	1,846	24,696
Light duty diesel trucks	16	464	364	147	285
Light duty diesel vehicles	37	1,041	722	243	583
Light duty gasoline trucks	185	959	15,095	20,346	196,767
Light duty gasoline vehicles	883	3,376	72,241	103,971	954,911
Marine	1,897	17,338	10,397	4,777	13,520
Motor cycles	3	6	117	670	1,950
Off-road use of diesel	3,881	4,087	48,846	4,198	14,365
Off-road use of gasoline	632	302	6,051	15,177	206,968
Railroads	569	731	10,542	512	4,209
Tire wear	9,500			192	
Category total	**23,615**	**36,387**	**242,731**	**161,385**	**1,459,604**
Incineration					
Other incineration	91	915	624	432	5,763
Wood waste incineration	4,355	40	396	4,355	51,466
Category total	**4,446**	**955**	**1,020**	**4,787**	**57,229**
Miscellaneous					
Application of surface coatings				40,356	
Cigarette smoking	892				356
Dry cleaning				2,548	
Fuel marketing				17,765	
General solvent use				77,112	
Structural fires	98		36	127	1,530
Category total	**990**		**36**	**137,908**	**1,885**
TOTAL FOR QUEBEC	**238,314**	**395,084**	**300,476**	**400,036**	**2,084,588**

NOTE: Due to rounding, totals may not add exactly.

* Industrial sectors such as Abrasives Manufacture were included in the Other Industries sector to protect the confidentiality of the information.

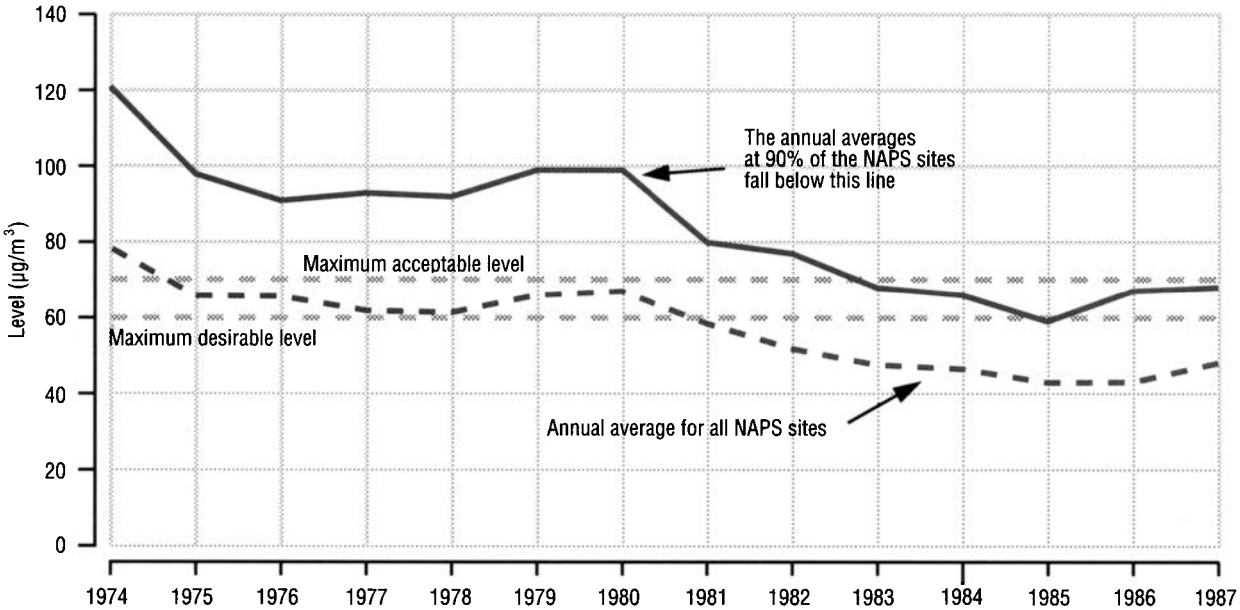


FIG. P-2 Trends in ambient air levels of total suspended particulate matter in Canada, 1974–1987. Note: Particulate matter is collected over a 24-hour period on every sixth day throughout the year. (Source: Environment Canada.)

Sulfur dioxide

About 70 percent of total sulfur dioxide emissions in 1985 in Canada came from industrial processes—69 percent of this was from copper, nickel, lead, zinc, gold, and aluminum production, and a further 21 percent was from oil and natural gas recovery and processing. Fuel combustion, mainly by power plants and industries, accounted for 28 percent of total Canadian emissions of sulfur dioxide. Between 1970 and 1985, sulfur dioxide emissions declined by almost 45 percent (Fig. P-3), mainly because of modifications to industrial processes and technology, the capture and use of sulfur dioxide to make commercially useful sulfuric acid, and the increased use of low-sulfur fuels.

Environmental levels of sulfur dioxide can adversely affect both human health and vegetation. In epidemiological and laboratory-controlled human health studies, effects on the lung and the induction of chronic lung disease have been recorded. Although no clear threshold has been identified, short-term exposures to sulfur dioxide at concentrations of up to 1 ppm have not induced severe or irreversible effects; however, significant reductions in lung function have been observed in healthy exercising adults after exposure to this level. Mild respiratory symptoms related to airway dysfunction and transient bronchoconstriction have also been observed in exercising asthmatic subjects.

In epidemiological studies, short-term exposures to sulfur dioxide that lasted a day or so have been correlated with deaths, although there was concomitant exposure to high particulate levels during these pollution episodes. Long-term or chronic exposures to levels of up to 50 ppb of sulfur dioxide–induced respiratory symptoms and disease (coughs and bronchitis), especially in young children and smokers.

The earliest sign of injury to vegetation is damage to foliage; other plant parts appear to be more resistant. The eastern white pine is a particularly sensitive

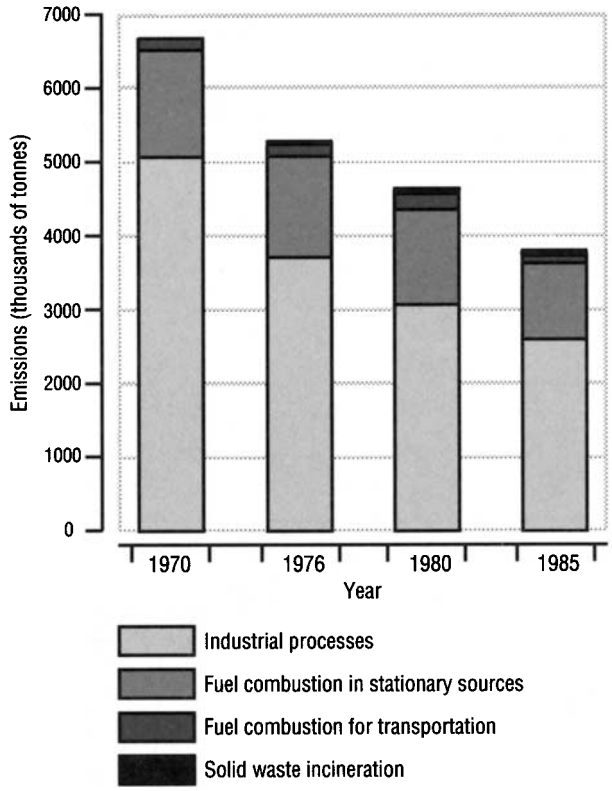


FIG. P-3 Sulfur dioxide emissions in Canada, 1970–1985. (Source: Environment Canada.)

species, showing signs of acute injury in a matter of hours at a sulfur dioxide concentration in air as low as 25–30 ppb. Many plant species are damaged within hours when exposed to a sulfur dioxide concentration between 100 and 1000 ppb, whereas some hardier plant species show acute damage only above 1 ppm.

Long-term injury to vegetation, usually observed as a yellowing of foliage, is generally assessed in terms of the average concentration of sulfur dioxide to which a plant is exposed over the growing season or over a one-year period, because sulfur dioxide concentrations can be quite variable from day to day. The response observed in plants is also strongly influenced by such environmental factors as sun, rain, wind, and drought. Studies of chronic forest damage suggest that effects are prominent when the average sulfur dioxide concentration is about 17 ppb but slight when the concentration is 8 ppb. Lichen species diversity and abundance are affected at 15–20 ppb.

The recommended air quality objectives were based on both human health effects and effects on vegetation. The maximum acceptable limits for a one-hour, 24-hour, and annual average were 340, 110, and 20 ppb, respectively; maximum desirable limits were 170, 60, and 10 ppb for the same time periods. These levels were retained following a review of the more recent literature.

In urban areas of Canada, the annual average level of sulfur dioxide measured at NAPS monitoring sites decreased by 54 percent between 1974 and 1986—from 13 ppb to 6 ppb—and the annual average levels at 90 percent of the NAPS monitoring sites are now well below the maximum desirable level annual air quality objective of 10 ppb (Fig. P-4). In 1986, the one-hour maximum desirable level

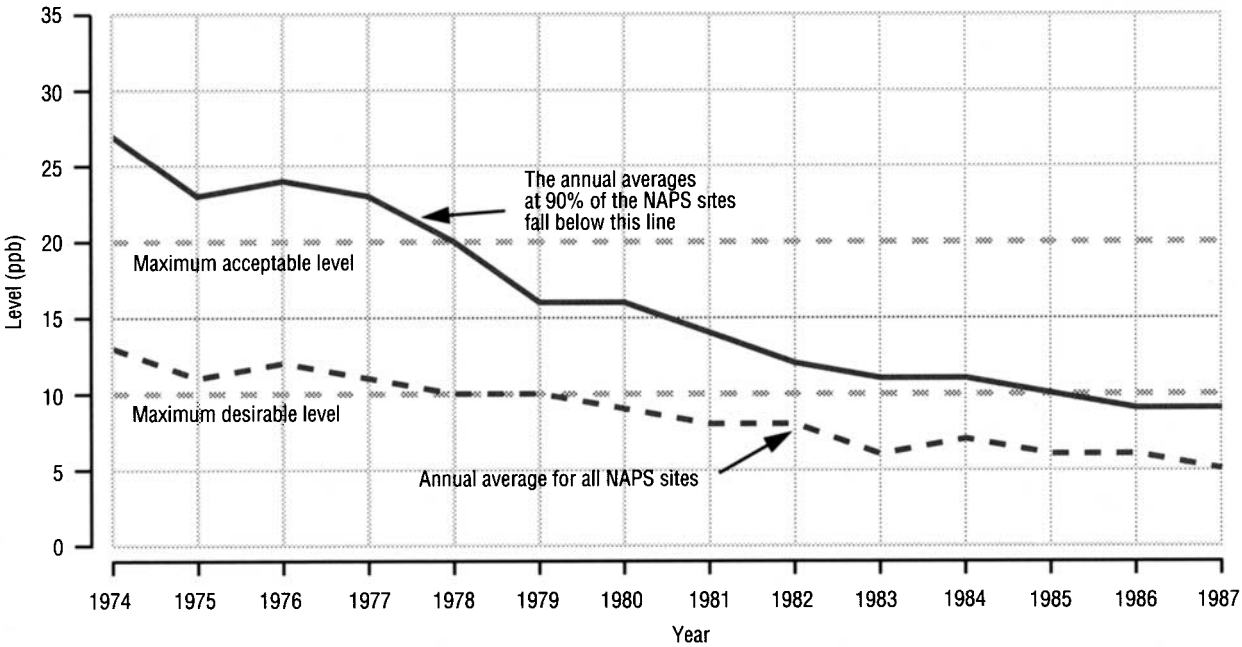


FIG. P-4 Trends in ambient air levels of sulfur dioxide in Canada, 1974–1987. (Source: Environment Canada.)

TABLE P-4 Air Quality Objectives for Sulfur Dioxide

Exposure Period	Maximum Desirable Concentration (ppb)	Maximum Acceptable Concentration (ppb)	Maximum Tolerable Concentration (ppb)
One hour	170	340	—
24 hours	60	110	310
One year	10	20	

objective of 170 ppb was met or bettered 99.9 percent of the time at 93 percent of the NAPS monitoring sites. Monitoring sites in Montreal, Quebec City, Rouyn, Trois-Rivières, Shawinigan, and Sudbury all recorded hourly average sulfur dioxide concentrations that exceeded the maximum acceptable level one-hour air quality objective of 340 ppb, but in all cases the objective was exceeded less than 1 percent of the time. At all but the Quebec City, Rouyn, and Shawinigan sites, however, the 24-hour maximum acceptable level air quality objective of 110 ppb was not exceeded in 1986. The 24-hour maximum tolerable level was exceeded 12 times in Quebec City in 1986.

See Table P-4 for air quality objectives for sulfur dioxide.

Carbon monoxide

Any combustion process where carbon-containing organic material is burned without sufficient oxygen will produce carbon monoxide. Motor vehicles, especially poorly tuned ones, are a major source of carbon monoxide because of the great number on the road at any given time. In fact, more than 66 percent of the carbon monoxide emitted in Canada in 1985 was from the internal combustion engines of motor vehicles, trains, aircraft, and boats, our principal means of transport (Fig. P-5).

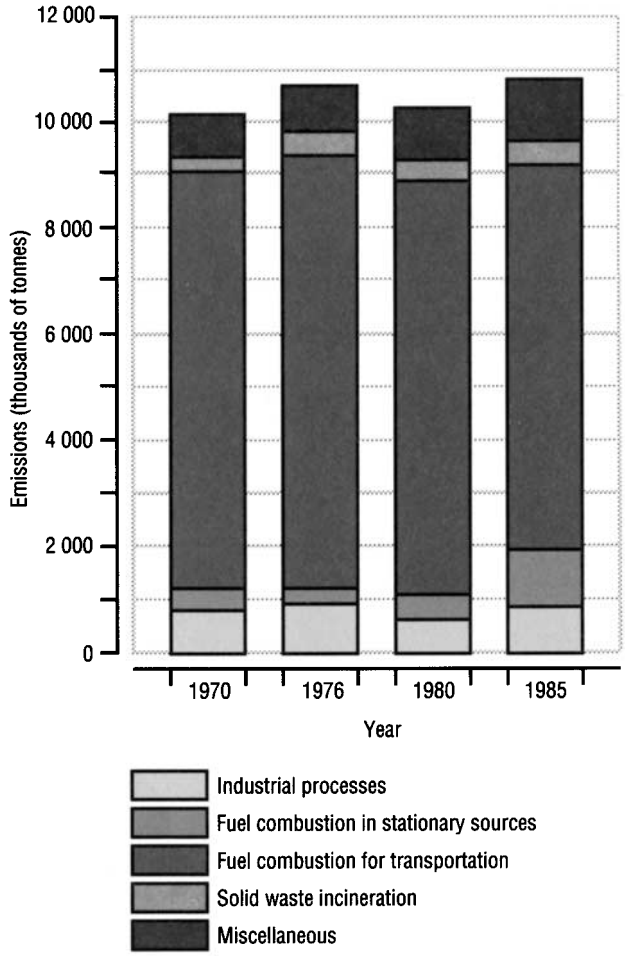


FIG. P-5 Carbon monoxide emissions in Canada, 1970–1985. (Source: Environment Canada.)

The amounts of carbon monoxide emitted from industrial processes, from the use of fuel for generating electric power and for heating homes, offices, and commercial and industrial structures, and from burning wood waste and debris from logging operations are each small in comparison with emissions from transportation, but they are nevertheless important. For example, the concentrated emissions of carbon monoxide from home heating with firewood, although less than 6 percent of total national emissions, are a concern quite distinct from carbon monoxide emissions from motor vehicles, and the potential health effects should not be underestimated.

The more than 10 million gasoline-powered automobiles and trucks that traveled Canada's roads and highways in 1985 emitted about 5.8 million tonnes of carbon monoxide, marginally less than the amount emitted in the early 1970s. Considering that in 1985 there were more than four million additional motor vehicles in use than there were in 1970—i.e., about 70 percent more—achieving a reduction in total emissions over this period is significant. Most of the credit must be attributed to the use of catalytic converters in order to meet federal motor vehicle emissions standards introduced in 1971. These standards were tightened in September 1987, which is expected to lead to even greater emissions reductions.

The toxic effects of carbon monoxide are due to its preferential combination with



FIG. P-6 Eutrophication causes loss of marine life. (Source: Environment Canada.)

the heme component of red blood cells to form carboxyhemoglobin, which reduces the capacity of the red blood cells to carry oxygen to the tissues. Because humans produce carbon monoxide in their bodies as a consequence of normal respiration, levels of carboxyhemoglobin of about 0.05–1.0 percent are found in healthy humans. In humans with hemolytic disease, the level is normally higher. Physiological effects are known to occur when carboxyhemoglobin levels reach 2.5–3.0 percent, and this induces a deficiency of oxygen in the tissues, increased oxygen debt, and even fatal myocardial infarction at exposures of about 5000 ppm for a few minutes or chronic exposure to about 14 ppm.

Chronic exposures of animals (rats and mice) to carbon monoxide have revealed other physiological and behavioral changes. These include an impairment in time discrimination and a consistent trace metal loss, but further work on this aspect is needed before firm conclusions can be drawn.

Special groups at risk when exposed to relatively high levels of carbon monoxide (>2.0%) include those with heart function and blood circulation problems, smokers, anemics, and, possibly, those living at high altitudes.

Only one-hour and eight-hour objectives were recommended to Environment Canada by the Federal–Provincial Committee on Air Pollution (1976); these were 13 and 5 ppm for the desirable levels, and 30 and 13 ppm for the acceptable levels, respectively. These recommendations remained unchanged in the more recent appraisal (1987), except for the eight-hour acceptable level, for which 11 ppm was recommended. Environment Canada is considering this recommended change.

Water pollution

Eutrophication. All lakes have a life cycle; it is perfectly correct to speak of a young lake or an old lake. Within any lake, there is a delicate balance among oxygen supply; animal, plant, and aquatic life; and sediments. As we saw with the nymphs in Lake Erie, if there are too many nutrients, such as phosphorus, the lake becomes over-fertilized. Aquatic life proliferates, places too much demand on the limited amounts of oxygen the water can hold, and then begins to die and settle to the bottom. (See Fig. P-6.) Phosphorus, from agricultural runoff and inadequately

treated sewage, is one of the major causes of overfertilization. Stream runoff and suspended matter in sewage also load large amounts of particles into the lakes. When these settle, along with the dead and decaying aquatic life that has been bred as a result of overfertilization, the sedimentation rate of a lake increases rapidly. Oxygen is exhausted and fish begin to die. The lake silts up and its aging process accelerates; when it reaches its final stages it becomes a swamp and at last, dry land. Lakes Ontario and Erie were already "middle-aged" when the white man arrived. It is estimated that in the first 50 years of the twentieth century, we aged Lake Erie the equivalent of 15,000 years.

Fortunately, once the alarm was sounded, Ontario awakened to the dangers. A program of provincially funded municipal sewage treatment plant construction was begun in the 1950s. This program continues today with the financial assistance of the federal government.

In Ontario, the battle against phosphorus and untreated municipal sewage has been largely successful. Treatment plants were built at a great rate across the province, and in the 1970s, the staggering sum of $991 million was spent or committed for sewage treatment plant construction. Rapid strides have been made in the United States, although some major centers, such as Niagara Falls, still lack proper treatment facilities. The progress that has been made in reducing phosphorus has been dramatic, and the rate of eutrophication has been reduced to something more near normal. Perhaps the greatest source of nutrients now entering the lake is agricultural runoff, with industrial pollution from the air, water, and land also significant. The Canada Water Act of 1972 limited the use of phosphates in household detergents, a major source of phosphorus pollution, to 5 percent by weight. However, other products, such as wetting agents, water softeners, and industrial cleaners, may still contain up to 45 percent phosphates by weight.

The fight that is being fought and is so far being won against phosphates and unnaturally fast eutrophication demonstrates that society and government have had the will, the desire, and the capacity to halt the deterioration of the Great Lakes. That same will must be applied to the other major problem facing the Great Lakes: the menace of toxic chemicals.

Toxic chemicals in the Great Lakes

The increasing detection of toxic chemicals in the Great Lakes (Figs. P-7 through P-10) is the latest challenge we face. Here again we find ourselves having to make up for years of neglect. Over 70,000 chemicals are in use today, and 35,000 have been classified by the EPA as definite or potential threats to human health. New concoctions are being developed at the rate of 1000 per year. Chemicals such as PCBs can be extremely dangerous if uncontrolled. There is a need for the development and establishment of efficient methods of disposal for the PCB wastes that are now in storage in Ontario and American jurisdictions. The problem is compounded by the fact that, once in the food chain, many chemicals accumulate in the bodies of organisms that eat or drink substances that have been contaminated by them. The higher a species is in the food chain, the more the effect is magnified. Humans are at the top of the chain.

With the development of sophisticated monitoring equipment, we have discovered that the Great Lakes, Erie and Ontario in particular, are contaminated with hazardous substances. Trace elements in extremely small quantities are now detectable. Unfortunately in many cases, it is virtually impossible to say how much of a given chemical is too much. The exhaustive and expensive testing necessary to determine the amounts of a given chemical that are harmful has only been carried out on a fraction of the substances now in use. Some chemicals are so harmful that

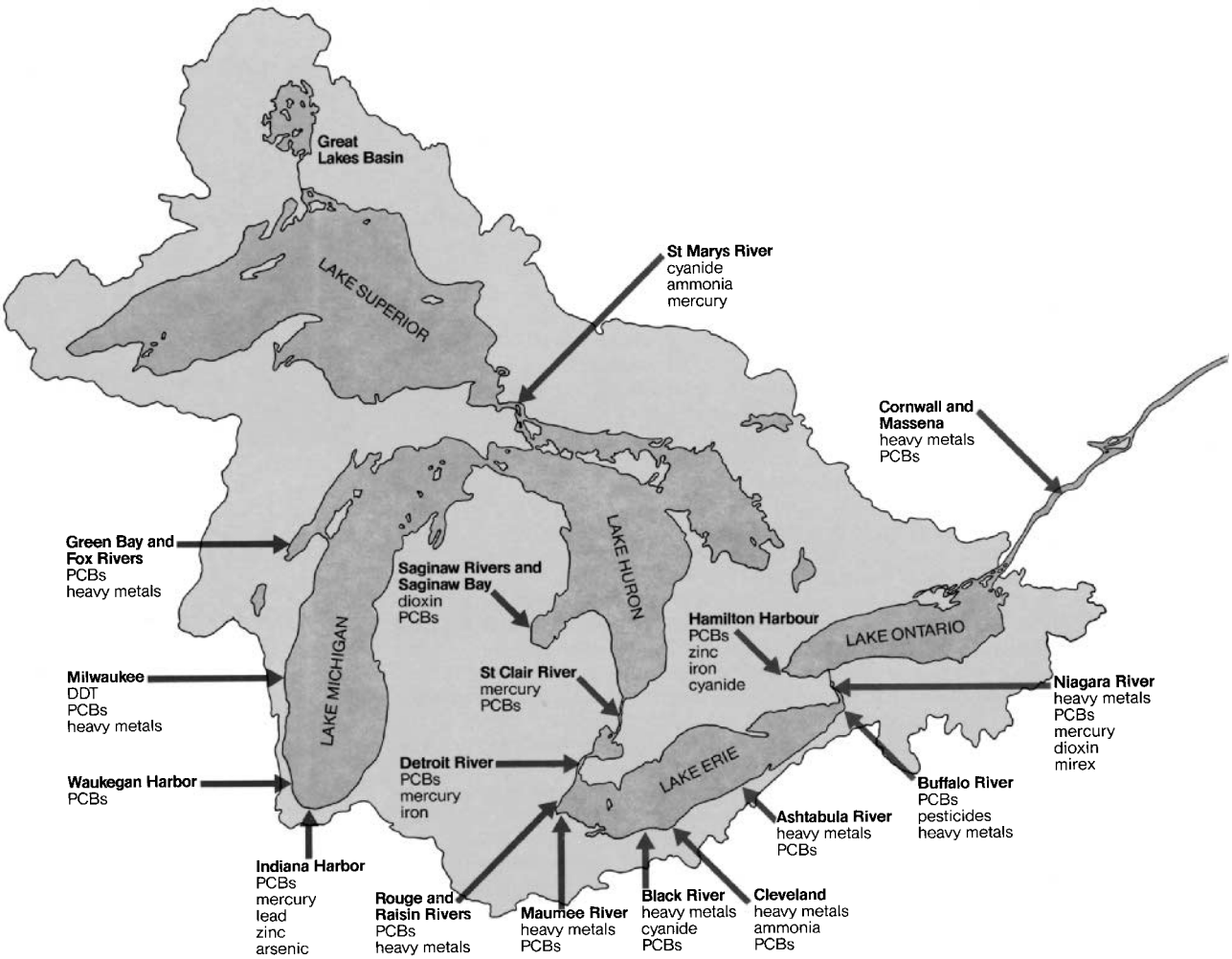


FIG. P-7 Toxic substances in the Great Lakes. (Source: Environment Canada.)

only a tiny bit, or a small accumulation, could cause serious illness or even death.

Furthermore, chemicals that are harmless or relatively harmless by themselves can become dangerous when they come in contact with other substances. These effects are extremely hard to predict, and there are so many potential combinations, that it would be impractical as well as prohibitively expensive for scientists to test all the possibilities. Far better to keep the lakes from becoming contaminated with hazardous substances in the first place.

The toxics problem is also international. It has been estimated that the overwhelming portion of the chemical contamination of the Niagara River comes from sources on the United States side of the border. But toxics also find their way into the Great Lakes from sources on the Canadian side. Agricultural pesticides from land runoff and chemicals from industry are examples.

The seriousness of the trouble, and the need for fast and effective action, has been



FIG. P-8 Illegal chemical dumping. (Source: Environment Canada.)



FIG. P-9 Monitoring water quality is a year-round job. (Source: Environment Canada.)

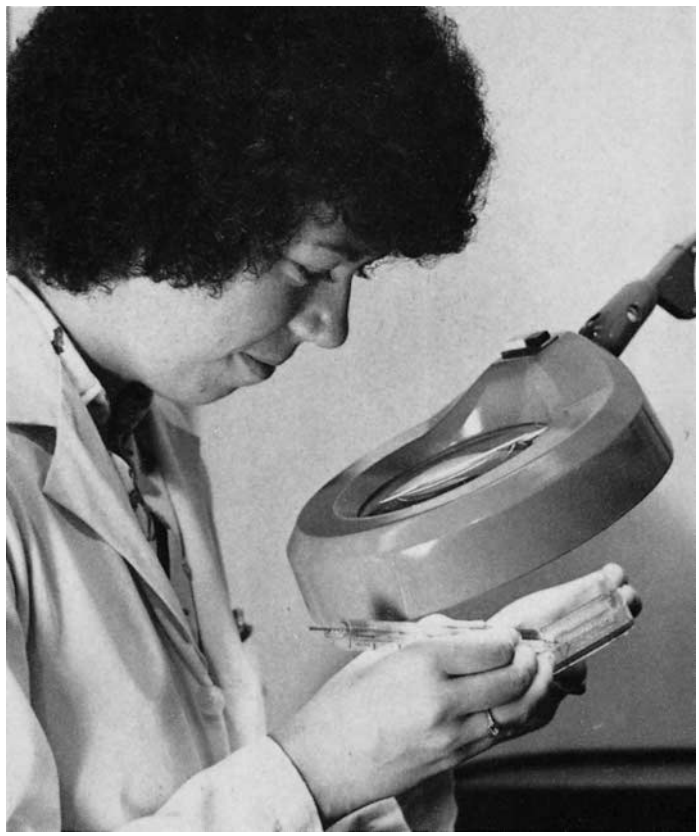


FIG. P-10 Monitoring water quality is a year-round job. (Source: Environment Canada.)

recognized by all. Canada has been in close consultation with the United States on such matters as the cleanup of the Niagara River. Both the federal and Ontario governments have developed or announced plans for the control and management of toxic chemicals. Inventories of toxic substances, crisis planning, cradle-to-grave tracking of toxics, and extensive investigative programs are a few of the measures being taken or advocated. Canada and Ontario together spend nearly $10 million per year monitoring and diagnosing the condition of the Great Lakes, with special attention paid to toxics. Great pains are taken to ensure that the United States has access to extensive Canadian environmental data, and the two nations cooperate through the Great Lakes International Surveillance Plan. Although drinking water supplies meet present guidelines, these guidelines cover only a small percentage of the many chemicals that occur in the Great Lakes. Clearly, the toxics situation cannot be allowed to deteriorate further.

The toxics issue must be solved in two basic ways. First, as much as possible of the hazardous wastes that already exist must be destroyed or recycled, and when necessary, safe methods must be used to store wastes that cannot be destroyed. Second, industrial wastes must be eliminated or reduced, *at their source*, to the fullest extent possible. Plants must be designed to produce no waste, or as little as possible; greater efficiency means less waste. This is the long-term answer to the problem—human ingenuity. Such things as closed-loop systems and waste exchange (whereby the wastes of one process become the raw materials of another) can minimize or eliminate entirely the need for the disposal of toxic wastes. Systems like this already exist, and they are economically feasible.

Society already has the technology necessary to reduce and eliminate toxics entering the Great Lakes. Whether we use it is a matter of will. If we want the benefits that chemicals can give us, then we must act responsibly in their use and disposal. If we don't clean up our act, we'll poison ourselves. It's as simple as that.

Demographic predictions. The present population in the Great Lakes basin is around 37 million; this is expected to double in the next 40 years. Sixty percent of Ontario's population now lives in the six major urban centers of Toronto, Hamilton, Ottawa, Kitchener-Waterloo, London, and Windsor, all within the Great Lakes watershed. It is forecast that by 2020 this will rise to 80 percent.

By the year 2020, the United States will require in the Great Lakes basin:

- For power generation, 15 times more land than at present.
- For power generation, 13 times more cooling water.
- Eight times more industrial water.
- Five times more irrigation water.
- Twice as much sewage capacity.
- Twice the present amount of land devoted to urban use.

Wastewater flows

Case study: British Columbia townships. Sewage treatment plants (STPs), which are operated by the Greater Vancouver Sewerage and Drainage District, discharge enough wastewater each year to fill B.C. Place Stadium 160 times. Flows have increased by 60 percent since 1976 for the Annacis plant and are expected to double by 2036. Flows from the Iona plant, which are now discharged to a deep sea outfall in Georgia Strait, are expected to remain at their current level, while steady growth is expected for the Lulu Island plant. Primary sewage treatment removes suspended particles from the waste stream and the remaining waste water is chlorinated in the summer months. STPs also produce sludges that can be contaminated with heavy metals.

Between Kanaka Greek and Hope, six municipal STPs discharge approximately 35,000 m^3/day of secondary-treated* effluent. Steady growth is expected for these areas. Throughout the Lower Fraser River Basin there are approximately 20 small private STPs treating effluent from schools, marinas, trailer parks, or other developments. Almost one-half the 1360 m^3/day of effluent discharges from these sources are to ground disposal systems. (See Figs. P-11 and P-12.)

Effluents from the Annacis and Lulu Island STPs frequently contain higher levels of contaminants than permitted by the provincial government. For the Annacis plant permit, noncompliance is most apparent for Biochemical Oxygen Demand (BOD),† toxicity, oil and grease, and dissolved oxygen. For example, in 1985 toxicity levels were exceeded 50 percent of the time for Annacis and 66.7 percent of the time for Lulu Island STPs. The toxic compounds identified in municipal STP effluent include un-ionized ammonia, cyanide, sulfides, chlorine, chloramines, phenols, anionic surfactants, heavy metals, and organic compounds. Table P-5 provides a

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* After primary treatment, secondary treatment involves using either anaerobic bacteria (which do not use oxygen) or aerobic bacteria (which use oxygen) to treat the sewage.

† BOD—the oxygen required for the biochemical breakdown of organic material and the oxidation of inorganic materials such as sulfides and iron.

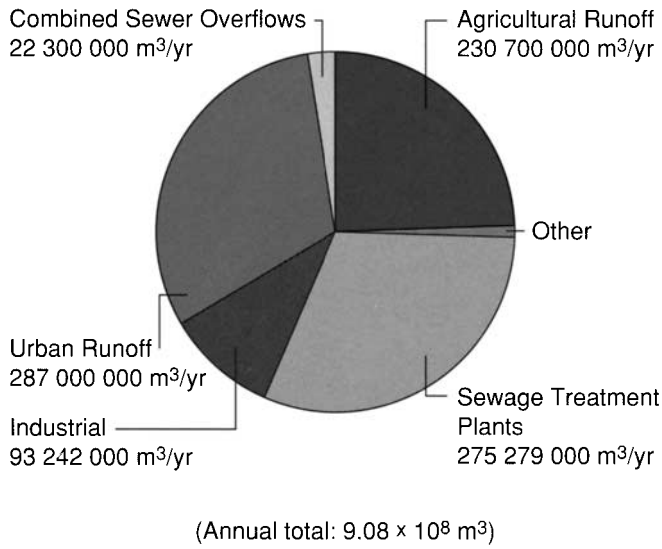


FIG. P-11 Summary of wastewater flows to the Fraser River estuary and Boundary Bay, 1987. (Source: Environment Canada.)

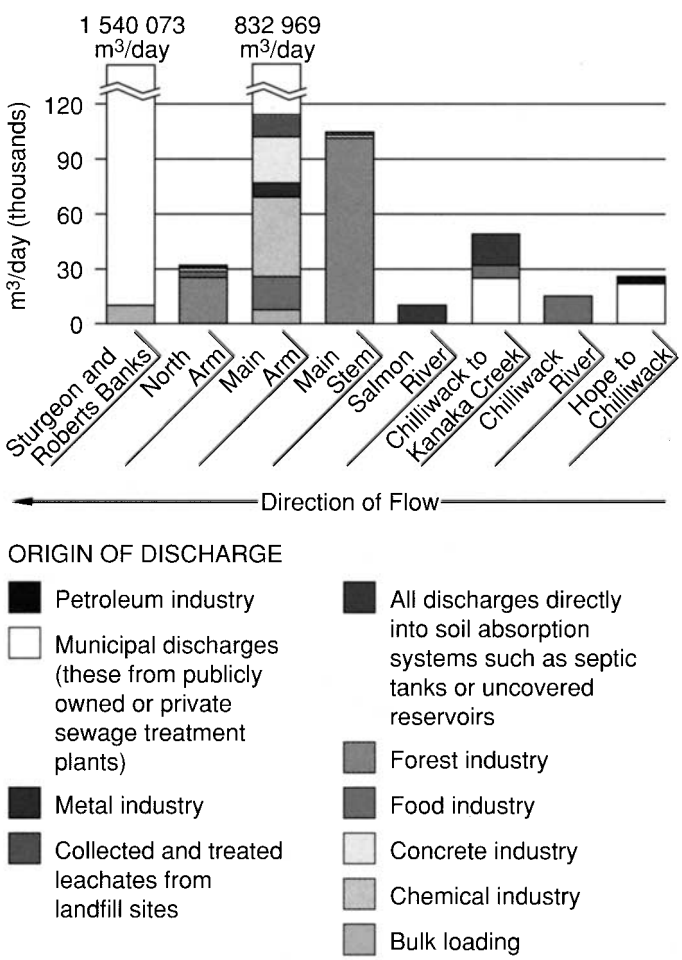


FIG. P-12 Distribution of discharges in the Lower Fraser River Basin authorized by B.C. Ministry of Environment Permits in 1987. (Source: Environment Canada.)

TABLE P-5 Sewage Treatment Plant Contaminant Loadings, 1985

Parameter (kg/day)	Iona	Annacis	Lulu
Discharge (m^3/day)	466,789	291,791	41,230
Oxygen demand	37,810	45,519	5,731
Suspended solids	26,607	20,717	2,639
Kjeldahl nitrogen	7,469	7,587	1,237
Ammonia	4,108	4,669	817
Fluoride	75	44	5.8
MBAS*	420	554	87
Sulfate	12,137	7,878	1,484
Calcium carbonate	33,609	30,930	4,535
pH	3,361	2,013	284
Phosphorus (diss)	840	905	136
Phosphorus (total)	1,354	1,314	219
Oil and grease	7,469	8,462	1,278
Phenol	14	15	1.6
Boron (diss)	75	85	13
Aluminum (diss)	bdl	bdl	33
Aluminum (total)	ai	233	115
Cadmium (total)	ai	ai	0.07
Chromium (total)	bdl	bdl	6.2
Copper (diss)	19	15	2.1
Copper (total)	47	41	6.6
Iron (diss)	135	236	41
Iron (total)	415	525	111
Lead (diss)	ai	3.8	0.7
Lead (total)	20	12	2.4
Manganese (diss)	23	20	2.5
Manganese (total)	28	29	3.7
Nickel (diss)	bdl	ai	4.9
Nickel (total)	bdl	ai	6.2
Zinc (diss)	37	29	6.2
Zinc (total)	56	50	14

NOTES:
Derived by multiplying finite effluent concentrations by the STP average reported flows for 1987.
ai: average indeterminate
bdl: below detection level
*MBAS: methylene blue active substances; ingredient in detergents and foaming agents.
SOURCE: Environment Canada.

summary of annual contaminant loadings and characteristics for the Annacis, Lulu, and Iona STPs.

Due to tidal conditions in the Fraser River, this effluent can pool and spread across the river within two hours at slack tide, exposing millions of juvenile salmon and eulachon larvae during downstream migrations. During low river flows, the effluent from Annacis STP, for example, can reside in the river for up to 1.7 days. Major concerns exist regarding the lethal and sublethal effects of the toxicity of the effluent on both anadromous and nonanadromous fish in terms of bioaccumulation, stress, disease, reproduction, feeding behavior, etc. Despite these concerns there are no techniques currently in place to link these effects to overall impacts on fish populations.

Industrial effluent. Authorized discharges from chemical, concrete, food, forest, gravel washing, metal fabricating and finishing, port industries, and other industrial sectors in the Lower Fraser River Basin total almost 300,000 m^3/day, 90

percent of which occur in the estuary. This is a drop from discharges of 351,571 m^3/day in 1973 probably due to industrial hookups to Annacis Island STP in 1975. Of 116 authorized waste management permits, 11 contribute about 80 percent of total industrial effluent flows.

Pollutant loadings include oil, grease, solids, metals, and organics. Total loadings are difficult to determine as permit requirements may not include all parameters, reporting periods and sampling methodology vary between permits, a few permit holders are in noncompliance situations, and unauthorized discharges may be occurring. Analysis of data for the 63 industrial permit holders on the Fraser River below Kanaka Creek show reported loadings of 4739 kg/day of BOD, 7226 kg/day of solids, 342 kg/day of oil and grease, and 5780 kg/day of nutrients. High priority industrial dischargers have been identified based on their flows and contaminant loadings.

Reference and Additional Reading

1. Soares, C. M., *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.

Portland Cement (*see* Cement)

Power Production; Power Production In-House; IPP; SPP

The acronym IPP generally refers to firms that existed for the sole purpose of investing in and building power plants and selling the power to a national governing body or other large customers. IPP ranks are swelling to include "small" power producers, whose production of power is secondary to their main purpose.

Small producers include large industrial entities, such as refineries and manufacturing plants, that buy their own power production machinery (sometimes to avoid expensive brownouts or outages) and make their own power. In most countries they can sell their excess power back to the national grid. The limits of this sale are generally set by the size of the distribution lines available. This small power producer generally gets less of a tariff for its power than it pays for national grid-supplied power. As such power producers increase, they lessen demand growth and therefore the required size of new, large power plants.

National power authorities traditionally move with a sluggishness that struggles to keep up with increasing power demand and changes in environmental legislation. However, the nature of the contemporary power business forces certain other optimization measures as the following paragraph indicates.

A power station in Dagenham, England, with both Alberta (Canada) power and English partners for owners, is an IPP. In anticipation of CO_2 (carbon dioxide) emissions legislation, the firm ordered a high-precision condition-monitoring system for its power-generation turbines. Their logic: the system would optimize fuel consumption and cut down on CO_2 emissions. Current technology made the cost of the system initially ordered unnecessarily high; nonetheless, the trend is clear.

Note also that, generally, IPPs have to have the mental flexibility to see the return on investment of such a system. The potential effect on the national power to IPP power production ratio in the future is evident. If national power producers do not react swiftly to changing environmental pressures, their profitability margins could decrease to the point where IPPs can further encroach on their territory.

One lesson learned from the severe ice storms suffered by Canada and the United States early in 1998 is that smaller IPP installations might prove less of an "Achilles heel" (weak link) to overall power demand than a few large national power plants.

Sagging national nuclear industries in Canada, the United States, and Japan are testament to overly optimistic life prognoses of nuclear fission reactors. They have been, and will continue to be, decommissioned. This can result in several smaller IPPs taking up the slack.

If the process plant does not want to "go it alone" to be an IPP, other willing partners may be available. IPP ranks are further being swelled by IPP joint-venture companies that can have one of the turbine manufacturers, such as Alstom [formerly ABB (Asea Brown Boveri)] or Siemens, as a major or controlling-interest partner. Interesting variations on a theme can be arranged contractually with original equipment manufacturers (OEMs). Alstom had a turnkey arrangement on the Kuala Langat, Malaysia, plant with the Genting Corporation, Malaysia, and the Lumut, Malaysia, plant with Segari Ventures, Malaysia. The pulp mill next to the Kuala Langat plant exchanges steam with its power-generation neighbor. Transmission and distribution systems in the vicinity also provide scope for minimizing hardware.

IPP Trends Globally

IPP conglomerates that include an OEM or large contractor, such as Enron, will continue to increase. Partners for many of these ventures include major oil and process firms. The advantage gained by joining forces with an OEM can include bargaining the terms of comprehensive maintenance contracts. If they team up with a major contractor, they may thus have negotiated a plant expansion for optimized dollars per unit of capacity. Alstom, for instance, is starting to increase its ownership of power facilities, even partially owned state or municipality ones, in the United States, such as the massive Midland plant. Alstom's participation in long-term comprehensive maintenance contracts in power projects, such as Deeside in the United Kingdom, serve to illustrate how entwined OEMs now are with the IPP sector. Quite apart from the return-on-investment figures that a "plain" investor might consider, the profit margin on spare parts and the markup on maintenance or construction services further add to the attraction of IPP projects for OEMs and contractors.

Deregulation of the power industry, increasing environmental legislation, and the increased difficulty of maintaining profit margins serve to accelerate the gradual turnover of national power authorities' territory to IPPs. The nuclear industries in Canada, the United States, and Japan are likely to provide further illustration of this fact in the near future.

More and more, "IPP" can mean "small IPP." As tax incentives for internal power production rise, some countries that were formerly opposed to SPPs are now lifting their objection. Singapore is one such example. The corresponding number of firms who then qualify to invest as IPPs increases correspondingly.

Oil and gas companies increasingly make their own power. They then become their own best customer. This trend is further stepped up as technology makes viable fuel selections of many of the "unwanted" by-products these facilities produce. The resultant economic benefits of producing their own power escalate further over time. An excellent example of this is seen under Stepper Motor Valves (a subsection of Control Systems) with the example of the PCS plant in Singapore.

Power Transmission

Power transmission is the act of taking power from a driving piece of equipment (such as a gas turbine, steam turbine, or motor) and transmitting it to a driven piece of equipment (such as a compressor or a pump). Power-transmission

equipment then includes gears and gearboxes, couplings, and other systems that transmit power from the "driver" to the "driven."

In this section, model numbers used by the information source companies will appear, as in other sections in this book. Care was taken to get source information from suppliers with the widest product ranges currently available, so the reader can then use this information as a basis for comparison with other OEMs being considered.

Gears*

Helical gears

Gears are associated with nearly every human activity in the modern world. They come in all sizes, shapes, and materials. They go by such names as spur, helical, bevel, hypoid, worm, skew, internal, external, epicyclic, and so on.

The following material is presented to assist an engineer who is not a gear specialist in determining the basic size and requirements of a gearset for one specific type of gearing: high-speed, high-power parallel-axis gears. The industry definition of high speed is 3600 rpm and/or 5000 ft/min pitch-line velocity. In this instance, *high power* means from 1000 to 2000 hp at the low end and upward of 50,000 hp at the high end. The kinds of applications that generally require high-speed gearing are those involving steam and gas turbines, centrifugal pumps and compressors, and marine propulsion equipment.

High-speed gears. Gears for high-speed service are usually of the helical type. They can be either single- or double-helical and can be used in either single or double stages of increase or reduction, depending on the required ratio. The ratio of a single stage is usually limited to about 8 to 1. There is a very small difference in frictional loss at the teeth, depending on whether the pinion or the gear is driven, but for all practical purposes no distinction need be made between speed increasers and speed reducers.

Most high-speed gearing operates at pitch-line velocities of 25,000 ft/min or less. At higher speeds, up to about 33,000 ft/min, special consideration must be given to many aspects of the gearset and housing. Speeds of over 33,000 ft/min should be considered developmental.

As gears go faster, the need for gear accuracy becomes greater. The following can be used for guidelines for high-speed gearing. Tooth-spacing errors should not exceed about 0.00015 in; tooth-profile errors, about 0.0003 in; and helix or lead error, as reflected by tooth contact over the entire face, about 0.0005 in. The usual range of helix angles on single-helical gears is between 12 and 18°. For double-helical gears, the helix is generally between 30 and 40°. Pressure angles are usually found between 20 and 25° (in the plane of rotation).

In addition to the requirement for extreme accuracy, a characteristic of high-speed helical gears that sets them apart from other helicals is the design objective of infinite life, which in turn results in fairly conservative stress levels.

Overload and distress. If a gearset is overloaded from transmitting more than the design power, or by being undersized, or as a result of misalignment, the teeth are likely to experience distress. The three most probable forms of distress are *pitting*, *tooth breakage*, and *scoring*.

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* Source: Demag Delaval, USA.

Pitting is a surface-fatigue phenomenon. It occurs when the hertzian, or surface, compressive stresses exceed the surface-endurance strength.

Tooth breakage is exactly what the name implies: sections of gear teeth literally break out. It occurs when the bending stresses on the flank or in the root of the teeth exceed the bending-fatigue strength of the material.

Scoring, sometimes called *scuffing*, is actually instantaneous welding of particles of the pinion and gear teeth to each other. It occurs when the oil film separating the teeth becomes so hot that it flashes or so thin that it ruptures, thereby permitting metal-to-metal sliding contact. The heat generated as the pinion and gear teeth slide on each other is sufficient to cause localized welding. These tiny welds are immediately torn loose and proceed to scratch the mating surfaces—hence the name *scoring*.

Neither pitting nor scoring causes immediate shutdown. If allowed to progress, however, they can produce a deterioration of the involute profiles in addition to producing stress risers. If permitted to continue too long, pitting or scoring can lead to tooth breakage.

Basic sizing. The basic sizing of a gearset, or what can be called the *preliminary design*, is based on resistance to pitting. Since the surface endurance strength is a function of the material hardness, preliminary sizing of a gearset is relatively simple. It should be understood, however, that the final design requires the efforts of a competent gear engineer to investigate and attend to such matters as:

1. The selection of materials and processing
2. The determination of the number of teeth on the pinion and gear, which is a function of the pitch, which in turn determines the tooth bending strength
3. An investigation of the scoring resistance of the gearset, which is a function of the gear-tooth geometry, the surface finish of the teeth, and the properties of the lubricant
4. Rotor proportions and bearing design, with particular interest in related vibration characteristics
5. Gear-case features, including such things as running clearances, proper drainage, venting, mounting, doweling, and, in particular, maintenance of internal alignment
6. The many system considerations such as lateral and torsional vibration, external alignment with associated forces and moments on shaft ends, torque pulsations, etc.

The American Gear Manufacturers Association's (AGMA) fundamental equation for surface durability (pitting resistance) of helical gear teeth is

$$s_c\left[\frac{C_L C_H}{C_T C_R}\right] \leqq \left[\frac{1/\pi}{\dfrac{1-\mu_p^2}{E_p}+\dfrac{1-\mu_G^2}{E_G}}\right]^{1/2} \times \left[\frac{W_t C_o}{C_v}\frac{C_s}{dF}\frac{C_m C_f}{I}\right]^{1/2} \tag{P-1}$$

where s_c = contact-strength number
C_L = life factor
C_H = hardness-ratio factor
C_T = temperature factor
C_R = factor of safety

μ_p, μ_G = Poisson ratio for pinion and gear
E_p, E_G = modulus of elasticity for pinion and gear
W_t = transmitted tangential load at pitch diameter
C_o = overload factor
C_v = dynamic factor
C_s = size factor
d = pinion pitch diameter
F = face width
C_m = load-distribution factor
C_f = surface-condition factor
I = geometry factor

Substituting appropriate values for high-speed gears and rearranging Eq. (P-1) results in

$$K \leqq [s_c \times 10^{-4}]^2 \times \frac{1.56}{C_o C_m} \tag{P-2}$$

where K is an index of hertzian stress. It is defined mathematically as:

$$K = \frac{W_t}{Fd} \times \frac{R+1}{R} \tag{P-3}$$

where R = ratio (D/d)
D = gear pitch diameter
W_t = tangential tooth load
$$= \frac{126{,}000 \times \text{hp}}{N_p \times d}$$
and
hp = transmitted, or design, horsepower
N_p = pinion, rpm

The term C_o, the overload factor, is accounted for by application service factors (SF), shown in Table P-6.

The term C_m, the load-distribution factor, accounts for maldistribution of load across the face width of the gearset due to lateral bending and torsional twisting of the pinion, thermal distortion of the pinion and/or the gear, and centrifugal deflection of the gear. If the length-diameter ratio (L/d) of the pinion is kept within reasonable limits, usually less than 2.2 for double-helical and 1.5 for single-helical gears, and proper attention is paid to cooling and gear-band deflection, the magnitude of the C_m factor will probably lie between 1.2 and 1.4. If the higher value is used in the interest of conservatism, Eq. (P-2) can be further simplified to

$$K \leqq \frac{(s_c \times 10^{-4})^2 \times 1.11}{\text{SF}} \tag{P-4}$$

By using Eqs. (P-3) and (P-4) and Tables P-6 and P-7, the basic size of a high-speed gearset can be determined in several minutes on a hand calculator.

Table P-7 gives values of s_c for commonly used gear materials. The rating of a gearset should be based on the softer of the two members, which is normally the gear. The Brinell hardness of the pinion in through-hardened gearsets is usually about 1.2 times that of the gear.

Arrangements. Figures P-13 and P-14 show sections through a typical industrial high-speed-gear unit. See Table P-6.

TABLE P-6 Service-Factor Values

Application	Service Factor		
	Prime Mover		
	Motor	Turbine	Internal-Combustion Engine (Multicylinder)
Blowers			
Centrifugal	1.4	1.6	1.7
Lobe	1.7	1.7	2.0
Compressors			
Centrifugal: process gas except air conditioning	1.3	1.5	1.6
Centrifugal: air-conditioning service	1.2	1.4	1.5
Centrifugal: air or pipeline service	1.4	1.6	1.7
Rotary: axial flow—all types	1.4	1.6	1.7
Rotary: liquid piston (Nash)	1.7	1.7	2.0
Rotary: lobe-radial flow	1.7	1.7	2.0
Reciprocating: three or more cylinders	1.7	1.7	2.0
Reciprocating: two cylinders	2.0	2.0	2.3
Dynamometer: test stand	1.1	1.1	1.3
Fans			
Centrifugal	1.4	1.6	1.7
Forced-draft	1.4	1.6	1.7
Induced-draft	1.7	2.0	2.2
Industrial and mine (large with frequent-start cycles)	1.7	2.0	2.2
Generators and exciters			
Base-load or continuous	1.1	1.1	1.3
Peak-duty cycle	1.3	1.3	1.7
Pumps			
Centrifugal (all service except as listed below)	1.3	1.5	1.7
Centrifugal: boiler feed	1.7	2.0	. . .
Centrifugal: descaling (with surge tank)	2.0	2.0	. . .
Centrifugal: hot oil	1.5	1.7	. . .
Centrifugal: pipeline	1.5	1.7	2.0
Centrifugal: waterworks	1.5	1.7	2.0
Dredge	2.0	2.4	2.5
Rotary: axial flow—all types	1.5	1.5	1.8
Rotary: gear	1.5	1.5	1.8
Rotary: liquid piston	1.7	1.7	2.0
Rotary: lobe	1.7	1.7	2.0
Rotary: sliding vane	1.5	1.5	1.8
Reciprocating: three cylinders or more	1.7	1.7	2.0
Reciprocating: two cylinders	2.0	2.0	2.3
Marine service			
Ship's service turbine-generator sets	. . .	1.1	. . .
Turbine propulsion	. . .	1.25	. . .
Diesel propulsion	. . .	. . .	1.35

Figures P-15 and P-16 show sections through a typical turbine-driven marine propulsion reduction gear. It will be noted that the high-speed pinions each mesh with two first-reduction gears, thereby splitting the power from each turbine. These twin-power-path gears, or so-called locked-train gears, are popular in the horsepower range of 30,000 shp and up.

Figures P-17 and P-18 show sections through a typical diesel-driven marine propulsion reduction gear. In this arrangement, each pinion is fitted with a pneumatically operated clutch that permits either engine to be operated singly or one engine ahead and one astern for fast maneuvering.

TABLE P-7 s_c Values

	Gear Hardness	s_c
Through-hardened	229 BHN	112,000
	248 BHN	117,500
	302 BHN	135,000
	340 BHN	152,000
Nitrided	55 R_c	207,000
	58 R_c	218,700
	60 R_c	226,800
	63 R_c	239,400
Case-carburized	55 R_c	230,000
	58 R_c	243,000
	60 R_c	252,000
	63 R_c	266,000

NOTE: BHN = Brinell hardness number; R_c = Rockwell number.

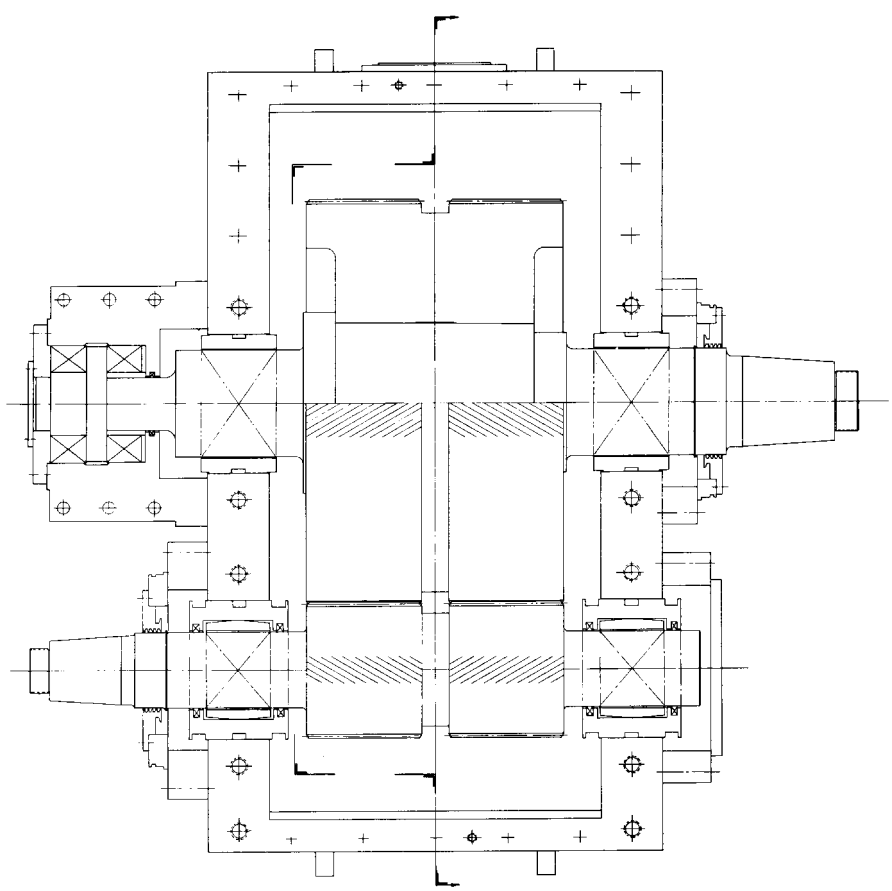


FIG. P-13 Plan cross section, typical industrial gear. (Source: Demag Delaval.)

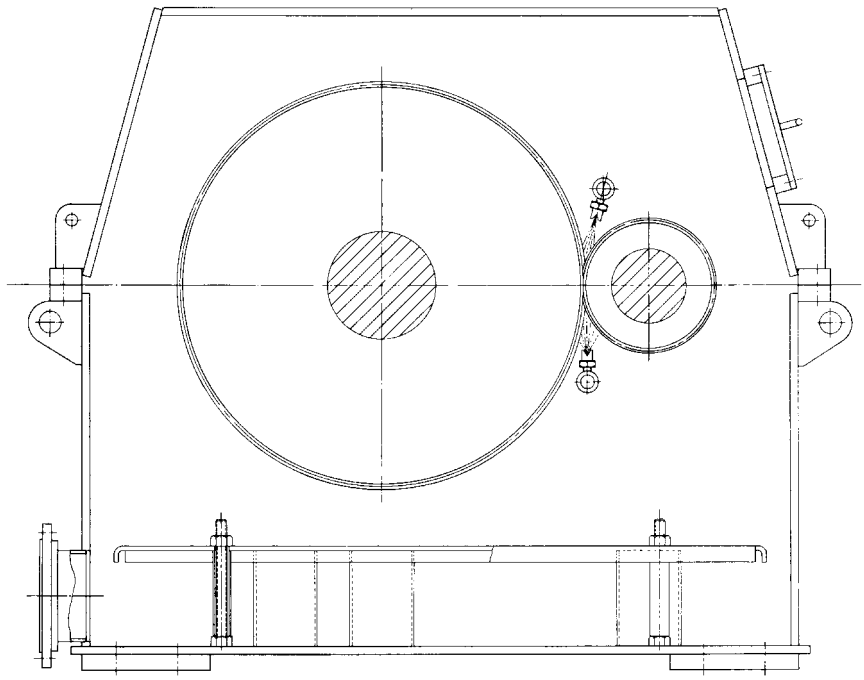


FIG. P-14 End cross section, typical industrial gear. (Source: Demag Delaval.)

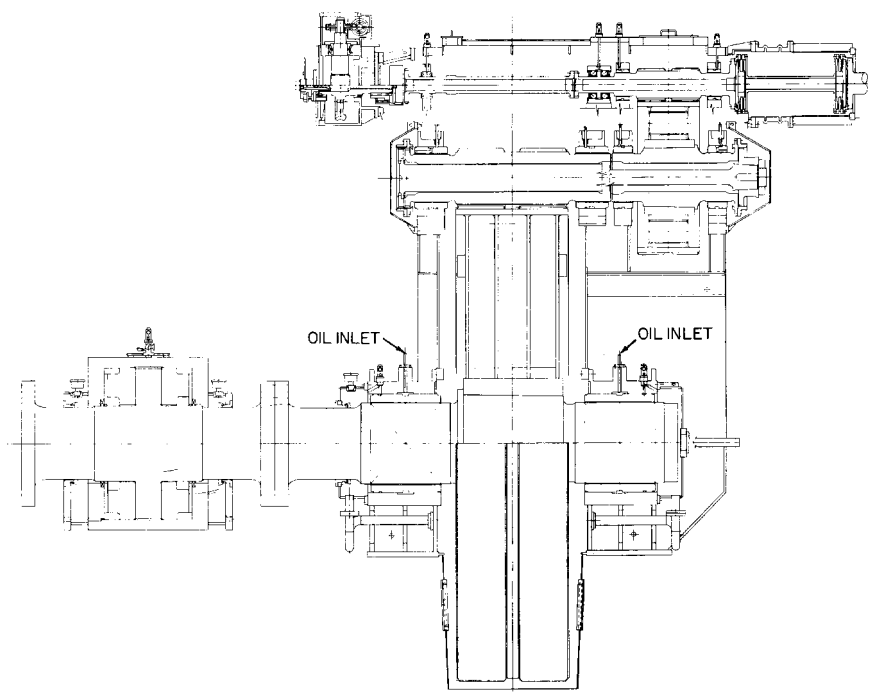


FIG. P-15 Plan cross section, typical locked-train reduction gear. (Source: Demag Delaval.)

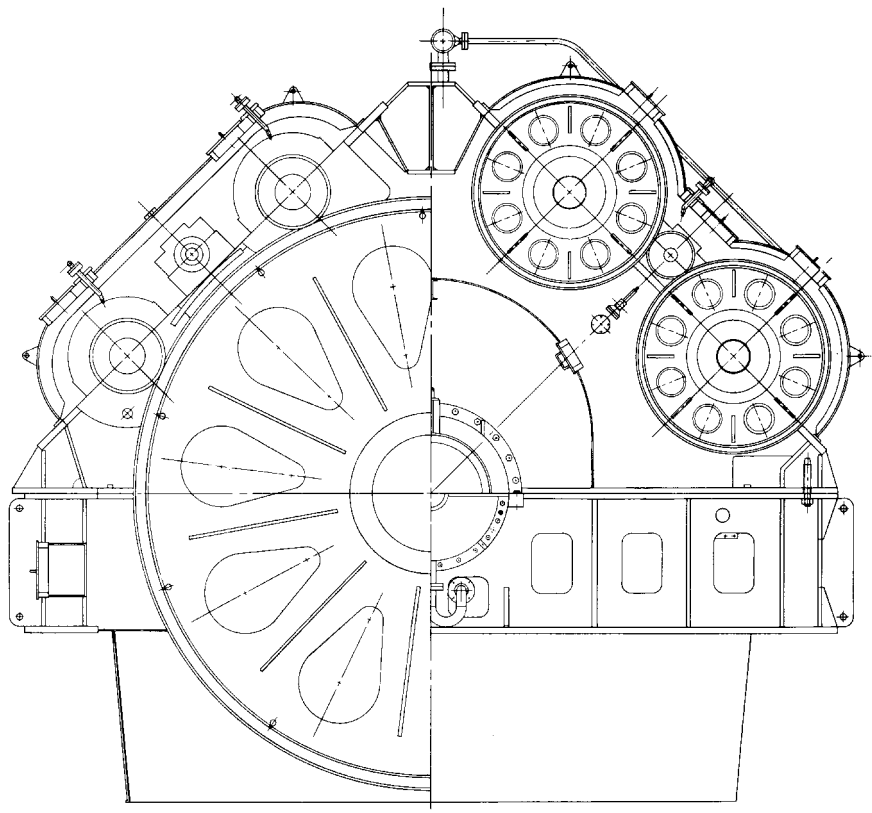


FIG. P-16 End cross section, typical locked-train reduction gear. (Source: Demag Delaval.)

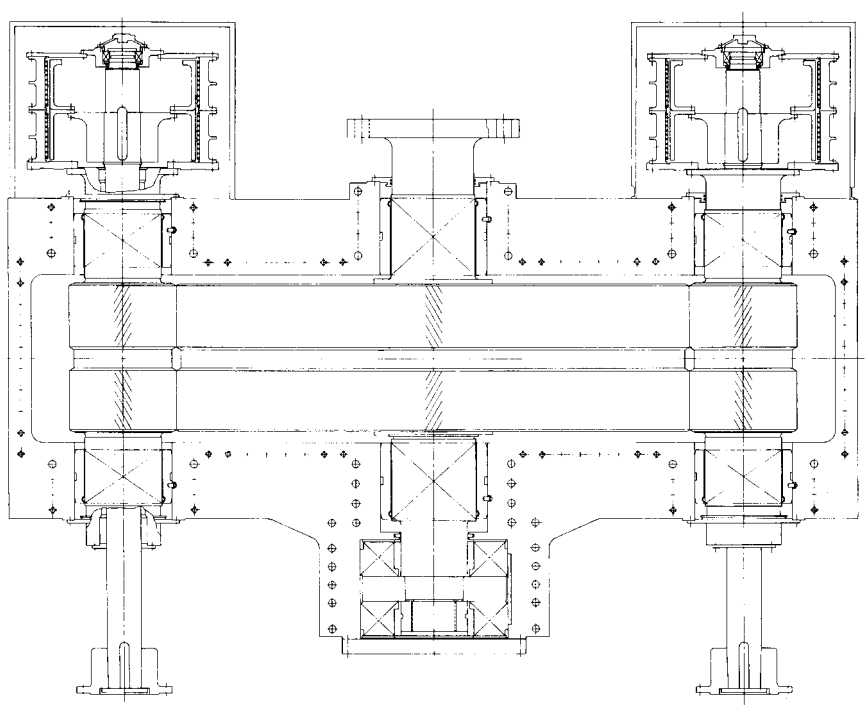


FIG. P-17 Plan cross section, typical diesel propulsion reduction gear. (Source: Demag Delaval.)

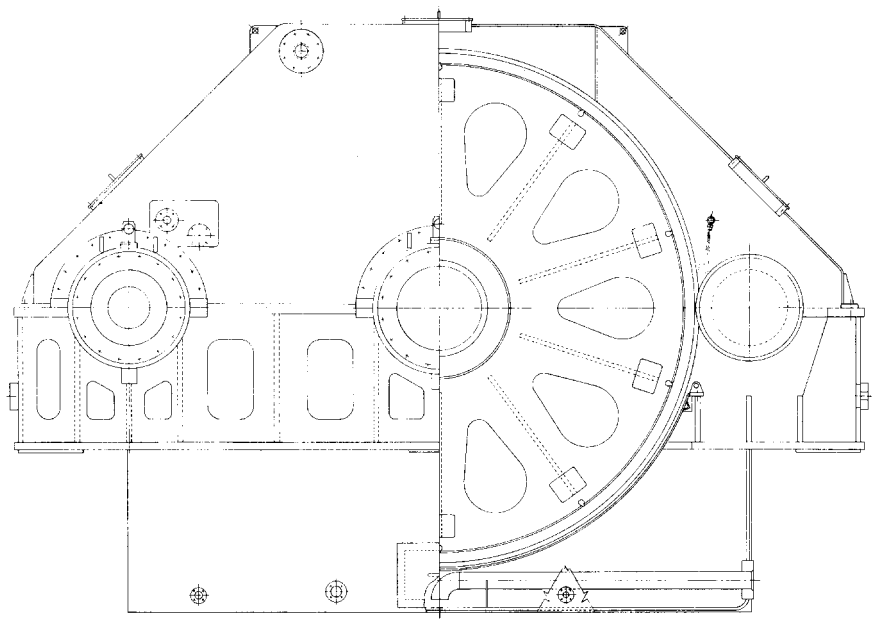


FIG. P-18 End cross section, typical diesel propulsion reduction gear. (Source: Demag Delaval.)

Horsepower losses. Prediction of gear-unit losses is an inexact science at best. The total power loss of a gear unit is made up of (1) the frictional loss in the oil film separating the teeth as they slide over one another, (2) bearing losses, and (3) windage and pumping losses.

Empirical equations have been developed for most types of gears to calculate these losses. Often rule-of-thumb estimates are as good as the calculations. Tooth-mesh losses usually amount to between 0.5 and 1 percent of the transmitted horsepower at each mesh. Bearing losses may vary a bit more, depending primarily on the bearing type, operating clearance, and sliding velocity. They usually fall into a range of 0.75 to 1.5 percent of transmitted power.

Windage losses depend primarily on the clearance between rotating parts and the housing, the smoothness of the surfaces, and the peripheral velocities.

Pumping loss, the displacement of the air-oil mixture from the tooth space as engagement takes place, is influenced by tooth size, helix angle, rotative speed, and location of the oil sprays. Losses of this type are the biggest variable and can fall anywhere from 0.5 to about 2 percent of transmitted power.

The most important consideration is that a realistic view be taken of gear losses when selecting a pump, cooler, and filters for the lubrication system. These should be large enough to do the job.

Lubrication. The oils normally used in high-speed-gear applications are rust- and oxidation-inhibited turbine oils in the viscosity range of 150 to 300 SSU at 100°F. As a general rule, the higher the pitch-line speed of the gear, the lower the viscosity oil required. In marine units, in which the propeller shaft turns at a relatively low speed, pitch-line speeds are frequently found below 5000 ft/min. In these cases, it is generally desirable to use a more viscous oil. The viscosity of the oils frequently found in turbine-driven propulsion plants is in the range of 400 to 700 SSU at 100°F. In diesel propulsion gearing, in which the engine and the gear are on separate systems, the viscosity of the gear oil is frequently in the range of 600 to 1500 SSU at 100°F.

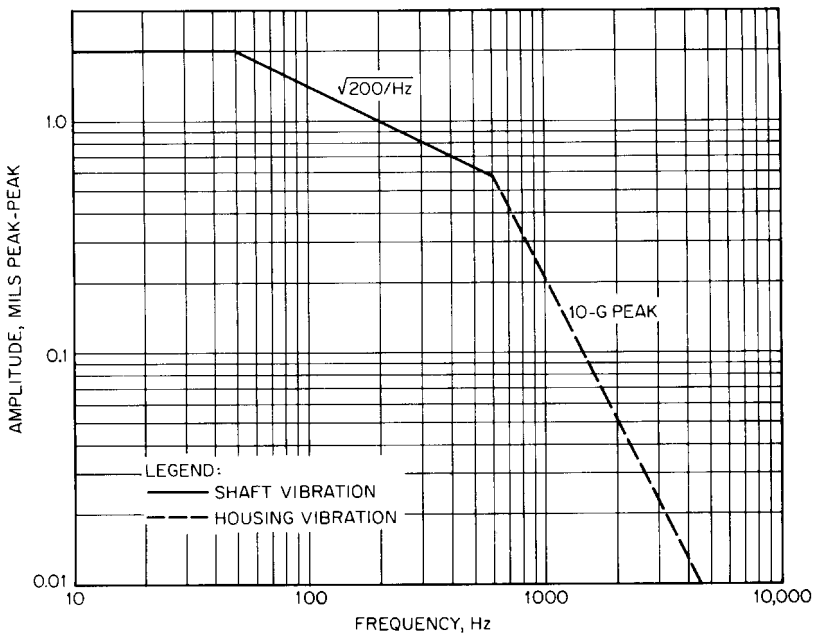


FIG. P-19 Acceptable vibration levels. (Source: Demag Delaval.)

Regardless of the application, the scoring or scuffing resistance of the gear teeth should be investigated. In many cases, it will be desirable to use an oil with appropriate extreme-pressure additives that greatly increase the antiweld or antiscoring characteristics of the lubricant.

Installation and maintenance. If a gear unit is correctly sized, properly installed, and properly maintained, it can be expected to last indefinitely. Proper installation includes (1) proper initial alignment, both internal and external, and (2) a rigid foundation that will not settle, crack, or elastically or thermally deform under operating conditions in amounts greater than the gear-alignment tolerance.

For those interested in additional information on systems considerations (overloads, system vibration, alignment, foundations, piping, and lubrication), AGMA Information Sheet 427.01, *Systems Considerations for Critical Service Gear Drives*, is recommended.

Proper maintenance consists primarily of providing a continuous supply of the correct lubricant at the right temperature, pressure, and condition. Obviously, alignment and balance must be maintained. Vibration monitoring is a good preventive-maintenance tool. Figure P-19 can be used as a guide for acceptable lateral-vibration limits. Additional information regarding vibration instruments, interpretation, tests, etc., may be found in AGMA Standard 426.01, *Specification for Measurement of Lateral Vibration on High Speed Gear Units*.

Worm gears

The use of high-speed drivers for efficient operation makes speed reduction necessary for many applications.

Worm-gear reducers are very compact, requiring less space than belts, chains, or trains of open gearing. The right-angle drive often permits compact placement of the driving and driven machines. Since three or more teeth are always in contact,

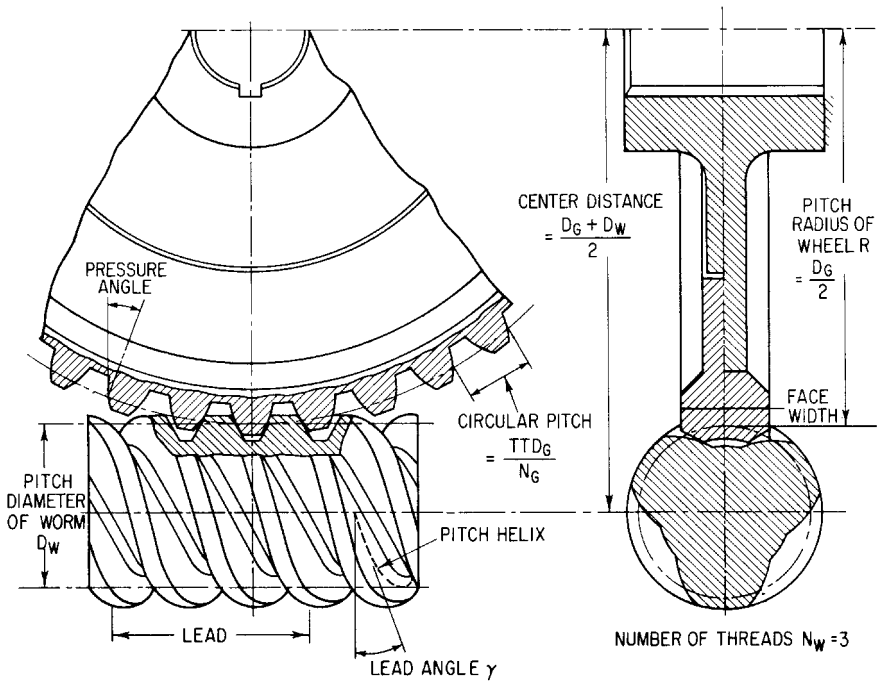


FIG. P-20 Worm gear terminology. (Source: Demag Delaval.)

there is an even flow of torque, which reduces vibration, prolongs the life of the driven machinery, and provides quiet power transmission. There are few moving parts (hence few bearings), and these are enclosed in a dustproof housing that contributes to long life and avoids danger of injury to workers.

Worm gearing consists of an element known as the *worm*, which is threaded like a screw, mating with a gear whose axis is at a 90° angle to that of the worm. The gear is throated and partially envelops the worm. The worm may have one or more independent threads, or "starts."

The ratio of speeds is determined by dividing the number of teeth in the gear by the number of threads in the worm. Since a single-threaded worm acts like a gear with one tooth and a double-threaded worm as a gear with two teeth, very large ratios can be designed into one set of gearing. Ratios between 3 to 1 and 100 to 1 are common for power transmission purposes, and even higher ratios are employed for index devices.

Mechanical elements. Dimensions of the worm and worm gear are defined as follows (see Fig. P-20):

Outer diameter of worm is the diameter of a cylinder touching the tops of the threads.

Pitch diameter of worm is the diameter of a circle that is tangent to the pitch circle of the mating gear in its midplane.

Outer diameter of gear is the diameter over the tips of the teeth at their highest points.

Throat diameter of gear is the diameter over the tips of the teeth at the middle plane that is perpendicular to the axis of the gear shaft and passes through the axis of the worm.

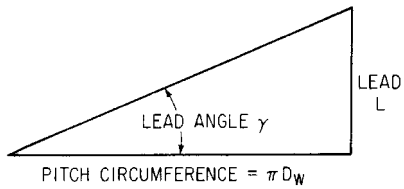


FIG. P-21 Lead angle. (Source: Demag Delaval.)

Pitch diameter of gear is the diameter of the pitch circle at the midplane of the gear that would roll upon the pitch line of the worm if the latter were used as a rack.

Circular pitch is the distance from a point on one gear tooth to the same point of the succeeding tooth measured circumferentially on the midplane pitch circle. It is equal to the axial pitch of the worm, that is, the distance from any point on a thread of the worm to the corresponding point on the next thread, measured parallel to the axis.

Lead of worm is the distance parallel to the axis of the worm from a point on a given thread to the corresponding point on the same thread after it has made one turn around the worm. If the worm has only one thread, this distance is equal to the circular pitch, but if the worm has multiple threads, it is equal to the circular pitch multiplied by the number of threads. It is the distance that a point on the pitch circle of the gear is advanced by one revolution of the worm.

One revolution of the worm advances the gear by as many teeth as there are threads on the worm. Therefore, the *ratio of transmission* is equal to the number of teeth on the gear, divided by the number of threads on the worm, without regard to the pitch.

Lead angle of the worm threads is the angle between a line tangent to the thread helix at the pitch line and a plane perpendicular to the axis of the worm. The pitch lines of the worm threads lie on the surface of a cylinder concentric with the worm and of the pitch diameter. If this cylinder is thought of as unrolled or developed on a plane, the pitch line of the thread will appear as the hypotenuse of a right-angled triangle, the base of which will be the circumference of the pitch circle of the worm and the altitude of which will be the lead of the worm. In Fig. P-21 the lead angle is γ, and the tangent of this angle is equal to the lead L divided by π times the pitch-line diameter D_w of the worm, $\tan \gamma = L/\pi D_w$.

Pressure angle is defined as the angle between a line tangent to the tooth surface at the pitch line and a radial line to that point.

Classification. A large number of arrangements are available, permitting flexibility in application to a wide variety of driven machinery. Some of the typical arrangements manufactured are shown in Figs. P-22 to P-28.

Motorized units may be furnished for:

Horizontal-shaft units
- Single worm reduction
- Helical worm reduction
- Double worm reduction

Vertical-output-shaft units
- Single worm reduction
- Helical worm reduction
- Double worm reduction

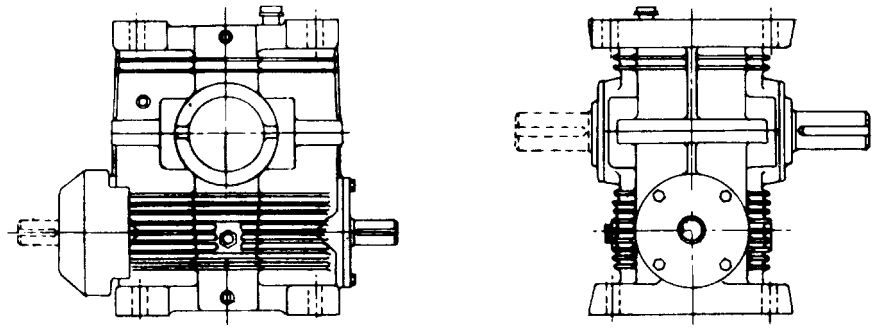


FIG. P-22 Single worm reduction. (Source: Demag Delaval.)

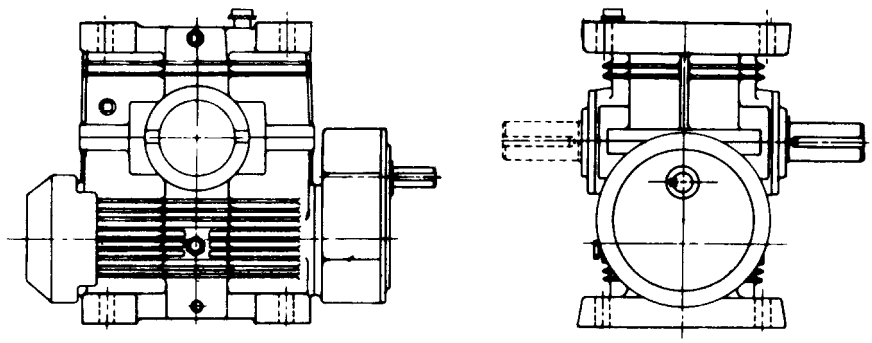


FIG. P-23 Helical worm reduction. (Source: Demag Delaval.)

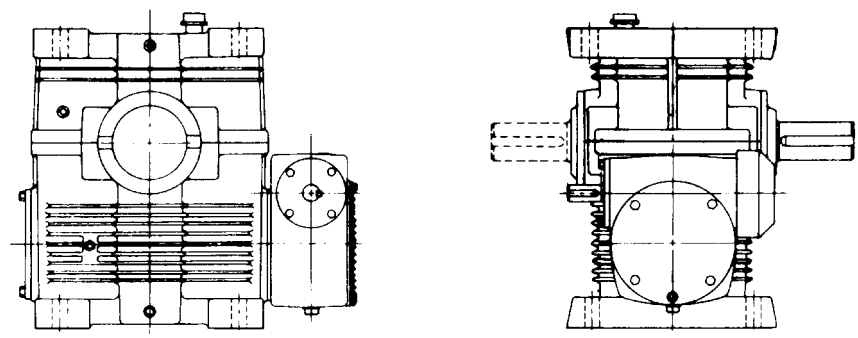


FIG. P-24 Double worm reduction. (Source: Demag Delaval.)

Shaft-mount units
- Single worm reduction
- Helical worm reduction
- Double worm reduction

Special reducers. Special reducers in various combinations are also available. An example is presented in Fig. P-29, which shows a large vertical-output-shaft unit with a single worm reduction having 38-in gear centers, which is used in pulverized-coal service.

Efficiency of worm gearset. To determine the approximate efficiency of a worm gearset in which the worm threads are of hardened and ground steel and the gear

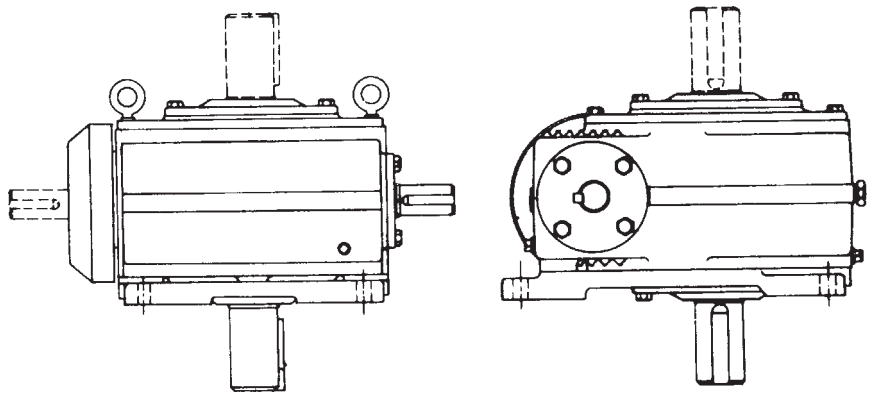


FIG. P-25 Vertical single worm reduction. (Source: Demag Delaval.)

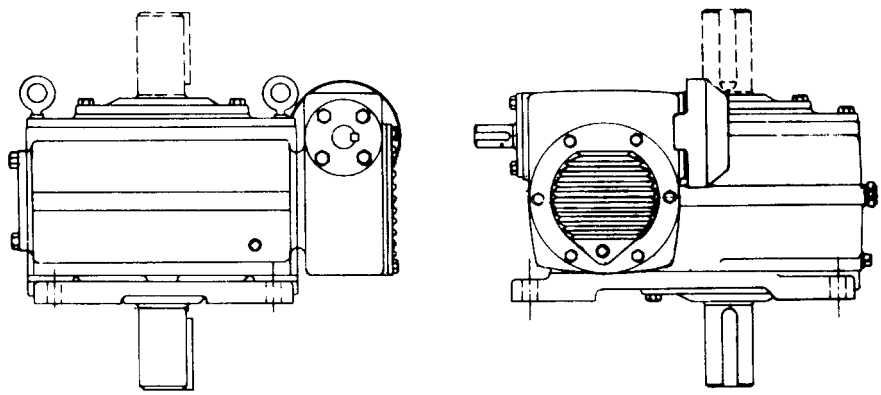


FIG. P-26 Vertical double worm reduction. (Source: Demag Delaval.)

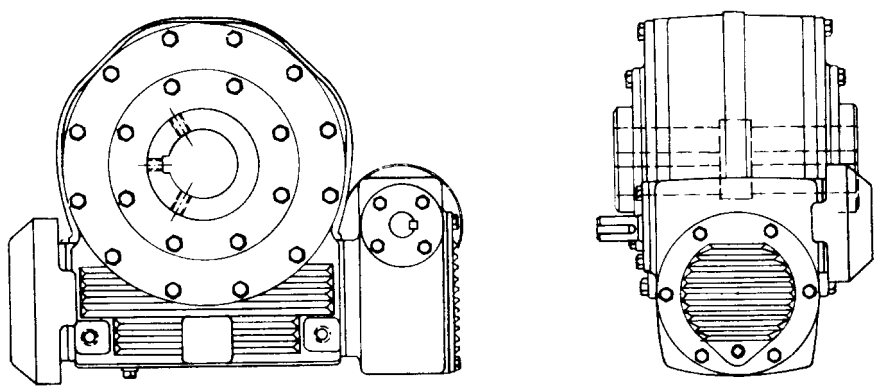


FIG. P-27 Double-worm-reduction shaft-mount unit. (Source: Demag Delaval.)

teeth of nickel bronze or phosphor bronze, lubricated with a steam-cylinder oil, Figs. P-30 and P-31 may be used. To use the coefficient-of-friction curve, calculate the rubbing speed of the worm from the following formula:

$$\text{Rubbing speed, ft/min} = \frac{\text{pitch diameter of worm} \times 0.262 \times \text{rpm}}{\cos \text{lead angle}}$$

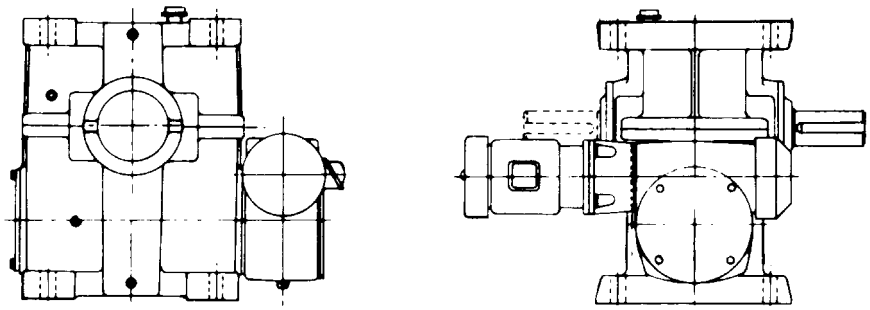


FIG. P-28 Motorized worm reduction. (Source: Demag Delaval.)

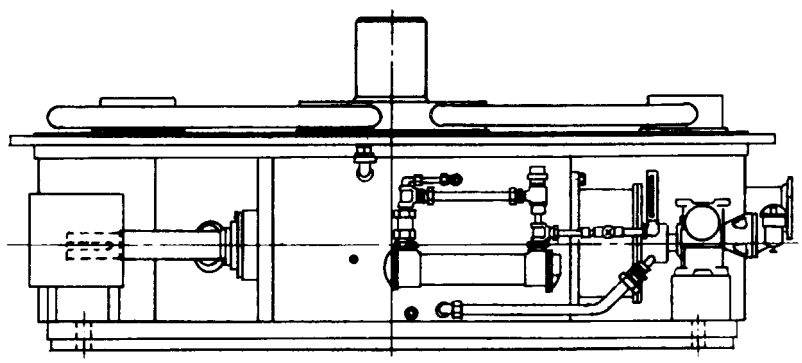


FIG. P-29 Large vertical-shaft single worm reduction. (Source: Demag Delaval.)

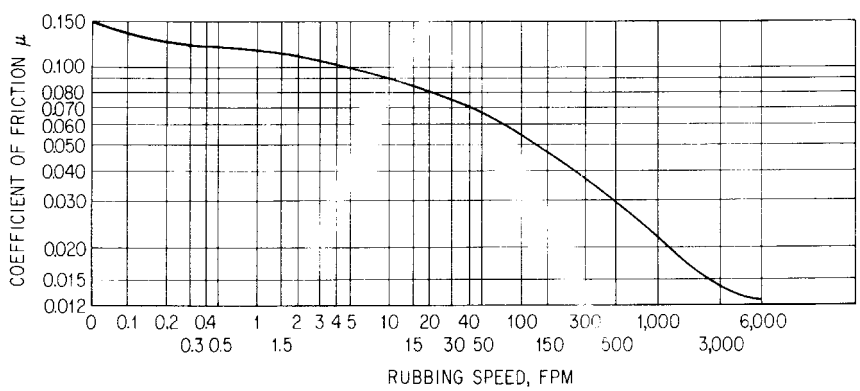


FIG. P-30 Coefficient-of-friction curve. (Source: Demag Delaval.)

(See Fig. P-21 for a definition of lead angle.) With this rubbing speed noted at the bottom of the diagram, read vertically upward until you intersect the coefficient-of-friction curve. Read the value of the coefficient of friction from the left-hand side of the diagram.

When the worm is the driver, enter the efficiency diagram with the lead angle of the worm at the bottom of the diagram. Read upward to the intersection of the curve with the correct coefficient of friction. The efficiency of the gearset may be read from the right-hand side of the diagram or the efficiency loss on the left-hand side of the diagram.

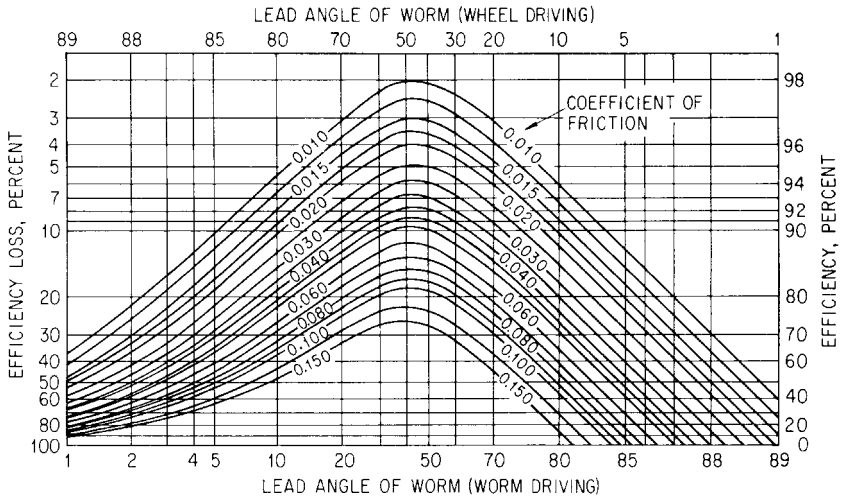


FIG. P-31 Efficiency diagram for worm gearing. (Source: Demag Delaval.)

When the gear is the driver, enter the efficiency diagram with the lead angle of the worm at the top of the diagram, reading down to the curve with the correct coefficient of friction. Find the efficiency as before.

These efficiencies, while approximate, are very close to the operating efficiency of the gearset alone. When the gearset is enclosed in a housing with bearings, seals, and oil reservoir, some allowance must be made for bearing loss, seal drag on the shaft, and churning of oil.

Self-locking gearset. A self-locking gearset is one that cannot be started in motion by applying power at the gear. Theoretically, this can be obtained when the lead angle of the worm is less than the friction angle. For normal static conditions the friction angle would be approximately 8°30′, and therefore it might be deduced that gearsets having a worm lead angle less than this value would be self-locking.

However, it is impossible to determine the point of positive self-locking for several reasons. The value of the static coefficient of friction varies considerably because of the effect of a number of variables. Furthermore, if a source of vibration is located near a self-locked gearset, a very slight motion might occur at the gear contact. Since the coefficient of friction decreases rapidly with an increase in rubbing velocity from the static condition, the friction angle may become smaller than the lead angle. Once this occurs, motion will continue and the gearing will accelerate under the action of the power applied to the gear.

Figure P-32 indicates the rapid increase in efficiency with increase in rubbing speed from the static condition for both the worm driving and the gear driving. For this particular example at a rubbing velocity of 500 ft/min, there are only a few points of efficiency difference between the two curves.

The best way to obtain locking is to use a brake, released electrically when the motor is started. With worm gears of high ratios, the braking effect need be only a fraction of full-load motor torque. A solenoid brake is usually best suited for this operation since the braking effect may be adjusted by weights that can be proportioned to stop the load gradually and avoid damage. Dashpots can be employed to ensure gradual setting of the brake.

Tooth form. The tooth form used by this information source is the involute helicoid. Figures P-33 and P-34 show the straight generating line tangent to the base circle and the convex axial section of thread.

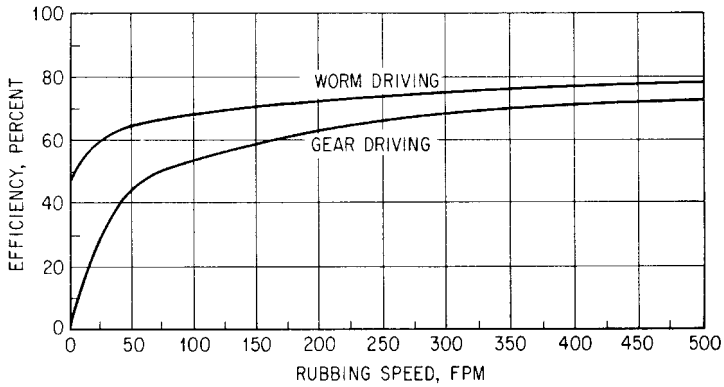


FIG. P-32 Comparison of efficiencies at tooth contact (ratio 50 on 20-in-center distance). (Source: Demag Delaval.)

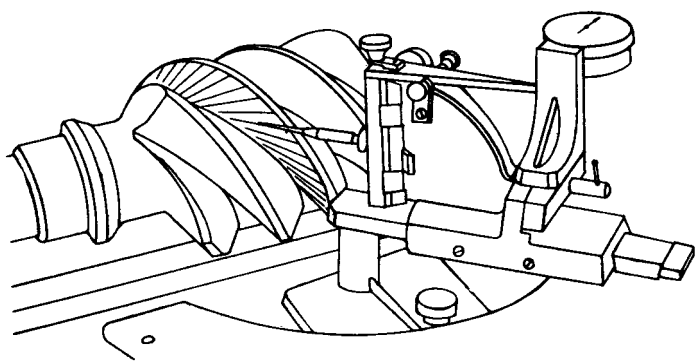


FIG. P-33 Generation of tooth form. (Source: Demag Delaval.)

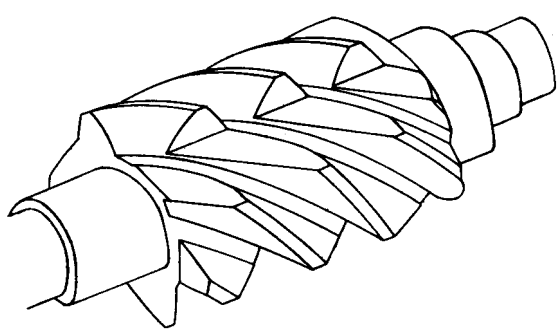


FIG. P-34 Convex axial section of thread. (Source: Demag Delaval.)

Worm-gear performance is judged in terms of load capacity, smooth, silent running, and high efficiency. The attainment of these goals requires accurate methods of producing and inspecting the worm and gear.

Since the involute helicoid worm is based on generation of a straight line tangent to the base circle, the accuracy of this line is very simple to check (Fig. P-35). This thread form lends itself to accurate manufacture, inspection, and interchangeability, as all worms can be checked to calculated measurable dimensions.

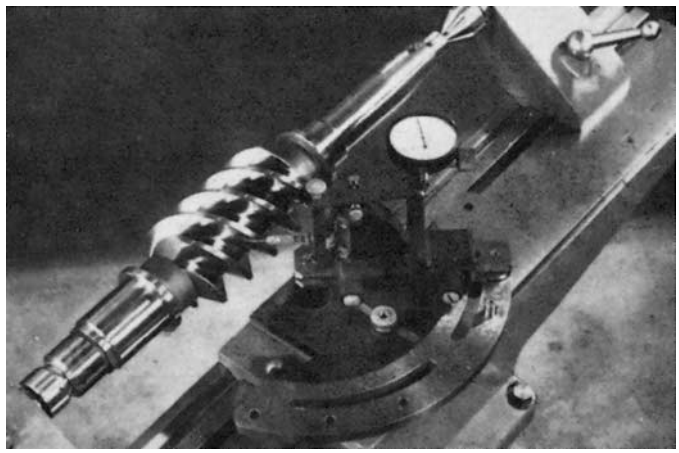


FIG. P-35 Inspection of tooth form. (Source: Demag Delaval.)

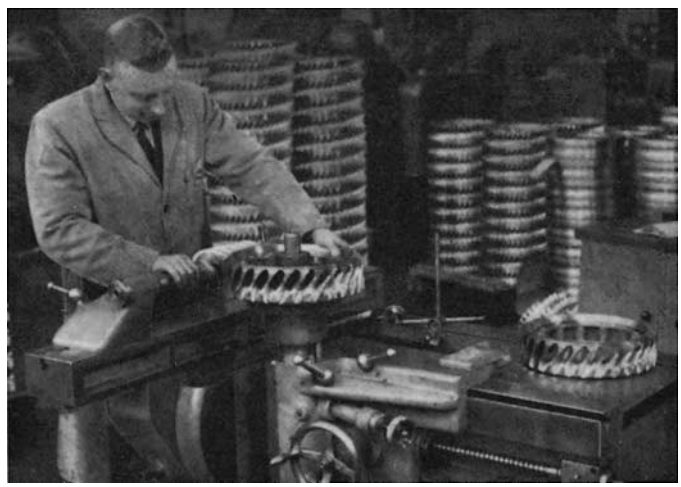


FIG. P-36 Checking with master worm. (Source: Demag Delaval.)

All wheels are checked with a master worm to ensure interchangeability and correctness of form (Fig. P-36).

Tooth contact. The involute helicoid thread form is a calculated form, and the theoretical contact is maintained more accurately and is more easily determined than that of any other worm thread, particularly a concave thread flank.

Figure P-37 shows theoretical "lines" of contact that exist between two worm threads and two gear teeth at a given angular position of the worm. As the worm rotates in the direction shown, these contact lines move progressively across the flanks of the worm and gear teeth and are inclined at an angle to the direction of sliding. This inclined effect is known to give a highly efficient form of surface lubrication and a low coefficient of friction as compared with a gear form in which the lines of contact are in the approximate direction of sliding. The contacting surfaces are always freshly lubricated and are not subject to the undesirable effects of double contact.

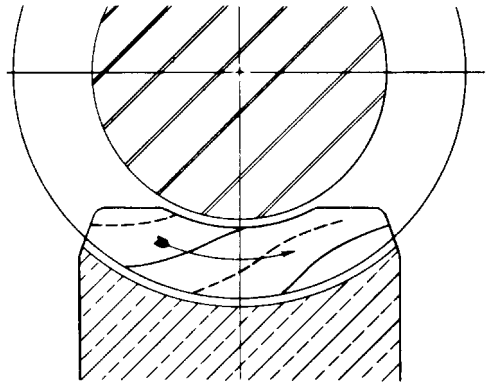


FIG. P-37 Tooth contact. (Source: Demag Delaval.)

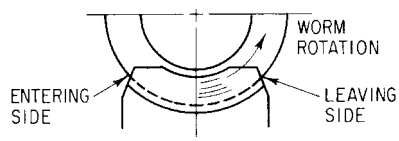


FIG. P-38 Tooth contact: good. (Source: Demag Delaval.)

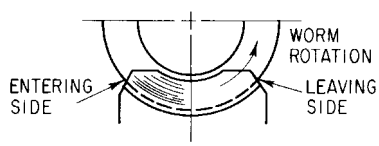


FIG. P-39 Tooth contact: poor. (Source: Demag Delaval.)

Depending on the relative radii of curvature between the two contacting surfaces and the load applied, these lines of contact actually have some width, thereby providing area contact. In spite of claims to full area contact, line contact occurs on all other thread forms including the double-enveloping thread form. Only the involute helicoid thread form provides the necessary control of the geometry of thread form in design and manufacture to obtain optimum contact conditions.

All gears, bearings, and housings deflect and distort to some extent when operating under load as compared with conditions under no load. A correction in the tooth-contact pattern is provided to ensure proper contact under loaded conditions. This correction is accomplished by producing gears with leaving-side contact as shown in Fig. P-38. This is the ideal contact pattern to aim for when assembling a worm gearset under a no-load condition. This contact pattern allows a lubricant-entry gap in tooth contact. When the gear deflects under load, the contact tends to move to a more central position on the bronze gear face, still allowing a lubricant-entry gap.

A contact pattern such as that shown in Fig. P-39 is the worst possible contact pattern under a no-load condition. This contact does not allow a lubricant-entry gap, and deflection under load will aggravate this condition. A gearset mounted in this manner may cause a temperature rise in oil 20 percent higher than that of the same gearset mounted as shown in Fig. P-38. The remedy is to move the gear axially

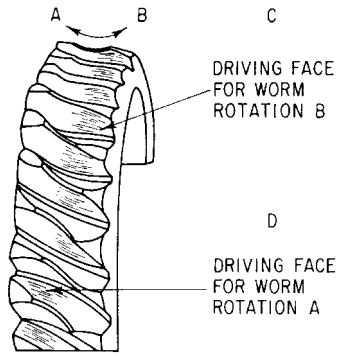


FIG. P-40 Driving face for worm notation. (Source: Demag Delaval.)

TABLE P-8 Minimum Recommended Number of Gear Teeth for General Design

Center Distance, in.	Minimum Number of Teeth
2	20
3	25
5	27
10	29
14	35
20	40
24	45

to the left (adjusting by shims or other adjustments provided) until a contact similar to that of Fig. P-38 is obtained.

When assembling worm gears that will run in both directions of rotation, it is necessary to consider both driving faces of the gear and to aim for contact as shown in Fig. P-40. When the worm is rotating in direction *A* in Fig. P-40, contact should be at *D* on the leaving side. When the worm is rotating in direction *B* in Fig. P-40, contact should be at *C* on the leaving side. For gears that will run in one direction only, it is necessary to obtain a contact pattern that is correct for the driving-side flank of gear teeth only.

Assembly adjustment. The gear should be mounted approximately on the centerline of the worm. The worm threads should be coated with prussian-blue dye. A section of the gear teeth should be coated with an orange-colored lead paste. The worm and gear should be rotated in both directions of rotation by hand. The blue markings from the worm threads will show the contact against the orange coating on the gear teeth. If the contact pattern is not as desired, the gear should be adjusted axially until a correct pattern is obtained.

Design considerations for worm and gearset. It is assumed that at the start of this design sequence the center distance for this gearset is known. (See Table P-8.)

The maximum number of teeth selected will be governed by high ratios of reduction and consideration of strength and load-carrying capacity.

Number of threads in worm. The minimum number of teeth in the gear and the reduction ratio will determine the number of threads for the worm. Generally 1 to 10 threads are used.

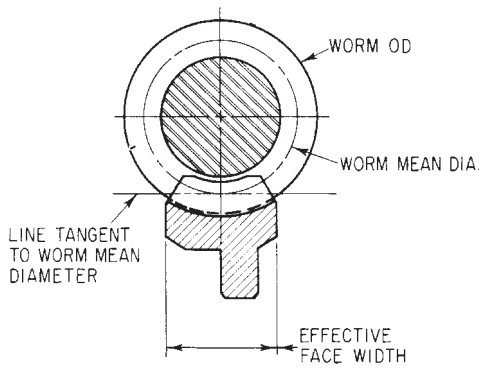


FIG. P-41 Gear-face width. (Source: Demag Delaval.)

Gear ratio

$$\text{Gear ratio} = \frac{\text{number of teeth in gear}}{\text{number of threads of worm}}$$

Pitch. Axial pitch of worm = circular pitch of gear. Keep the fraction simple so that accurate factoring can be used to determine change gears.

Worm pitch diameter. The pitch diameter of the worm is assumed to be at the mean working depth of the worm thread. The following factors should be considered when selecting worm pitch diameter:

1. Smaller pitch diameters provide higher efficiency and reduce the magnitude of the tooth loading.
2. The root diameter that results from pitch-diameter selection must be sufficiently large to prevent undue deflection and stress under load.
3. For low ratios the minimum pitch diameter is governed by the desirability of avoiding too high a lead angle. Lead angles up to 50° are practical.

Gear pitch diameter

$$\text{Gear pitch diameter} = 2 \times \text{center} - \text{pitch diameter of worm}$$

Recommended pressure angle. For general usage, pressure angles from 20 to 25° are common. Smaller values of pressure angle decrease the separating force, extend the line of action, making the amount of backlash less sensitive to change in center distance, and are used in index gearing. Larger values of pressure angle provide stronger gear teeth and assist in preventing undercutting of teeth with large lead angle. They are used in extremely heavily loaded applications.

Gear-face width (Fig. P-41). Maximum effective face width is the length of a line tangent to the mean worm diameter, to a point at which the outside diameter of the worm intersects the gear face. Any face width larger than this effective face width is of very little value and is wasteful of material.

Gear-throat diameter = gear pitch diameter + 2 × gear addenda. Gear outside diameter = gear throat diameter + 1 addendum of worm rounded off to the nearest fraction of an inch.

Gear blank under rim diameter (Fig. P-42)

$$h_t = \text{tooth depth of gear}$$

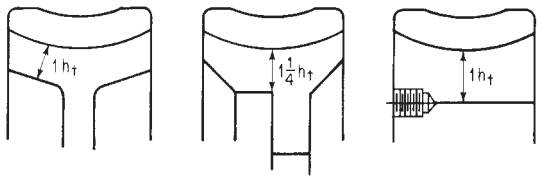


FIG. P-42 Gear-blank shapes. (Source: Demag Delaval.)

TABLE P-9 Bearing Loads

Resulting from	Bearing No. 1	Bearing No. 2	Bearing No. 3	Bearing No. 4
P	$Pa/L_W = P_1$	$Pb/L_W = P_2$	$Pr_G/L_G = U_3$	$Pr_G/L_G = U_4$
S	$Sa/L_W = S_1$	$Sb/L_W = S_2$	$Sd/L_G = S_3$	$Sc/L_G = S_4$
T	$Tr_W/L_W = U_1$	$Tr_W/L_W = U_2$	$Td/L_G = T_3$	$Tc/L_G = T_4$
Radial load	$\sqrt{P_1^2 + (S_1 - U_1)^2} = R_1$	$\sqrt{P_2^2 + (S_2 - U_2)^2} = R_2$	$\sqrt{T_3^2 + (U_3 - S_3)^2} = R_3$	$\sqrt{T_4^2 + (S_4 - U_4)^2} = R_4$
Thrust load	. . . . .	T	. . . . .	P

Underrim dimension for bronze gear block

= gear-root diameter – 2 to $2^1/_2 \times$ gear-tooth depth

Worm face

Minimum worm face

$$= z\sqrt{(\text{gear-throat diameter}/2)^2 - (\text{gear pitch diameter}/2 - \text{gear addendum})^2}$$

Allowable shaft stressed. All shafting in accord with AGMA Practice 260.01, March 1953.

Allowable bolt stressed. All bolts in accord with AGMA Practice 255.02, November 1964.

Bearing loading. All bearings selected in accord with AGMA Practice 265.01, March 1953. (See Figs. P-43 and P-44.)

Ball and roller bearings are selected on the basis of supporting loads equal to the maximum basic rating of the gear reducer and allow a minimum bearing life of 5000 h or an average life of 25,000 h. (See Table P-9.)

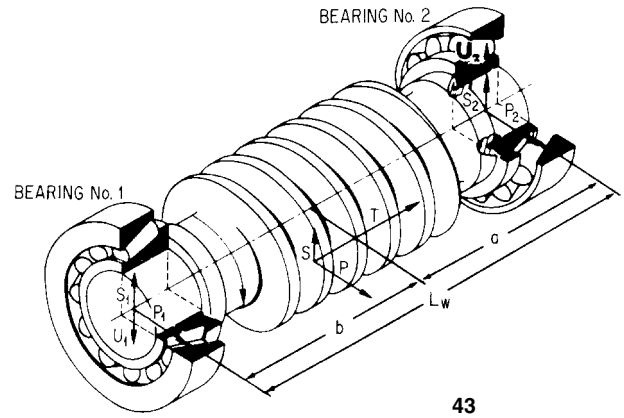
Performance

Mechanical ratings of cylindrical worm gears. The practice for this rating follows AGMA Practice 440.03, September 1959.

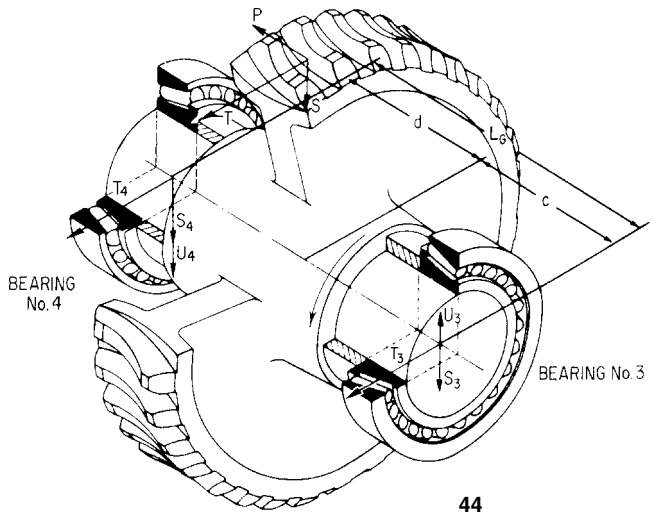
The ratings that are cataloged according to this practice are *wear* ratings that the gearset will satisfactorily permit, at the load shown, provided the driven machine has a uniform load requirement free of shock loading, 10 h/day. This is the basic rating by which worm-gear drives are selected, subject to thermal limitations.

Service factors are applied to this basic rating to factor the wear rating for shock loading or intermittent service.

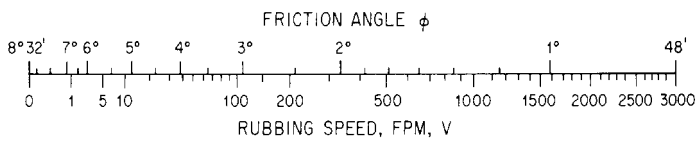
Thermal ratings of cylindrical worm gears. Thermal ratings above 100- to 200-rpm worm speed represent the input horsepower and output torque that will provide a stabilized 100°F oil-temperature rise over ambient air temperature when the machine is operated continuously. For example, if the ambient air temperature is 70°F, a reducer carrying rated thermal horsepower will operate with an average oil



43



44



Principal forces and bearing loads in a worm (FIG. P-43) and gearset (FIG. P-44). D_w = pitch diameter of worm, in; r_w = pitch radius of worm, in; r_G = pitch radius of gear, in; γ = lead angle of worm, °; P = tangential force on worm, lb; Q = torque input to worm, in · lb; S = separating force, lb; T = axial thrust of worm, lb; NPA = normal-pressure angle: ϕ = friction angle for worm driving; rpm (r/min) = worm speed; V = rubbing speed, fpm (f/min); $P = Q/r_w$; $S = P$ tan NPA/sin($\gamma + \phi$); $T = P$/tan($\gamma + \phi$); $V = 0.262\ D_w$ rpm/cos γ. (Source: Timken Roller Bearing Company.)

temperature of 170°F. Since normal worm-gear lubricants will deteriorate rapidly, require frequent replacement, and may not support the gearmesh loads when the machine is operating continuously at 210 to 220°F, the practical maximum ambient air temperature for worm-gear reducers carrying full thermal rating horsepower is 100°F.

For operation at higher ambient-air temperatures, a larger unit with a higher thermal rating must be selected for continuous operation, or a cooling system must be employed. For example, if a unit is to operate in an ambient-air temperature of 150°F, the increase in oil temperature must be limited to 50°F to keep the oil

TABLE P-10 Basic Lubricant Recommendations (AGMA)

	Size 60 Units and Smaller, Ambient Temp., °F		Size 70 Units and Larger, Ambient Temp., °F	
Worm Speed, rpm	15–60	50–125	30–60	50–125
Up to 400	7 Comp.	8 Comp.	7 Comp.	8 Comp.
Above 400			7 Comp.	7 Comp.

temperature from rising above 200°F. This means that the heat generated in the reducer must be one-half of the heat generated when the machine is operating at the catalog thermal rating; or since bearing and oil losses remain constant for a given speed, applied horsepower must be less than one-half of the catalog thermal rating.

For operation at ambients of less than a maximum of 100°F or when artificial or natural air drafts are present, catalog thermal ratings can be exceeded. For a proper evaluation, all data on ambient conditions should be determined.

Allowable starting load. Worm-gear reducers have a momentary overload strength rating + 300 percent of mechanical-wear rating. Peak starting load of the driven machine should not exceed 300 percent of the mechanical-wear rating.

Lubrication

General. Because of the nature of worm-gear sliding and rolling action, lubricants used for other types of gearing are not satisfactory. All units are shipped without oil, but reducer instructions and lubrication nameplates refer to the use of AGMA lubricants. Generally speaking, suppliers of *industrial lubricants*, *not* service stations, should be contacted and should be able to supply suitable lubricants from stock to meet these AGMA specifications (Table P-10). The units should be filled with the proper lubricant before operating.

These lubricants are basically a steam-cylinder oil. A list of trade names of the various manufacturers of oils that meet the AGMA 7 Compounded and AGMA 8 Compounded specifications is maintained by Delaval. These lubricants are basically petroleum-base oils but with 4 to 5 percent acidless tallow additives that provide additional film strength. They are heavy oils, much heavier than normal motor oils. The viscosity of AGMA 7 Compounded is approximately 135 SSU at 210°F, and that of AGMA 8 Compounded is approximately 150 SSU at 210°F. This heavy viscosity plus the plating action of the additives on the worm and gear contact surfaces is required to ensure the long trouble-free life that the gearing is designed to provide.

Lubricants not recommended. The following lubricants should never be used for worm gearing:

1. Ordinary motor oils, no matter what their viscosity.
2. Automotive rear-end oils.
3. Extreme-pressure lubricants containing compounds of sulfur or phosphorus. It may be claimed that these lubricants are noncorrosive to steel, but they are extremely corrosive to bronze and will not provide the necessary plating action required.
4. Greases of any kind. These do not flow sufficiently to provide the necessary cooling.

TABLE P-11 Cold-Weather Lubrication

For Min. Ambient Temp., °F	Use a Mild Extreme-Pressure Oil Containing Lead Naphthanate and Having a Viscosity of
0	120 SSU at 210°F
–10	100 SSU at 210°F
–20	75 SSU at 210°F
–30	53 SSU at 210°F

Cold-weather lubricants. If ambient temperatures below 15°F are expected, a winter, or cold-weather, lubricant must be selected, since the AGMA 7 Compounded or 8 Compounded will solidify and the motion of the gears will channel the solidified oil until no lubricant is present at the gear mesh. For this condition, a minimum ambient temperature to be expected must be estimated and a reputable supplier consulted to recommend an oil with a channel point well below the expected minimum ambient temperature. This will require a lighter-viscosity oil, but the oil should still contain additives. The best selection is usually the mild extreme-pressure oils containing lead naphthanate with the viscosities shown in Table P-11. The lubricant should be changed to the heavier-viscosity oils when the ambient temperature again goes above 15°F.

Frequency of oil changes. The frequency of oil changes varies with the type of service. After the initial 50 to 100 h of running, a change should normally be made to remove the particles of bronze burnished off the gear during the run-in period. Thereafter, a general rule is that the oil should be changed every 6 months of normal service and every 3 months of severe service. However, if the unit is in a dusty or moist atmosphere, dirt or water accumulation in the oil reservoir may require more frequent changes. Many oil suppliers will test a lubricant after a period of use free of charge and determine its useful life for a specific application.

Procedure for long shutdown periods. If the unit is to be idle for any length of time, particularly outdoors, something must be done to prevent rusting of the bearings, gears, and other internal parts. The easiest solution is usually to fill the unit completely with clean oil. Of course, before the unit is started again, the oil should be drained and refilled to its proper level.

Installation and operation

Installation. Normal good practice must be followed when handling the unit, choosing a foundation, checking alignment, and mounting couplings, pulleys, gears, sprockets, etc. Couplings should be pressed or shrunk on the reducer shafts. Do not drive couplings on shafts, as this may damage the bearings and also cause the shafts to spring. This, in turn, may result in failure of the bearings, vibration, and oil leakage. Sprockets, pulleys, and pinions should be mounted as close to the case as possible in order to avoid undue bearing-load and shaft deflection.

Operation. The unit is shipped from the factory *without oil* but is slushed internally with a rust-preventive compound, which need not be removed because it is oil-soluble. *Make certain that the reducer is filled to the correct level before start of operation in accordance with lubrication specifications*. The unit must be filled to, but not above, the oil-level gauge. The oil level will, of course, change with the mounting arrangement. It should be checked periodically and only at a time when the unit is not operating. A dipstick is provided in the oil-level gauge.

All units are subjected to test before shipment, but it takes additional hours of running under full gear load to attain highest efficiency. The gear may, if necessary, be put to work immediately on full load, but if circumstances permit, it is better for the ultimate life of the gear to run it under gradually increasing load. Immediate application of full load concentrates high unit pressures on tooth surfaces. When new driven equipment requires operation to achieve freedom and minimum friction loss, use precaution in the early stages of operation to prevent the reducer from taking an overload. When overload tests are specified on a machine before it is shipped, it is better to make preliminary runs under part load before building up to full load and overload. A reasonable running-in procedure is half load for a few hours, building up to full load, in two stages if possible.

Temperature rise on the initial run will be higher than that eventually attained after the gear is fully run in.

Some slight wear and/or pitting of the bronze gear teeth may be observed after a short period of initial operation. This condition is normal, as some initial wear is necessary for the hardened-steel worm to seat itself properly with the bronze gear.

Product Application Case 1: High-Speed Gears—New Developments*

High-speed gears are gears operating at high pitch line velocities up to 240 m/s and high power. The requirement to transmit significant power at extreme speeds, i.e., to have high speed *and* load on the bearings and the toothing, led to some new developments for such components.

A back-to-back test bed has been installed for research purposes. The gears transmit a power of 30,000 kW at 6380/15,574 rpm nominal speed, with the capacity of 120 percent overspeed.

A new type of radial tilting pad bearing designed by this OEM will be discussed. The face width to diameter ratio is 1.4, which enables the bearing to take higher rotational speeds at reasonable specific loads. Further, a new design for a double or multiple tilting pad thrust bearing is explained. The bearing can be used in any kind of machinery that is exposed to high axial thrusts at very high speeds or where an utmost compact design is required due to limited space.

Compressors and other turbomachines are constantly developed and improved to run at higher speeds with more power. The gear manufacturer therefore has to design gears that can transmit high toothing forces at very high rotational speeds. But the toothing and the bearings are bound to certain limits for thermal and mechanical load that must not be exceeded for safe operation. These limits have been elevated over the last years by continuous development of toothing and bearings; see Fig. P-45 for the pitch line velocity.

To solve the basic design problem of high speed and power, there are two general possibilities: to design a gear with power split, i.e., with two or more power paths or to increase the limits for toothings and bearings by development of these components.

Gears with power split reduce the load on bearings and toothing and allow the designer to stay within well-known limits. On the other side, such gears have a more complex layout, they have more tooth meshes and more bearings and they need some mechanisms for a correct power split. These mechanisms can be quill shafts, self-adjusting bearings, or others. The costs to build such a complex gear must be almost twice as much as for a simple two-shaft gearbox.

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* Source: MAAG Gear Company, Switzerland.

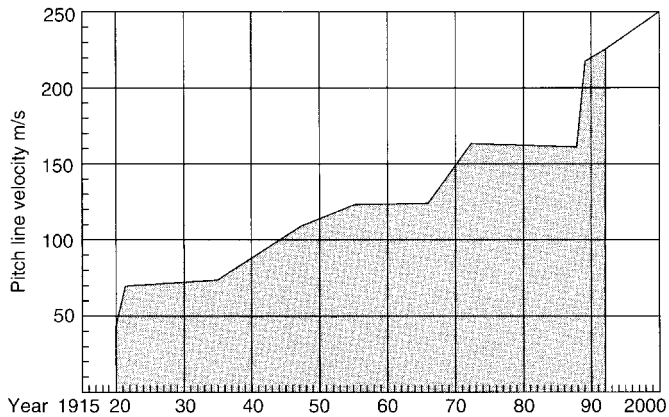


FIG. P-45 Development of toothing pitch line velocity since 1920. (Source: MAAG Gear Company.)

Therefore, the question is put: How can toothing and bearings be further developed in order to allow higher limits for load and speed? More power and speed could be transmitted in simple, reliable, and cost-effective two-shaft gear designs.

In 1985, this OEM started a research program to investigate bearings and toothings. Gears with high power and speed were designed to be carefully tested on a special back-to-back test bed. During the design phase, it was recognized that before the toothing, the pinion bearings reached their load limits. New radial and axial tilting pad bearings had to be developed for these gears to allow for safe operation with maximum white metal temperatures below 130°C. The design of these new bearings as well as the test results under full load are presented.

"Back-to-back" test bed

The mentioned back-to-back test bed is shown in Fig. P-46. It consists of two identical gearboxes that are mechanically coupled with a torque meter device on the low-speed shafts and with a special highest speed toothed coupling on the high-speed shafts.

With the single helical toothing, the gears can be loaded up to full load just by applying an axial force on the wheel. The axial shifting causes a rotary movement and the wanted closed torque circuit between the two gears is established. The axial force is produced by means of hydraulic pistons. With a variable hydraulic pressure on these pistons, any desirable load between zero and full load can be achieved. As driving power, only the total losses of the two gears have to be provided.

Some technical data of the back-to-back test bed:

Nominal power: P = 30,000 kW

Nominal speed: n = 6380/15,574 rpm

Overspeed 120%: n = 7656/18,689 rpm

Nominal pitch line velocity: v = 200 m/s

Overspeed plv.: v = 240 m/s

Center distance: a = 422 mm

Two sets of gearwheels are tested: one with a helix angle of 13°, the other with 19°. The rotors are equipped with strain gauges at the tooth root and with

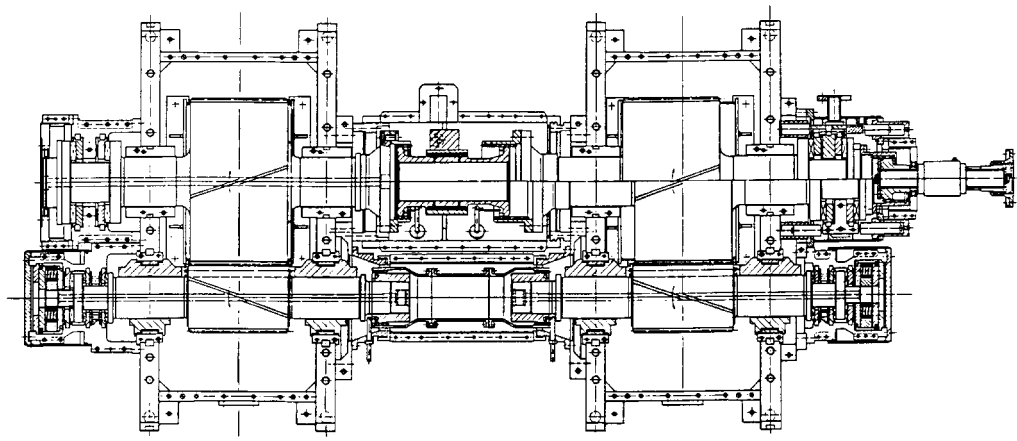


FIG. P-46 Back-to-back test bed. (Source: MAAG Gear Company.)

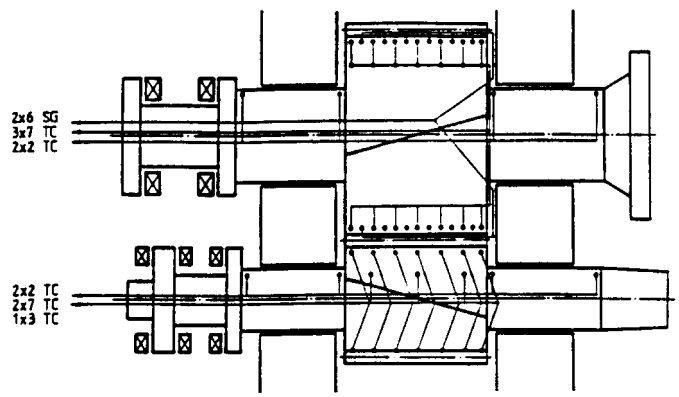


FIG. P-47 Instrumentation of test gear rotors. (Source: MAAG Gear Company.)

thermocouples near the toothing and journals. With these instruments, thermal distortions due to unequal temperature distribution and actual load distribution across the face width of the toothing are measured. The bearings are equipped with thermocouples in the hottest zones in order to determine maximum white metal temperatures and thermal deformation. Figure P-47 shows the instrumentation of the test gear rotors.

All shafts equipped with instruments are hollow to take all the cables which are connected via special high-speed slip rings to the static analysis instruments. Rotor vibrations are to be surveyed by pic-up's; all other instruments and sensors are as for a common industrial platform gear. The wheel of the slave gear is not equipped with sensors, but the hydraulic axial force device and the input motor drive are connected to it. At the input shaft, the total losses of both gears can be measured by means of a torque meter coupling.

The toothed coupling between the two pinions is a specially designed coupling for highest speeds. Its weight and overhang have been minimized in order to satisfy lateral critical speed requirements of the pinions. Without this coupling it would not have been possible to find a satisfactory solution for save operation at these speeds. Every other type of coupling, for example a disc coupling, has more weight and more overhang and is therefore only of limited use for extreme high-speed gears.

TABLE P-12 Comparison of Important Design Parameters of Conventional and New Bearing Design for Back-to-Back Gears

	Conventional	New Design
D (mm)	170	150
B/D	1.0	1.4
p (N/mm^2)	3.0	2.8
v (m/s)	139 (167)	122 (146)
T_{max} (°C)	132 (141)	121 (130)

Values in parentheses refer to overspeed 120%.

Due to thermal expansions and friction in the toothed coupling, additional axial forces will act on the gears. Earlier back-to-back tests showed that for the unloading of such a unit, it is not sufficient just to release the axial force. The gears had to be unloaded by applying an axial force in the reverse direction in order to overcome the friction in the toothed couplings. The friction in toothed couplings is existing and produces axial reaction forces that cannot be neglected.

This led to the knowledge that extreme high-speed gears, which must be equipped with toothed couplings for lateral critical speed reasons, should have a single helical toothing that is not affected by additional external thrusts.

On a double helical toothing, an external thrust would act on one helix only, which is considered a worst-case situation for the toothing.

Radial tilting pad bearings

Designs. For high-speed gears, white metal–lined slide bearings are commonly used. The known limits for such types of bearings are as follows:

- Specific load 3,2 . . . 4 N/mm^2
- Maximum white metal temperature 130°C
- With circumferential speeds above 90–100 m/s, tilting pad bearings should be used in order to avoid bearing instabilities due to oil whip

During the design phase of the back-to-back gears, it became evident that the pinion bearings could not be realized with a conventional design. A new design had to be found in order to keep the white metal temperatures within specified limits. The solution is a specially developed tilting pad bearing with the following design features:

- Ratio width/diameter for the main pad: 1:4
- Three pads, one main pad and two auxiliary pads
- The main pad has a circumferential groove in the center to evacuate the hot oil

Materials and fabrication methods are the same as for conventional bearings.

Table P-12 compares important design parameters of conventional and new bearing design for back-to-back gears. The general design of the bearing is shown in Fig. P-48. The circumferential groove is important for reduction of thermal deformations due to temperature gradients over the face width of the bearing.

Test results. The back-to-back gears have been run at full load and up to 120 percent speed for many hours. The bearings, as well as the toothing, have been

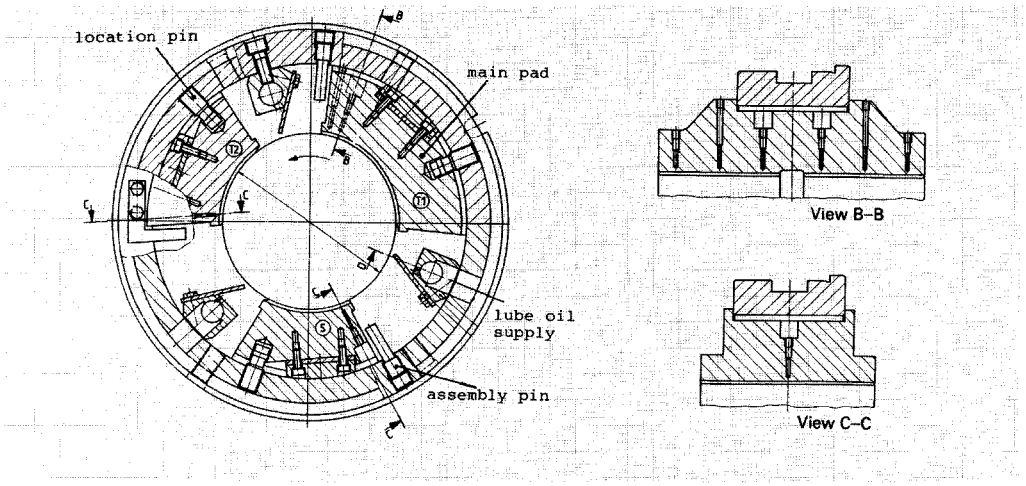


FIG. P-48 MAAG radial tilting pad bearing. (Source: MAAG Gear Company.)

operating very satisfactorily during the whole testing period. Careful inspection after test runs did not show any sign of wear or damage. This is not surprising since bearings and toothing are designed for infinite life.

The radial tilting pad bearings were equipped with thermocouples in the hottest area across the full width of the main pad (see Fig. P-49).

Measured maximum white metal temperatures are shown in Fig. P-50.

An analysis of the measured white metal temperatures and other test results leads to the following conclusions:

- The maximum values have always been lower than 125°C, at full load and full speed
- Measured values are very close to the calculated mean temperature of 121°C
- The circumferential groove stabilizes the temperature gradient at a lower level
- At overspeed 120 percent, i.e., with a circumferential speed of 146 m/s, no signs of oil whip occurred; the stability of the proposed tilting pad bearings is, as expected, very good
- Pinion lateral vibrations have always been below 1.1 mils, even at no load and overspeed; the lateral vibrations behavior of pinion and toothed coupling is good, the damping of the bearings is satisfactory

Deformation analysis. An important question for the development of a wide bearing is the heat distribution across its width and the resulting deformations. If these deformations are too large, the outer parties of the bearing will not take significant load and the heat will be concentrated in the center. This will result in overload for the bearing and consequently lead to damage. The optimal compromise for the B/D ratio must be found. After theoretical and experimental investigations, a 1.4 ratio was determined. With finite-element analysis, the deformations of the bearing were calculated. The stationary temperature distribution in the steel body of the bearing and the pressure distribution in the oil film are acted as loads on the main pad. Figure P-51 summarizes in a compromised sketch the complete results of the FE-analysis.

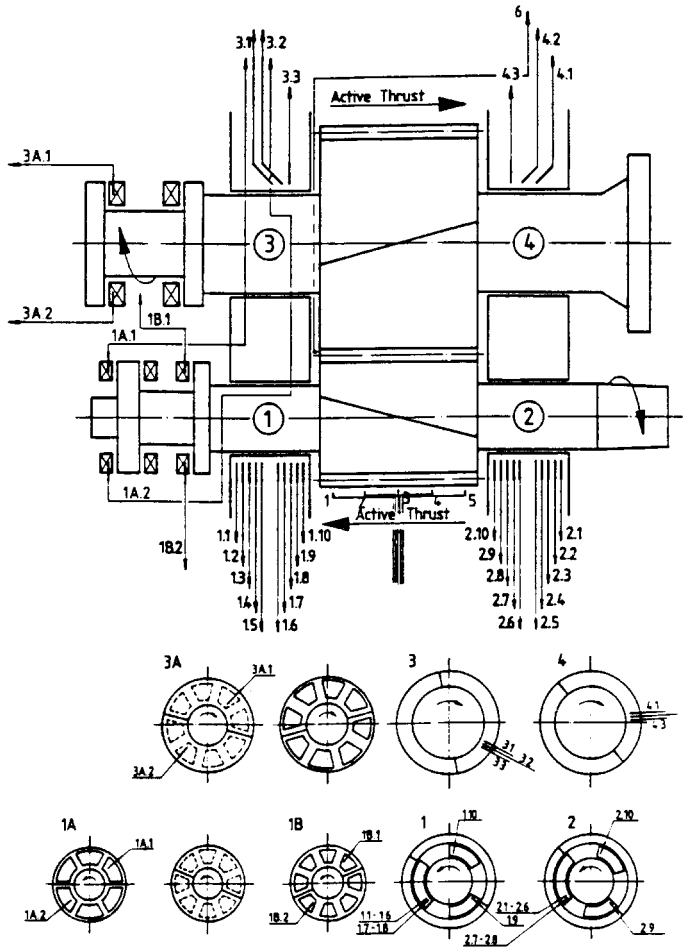


FIG. P-49 Instrumentation on bearings. (Source: MAAG Gear Company.)

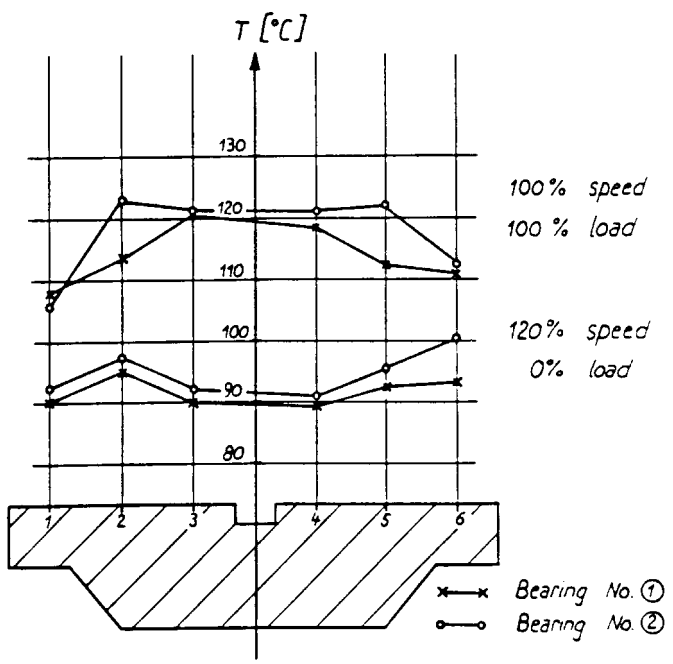


FIG. P-50 Measured white metal temperatures. (Source: MAAG Gear Company.)

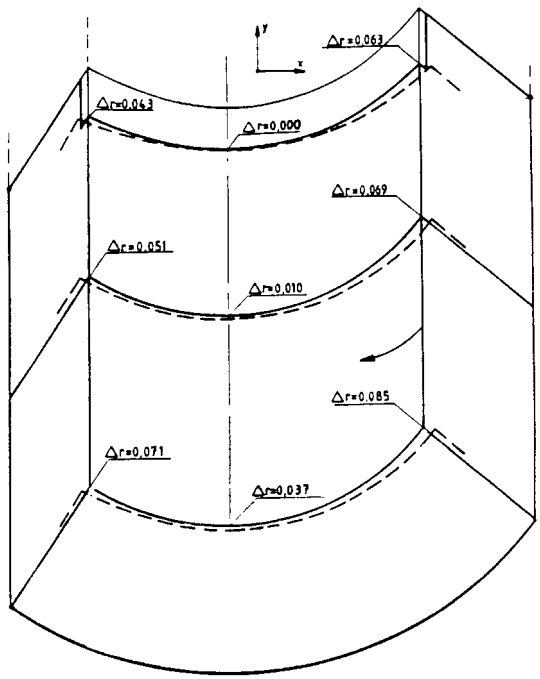


FIG. P-51 Finite-element-analysis radial deformations under thermal and mechanical load. (Source: MAAG Gear Company.)

The following conclusions can be drawn:

- 80 percent of the combined deformation is caused by unequal temperature distribution and only 20 percent by mechanical bearing load. It is, for this reason, of utmost importance to keep the temperature in the bearing low and uniform.
- The maximum deformation at the outer end of the bearing is approximately 0.08 mm. Supposing a minimum oil film thickness of 0.03 to 0.04 mm under load, it is evident that even the outer areas of the main pad will contribute to take load.

Axial tilting pad bearings

Design. High-speed gears are often designed with a single helical toothing, i.e., toothing thrust is to be compensated. In addition, external thrust from couplings can act on the high-speed shaft. The helix angle of the toothing must be high enough to limit the heat generation in the gear mesh. Thus, the question is how to absorb high axial forces at high speeds. The problem cannot be solved with a single axial tilting pad bearing or by shrunk-on trust collars because of load limits and centrifugal forces.

For the described back-to-back gears, two axial bearings have been arranged in series. It was possible to find a mechanism that allows for any power split between several axial bearings. These mechanisms are described in detail in the international patent description.

The basic principle is explained with Fig. P-52: On the upper half, the unloaded axial bearings are shown; below is the same arrangement under axial thrust. Each axial bearing (5) is supported by a ring (6) that rests on an arrangement for load adjustment (7, 8). These arrangements are supported by a rigid casing (9) that is connected to the main (gear) casing. The arrangement for load adjustment consists

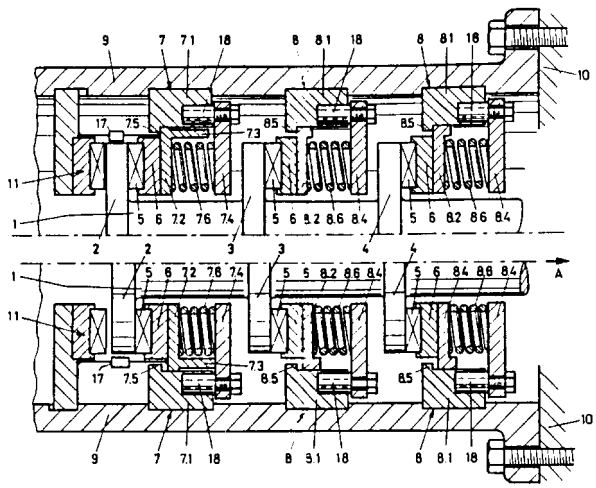


FIG. P-52 Multiple thrust bearing principle. (Source: MAAG Gear Company.)

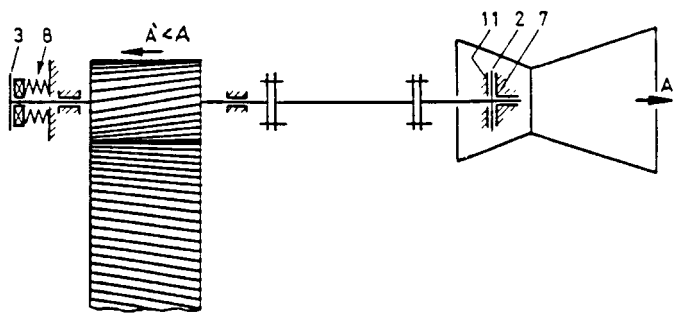


FIG. P-53 Thrust splitting between gas turbine and gearbox. (Source: MAAG Gear Company.)

of movable pistons that are preloaded by compressed springs. The preload for these springs can be chosen according to the load limit of the used axial bearing or to the preferred stage load for certain operating conditions. The piston of the last arrangement (7) has a stop (7.3) that limits the total axial mobility. The axial bearing clearances of the various stages are different; they increase from the inner to the outer bearings. With increasing axial thrust, the inner bearing starts to take load first until its spring preload limit is reached. An axial movement takes place until the second bearing has load and so on. With full axial thrust, each bearing has load according to the predetermined load sharing.

There are many different variations of the described basic principle. A very interesting one is the thrust split between rigidly coupled machines, for example, in the Fig. P-53 turbine and gearbox. Here, the toothing thrust helps to unload the turbine's axial thrust bearing. In addition, a well-defined part of the total thrust is absorbed with a bearing in the gearbox, without hindering the shafts to expand thermally. The bearing clearances have to be adjusted in a way that the gear's axial bearing is loaded even with maximum shaft expansion. The stationary bearing (7) can be positioned at the cold end of the turbine that helps to keep the thermal expansions to a minimum.

The described arrangement for axial bearings can be used not only for gears but principally for any rotating machines where high thrusts at high speeds must be

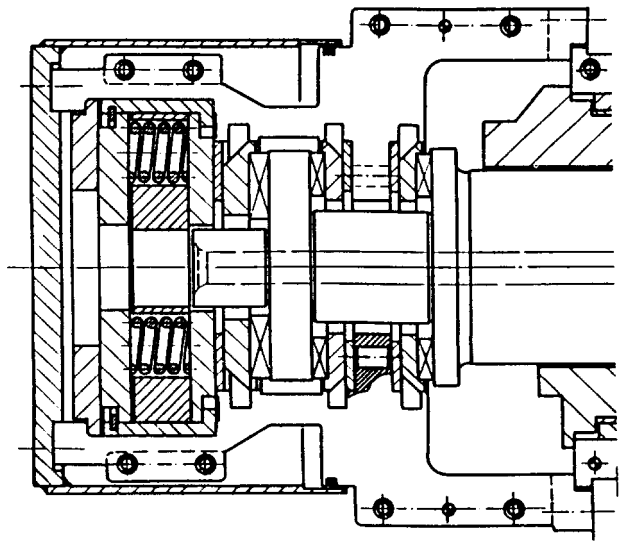


FIG. P-54 Double thrust bearing design of test gear. (Source: MAAG Gear Company.)

absorbed. It is of further interest that with a correct design, the power losses of two small bearings in series are lower than of one big bearing alone.

Test results. For the pinions of described back-to-back gears, the axial bearings have been designed according to the load-sharing principle (see Fig. P-54).

The toothing thrust is acting outward and the outer bearing will be loaded first with increasing power transmission.

Technical data are as follows:

Pinion speed: 15,574 rpm

Helix angle: 19°

Total axial thrust: 81′000 N

Preload of springs: 50′000 N

Specific bearing loads: 3.5/2.35 N/mm^2

Circumferential bearing velocities: 137/144 m/s

Bearing power losses: 34/31 kW

It would not be possible to take such a high total thrust with a single bearing at this speed. For the case of torque reversing, a "back" bearing is installed. Oil supply and instrumentation with thermocouples in the pads are realized exactly the same way as for a conventional single bearing design. The natural frequency of the preloaded piston springs was chosen to be different from potential exiting frequencies.

The results from back-to-back testing can be summarized as follows:

- No wear, no running tracks
- Maximum white metal temperatures well within permissible limits, as calculated in advance
- No axial vibrations
- Free mobility of piston, no running tracks

From these results after many hours of operation, one can conclude that the axial load sharing according to the described principle works.

Summary and outlook

The design of high-speed gears with two shafts is limited by the bearings and the toothing. Often the bearing limits are reached before the toothing becomes critical.

New radial tilting pad bearings have been developed. With these bearings, the rotor speed can be further increased at the same bearing load. In order to absorb high axial thrust at high speed, a new principle has been developed that allows for several axial bearings arranged in series. The load sharing between the stages can be chosen to any desirable values.

Both bearing designs have been tested on a back-to-back test bed with pinion speeds of more than 17,000 rpm and power up to 30,000 kW. These tests have proven that both designs are ready for industrial application.

What is the possible increase in speed? Assuming a maximum white metal temperature of 130°C and load, specific load, and circumferential bearing velocity to be constant, one can find the following relation:

$$n \text{ (increased)} \approx \sqrt{1.4} \cdot n \approx 1{,}18 \cdot n;$$

where n = max. perm. speed with "conventional" bearings.

This means that with the same thermal load on the pinion bearings, the pinion speed can be increased by approximately 20 percent when tilting pad bearings of the new type are used. Consequently, the range of high-speed gears based on a cost-effective two-shaft design is considerably increased.

Synchronous Clutch Couplings and Applications*

Synchronous clutch couplings (see Fig. P-55) are required today in a wide range of applications. These include, but do not exclusively consist of, applications in the following.

Power generation

- Alternator drives
- Peaking power stations
- Air storage power stations

Energy recovery, combined cycle technologies, cogeneration and others

- Connecting expander turbines to main drives in petrochemical plants
- Blower drives in nuclear power stations used during starting sequence
- Starting device for gas turbines
- Automatic turning gears

Synchronous clutch couplings are couplings that engage and disengage automatically. They are capable of engaging automatically at any speed within the operating range as soon as the driving machine overruns the driven machine. Basically, the synchronous clutch coupling is a disengagable coupling equipped with a mechanism (the synchronizing mechanism) that detects synchronism of both shafts and initiates the engaging movement.

The synchronous clutch coupling consists of two main elements:

* Source: MAAG Gear Company, Switzerland.

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